



Certifiable Trajectory Optimization for Rigid-Body Systems on Lie Groups via Semidefinite Relaxation

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Problem description:

Nonlinear discretized dynamics and nonconvex geometric constraints yield a highly nonconvex optimization landscape; in contact-rich regimes, the discontinuity of contact forces further degrades it [4]. Classical approaches typically rely either on nonlinear programming (which is highly sensitive to initialization) [5] or on mixed-integer formulations [1] (which incur worst-case exponential complexity).

Locomotion requires reasoning about nonlinear dynamics, discontinuous contacts, and nonconvex geometry (self/obstacle avoidance). These properties create hard optimization landscapes and brittleness to initialization. In contrast, Moment-SOS semidefinite relaxation offers: (1) a convex lower bound that certifies suboptimality; (2) principled extraction of candidate trajectories; and (3) natural ways to exploit problem sparsity (chain-like temporal structure and robot-specific sparsity from kinematic chains and separable contacts) [2, 6]. Specialized first-order SDP solvers (including GPU-based ADMM variants) make these relaxations practically viable [3]. This thesis translates these ideas into legged locomotion on real hardware, using a certifiable trajectory optimization framework for locomotion tasks via sparse semidefinite relaxation.

Tasks:

- Review relevant literature on Lie-group based dynamics modeling for legged locomotion, including contact modeling and polynomial optimization methods on manifolds.
- Derive and formulate discrete-time Lie-group variational dynamics and express objectives and constraints as polynomial polynomialized optimization problems exploiting system structure.
- Develop sparse Moment-SOS / semidefinite relaxations that leverage temporal, kinematic, and contact sparsity for efficient and certifiable trajectory optimization.
- Evaluate certifiable solvers with trajectory extraction and validate the approach in simulation with respect to performance, robustness, runtime, and certification gap.

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Abstract

Robotic systems composed of coupled rigid bodies are characterized by non-linear dynamics and nonconvex rotational and geometric constraints, making reliable trajectory optimization challenging. Semidefinite programming (SDP) is used to construct a convex relaxation of the original polynomial optimization problem. This thesis combines Lie group variational integrators (LGVIs) with sparse second-order Moment-SOS semidefinite relaxations to obtain structure-preserving polynomial models and analyze certificates of global optimality. LGVIs are derived and numerically validated for a controlled 3D pendulum on $SO(3)$ and an underactuated Acrobot represented in maximal coordinates on $SO(2)$. The Acrobot model is algebraically reduced from $17N+3$ to $13N-1$ decision variables while preserving quadratic dynamics.

For the 3D pendulum, all evaluated target rotations produce numerically rank-one relaxations and feasible extractions whose objective values agree with the SDP lower bounds, certifying global optimality. For the Acrobot, the 0° and 45° targets are globally certified, whereas the 90° , 135° , and 180° cases produce higher-rank moment matrices and infeasible full-horizon extractions. A problem-specific multi-body clique construction achieves comparable relaxation quality while reducing the solver runtime by a factor of approximately three to five.

For non-tight Acrobot problems, an SDP-based model predictive control implementation repeatedly applies the first extracted control and replans from the simulated state. With a $15Nm$ control bound, configurations with matched prediction and model discretization time steps stabilize 94 of 100 sampled initial states, compared with 34 of 100 under a time-discretization mismatch. The results demonstrate that the proposed framework can certify globally optimal trajectories for tight problems and generate feasible simulated trajectories for harder multibody problems. However, the computation times restrict the method to offline applications.

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Chapter 1

Introduction

The future of robotic systems trends towards autonomous operations in unstructured environments. Thereby, reliable and dynamically accurate motion planning is a fundamental requirement for the operation of robotic systems. Essential applications are search and rescue operations, naturally hazardous to humans, in which especially legged robots have the potential to gain access and precisely navigate through complex terrain (see Figure 1.1(a)) [LWLM07]. But also in industrial contexts, robots assembling various components require careful and precise motion planning. For example, Figure 1.1(b) illustrates a thermal processing application [The22]. Both applications consist of different tasks and constraints. However, both systems pose the same fundamental problem. Several rigid bodies must be moved from an initial configuration to a desired configuration by admissible control inputs, while their rotations, joint connections, nonlinear dynamics, and actuator limits must be satisfied simultaneously. Thereby, generated trajectories can be restricted to avoid collisions, while being dynamically feasible [KLY25].

In addition to a large number of system variables, rigid-body attitudes belong to rotation groups, considered as Lie groups, rather than to a Euclidean vector space. Joint constraints couple the configuration of neighbouring bodies, and the equations of motion couple states and controls over time [TJVG25]. Consequently, even a finite-horizon problem without contact or obstacles generally contains nonlinear equalities and nonconvex rotation constraints. Its feasible set can have several disconnected regions, and its objective can contain many local minima [KXS⁺26]. Moreover, the continuous-time dynamics are high dimensional as state and control trajectories are functions of time. So any numerical solution must discretize these problems. Nevertheless, classic global optimization methods simplify the dynamics of robotic systems by approximating them with simpler models [TJVG25]. Thereby, the discrete dynamics can differ from the continuous dynamics through loss of geometric structure and manifold drift, which results in deviations from the true solution and numerical infeasibility [MW01].

Currently, nonlinear programming is one of the most common approaches for computing robot trajectories. Starting from an initial guess, these methods repeat-



(a) Legged locomotion



(b) Industrial robot

Figure 1.1: Examples of robotic systems composed of coupled rigid bodies: (a) legged locomotion taken from [San22] and (b) industrial manipulation taken from [The22].

edly adjust the states and control inputs to find a trajectory with a low objective value that satisfies the given constraints [Bet98, Ted23]. Sequential convex methods, such as TrajOpt, follow a similar idea but replace the original difficult problem with a sequence of simpler convex problems [SDH⁺14]. Both approaches can find useful motions efficiently, even for relatively large robot models. However, the final result may depend strongly on the initial guess and provides a feasible solution without a guarantee of optimality. Sampling-based methods instead explore many possible configurations and connect them to construct a path. Although some of these methods are asymptotically optimal, a path obtained after a finite computation does not provide a certificate of global optimality [KF11].

To address the limitations of local methods, this thesis investigates semidefinite programming (SDP) for multibody systems. An SDP is a convex optimization problem formulated using matrix variables. Instead of solving the original nonconvex motion-planning problem directly, a semidefinite relaxation constructs a convex approximation that can be solved reliably and provides a lower bound on the globally optimal objective value [VB96, Las01]. In contrast, a feasible trajectory, for example generated by a local optimization method such as NLP or TrajOpt, provides an upper bound. If the suboptimality gap between the SDP lower bound and the feasible-trajectory upper bound is within the specified numerical tolerance, the trajectory is *certified* as globally optimal for the considered discrete motion-planning problem [TJVG25].

However, realistic trajectory optimization also requires a discrete model that represents the rigid-body dynamics accurately. Lie group variational integrators (LGVIs) discretize the equations of motion directly on the configuration manifold and update rotations through the Lie-group structure [MW01, Lee08]. Consequently, the numerical rotations remain on the rotation group. LGVIs also preserve important geometric properties of the mechanical system, such as its symplectic structure and energy conservation over a long time duration [TJVG25].

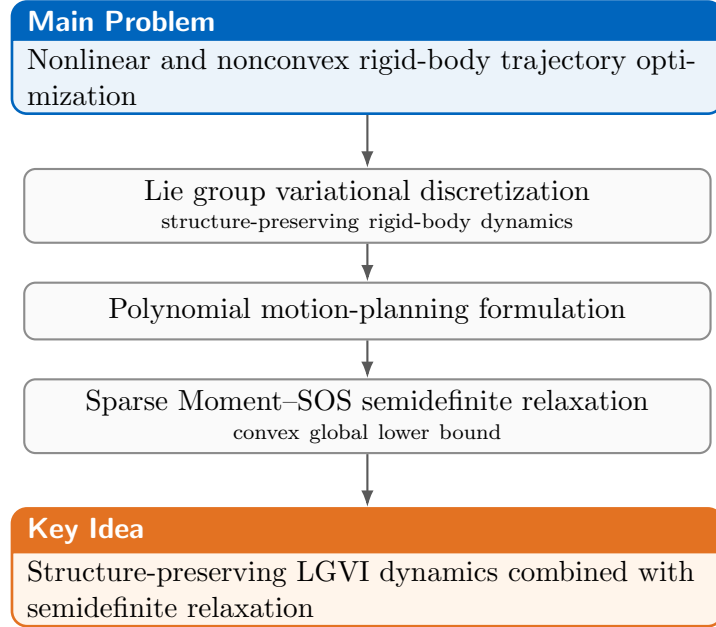


Figure 1.2: Framework of this thesis. Lie group variational discretization provides structure-preserving rigid-body dynamics, while sparse semidefinite relaxation provides a convex lower bound for the resulting polynomial motion-planning problem.

Therefore, the key idea of this thesis is to combine these two approaches. For the considered rigid-body systems, the LGVI dynamics can be expressed through low-degree polynomial equations. The resulting polynomial motion-planning problem can then be approximated by a Moment-SOS semidefinite relaxation [TJVG25]. Thereby, the LGVI provides a structure-preserving model of the rigid-body dynamics, while the SDP provides a global lower bound and certifies global optimality. This is the case when the relaxation is tight and a feasible trajectory can be extracted. The complete concept is summarized in Figure 1.2.

This thesis investigates a 3D pendulum as a single-rigid-body system and an Acrobot as a constrained and underactuated multibody system. The main contributions of this thesis are:

- The explanation of an LGVI for the controlled and unforced 3D pendulum on $\text{SO}(3)$, providing a structure-preserving discrete model of a single rigid body.
- The derivation of a constrained LGVI for the Acrobot in maximal coordinates on $(\text{SO}(2) \times \mathbb{R}^2)^2$. Thereby, the positions and rotations of both rigid links are represented separately and connected through holonomic joint constraints.
- The numerical implementation and validation of both LGVI models, including the preservation of the rotation and joint constraints, and the conserved long-term energy behavior.

- The formulation of the 3D-pendulum polynomial optimization problem (POP) as a polynomial optimization problem and its solution through a sparse Moment-SOS semidefinite relaxation. This is followed by an assessment of relaxation tightness, extraction feasibility, and the availability of global-optimality certificates.
- The formulation of the Acrobot POP using a reduced polynomial model, problem-specific sparse SDP structures and its solution through a sparse Moment-SOS semidefinite relaxation. Finally, the evaluation of the extracted trajectories and their global optimality certificates. Thereby, the evaluation is extended to a closed-loop implementation.

1.1 Problem Statement

The framework illustrated in Figure 1.2 addresses the following general motion planning problem. Thereby, the objective is to optimize the path of a rigid-body system from an initial configuration towards a desired configuration, while respecting the system's dynamics, geometric structure, joint connections, and physical limits. The motion is considered over a finite horizon divided into N intervals. Let $\mathbf{Y}_k$ describe the configuration and discrete motion of the system at step k , let $\mathbf{U}_k$ denote the applied control input, and let $\boldsymbol{\lambda}_k$ contain joint-constraint multipliers when several bodies are connected. The rigid links are indexed by $i \in \mathcal{B} := \{1, \dots, n_b\}$. At a high level, the corresponding motion-planning problem can be written as

$$p^* = \min_{\{\mathbf{Y}_k\}_{k=0}^N, \{\mathbf{U}_k, \boldsymbol{\lambda}_k\}_{k=1}^N} \Psi(\mathbf{Y}_N) + \sum_{k=0}^{N-1} \psi(\mathbf{Y}_k, \mathbf{U}_{k+1}, \boldsymbol{\lambda}_{k+1}), \quad (1.1a)$$

$$\text{s.t. } \mathbf{Y}_0 = \mathbf{Y}_{\text{init}}, \quad (1.1b)$$

$$\mathcal{D}_i(\mathbf{Y}_{k+1}, \mathbf{Y}_k, \mathbf{U}_{k+1}, \boldsymbol{\lambda}_{k+1}) = 0, \quad (1.1c)$$

$$(\mathbf{Y}_{k+1}, \mathbf{Y}_k, \mathbf{U}_{k+1}, \boldsymbol{\lambda}_{k+1}) \in \mathcal{A}_k, \quad (1.1d)$$

for $k = 0, \dots, N-1$. The objective combines the terminal cost Ψ with the running cost ψ . Additionally, the constraints $\mathcal{D}_i = 0$ represent the discrete rigid-body dynamics of link i . For this compact statement, the admissible set $\mathcal{A}_k$ collects the Lie-group constraints, inequality constraints, and control bounds that are written separately as $\mathcal{Y}_i$, $\mathcal{H}$, and $\mathbf{U}_{\min} \leq \mathbf{U}_{k+1} \leq \mathbf{U}_{\max}$ in Chapter 5. Therefore, (1.1) summarizes the physical requirements of the problem, generalizing it to any robotic system [KXS⁺26].

Generally, the objective appears simple. However, for many robotic systems composed of many rigid bodies, the equality constraints $\mathcal{D}_i$ are nonlinear, and the rotation constraints and equations of motion make the feasible set nonconvex. The LGVI formulation is useful because the considered dynamics can be expressed by low-degree polynomials and have energy-conservation properties [TJVG25]. After

collecting all scalar decision variables in $\mathbf{z} \in \mathbb{R}^{n_z}$, (1.1) takes the standard polynomial form

$$p^* = \min_{\mathbf{z} \in \mathbb{R}^{n_z}} p(\mathbf{z}) \quad \text{s.t.} \quad h_i(\mathbf{z}) = 0, \quad i \in \mathcal{E}, \quad g_j(\mathbf{z}) \geq 0, \quad j \in \mathcal{J}, \quad (1.2)$$

[Yan24]. Here, p^* is the globally optimal value, h_i are polynomial equality constraints, and g_j are polynomial inequality constraints, with their respective sets $\mathcal{E}$ and $\mathcal{J}$. This polynomial optimization problem remains nonconvex. At a selected relaxation order κ , a Moment-SOS semidefinite relaxation replaces (1.2) by a convex problem and returns a global lower bound γ_κ^* . In contrast, every verified feasible trajectory $\hat{\mathbf{z}}$ provides an upper bound through its objective value. Therefore, the globally optimal value p^* of the objective is bounded by

$$\gamma_\kappa^* \leq p^* \leq p(\hat{\mathbf{z}}). \quad (1.3)$$

If the lower and upper bounds agree within specified numerical tolerances, the feasible trajectory is certified as globally optimal [Las01].

Therefore, the central research question of this thesis is as follows:

What are the capabilities and limitations of combining structure-preserving Lie group variational integrators with semidefinite relaxation to obtain feasible and certifiably optimal trajectories for single- and multi rigid-body systems?

This question is investigated by first deriving structure-preserving LGVI models, then formulating the resulting discrete dynamics as polynomial optimization problems, and finally evaluating the SDP relaxations in terms of feasibility, tightness, certifiability, and computational cost.

1.2 Related Work

Trajectory optimization and global planning Trajectory optimization represents a robot motion by a finite sequence of states and control inputs. Direct nonlinear programming methods are widely used because they can solve large problems efficiently, described in [Bet98, Ted23]. Sequential convex methods such as TrajOpt, from [SDH⁺14], similarly replace the original problem by a sequence of simpler subproblems. These approaches can produce useful motions fast, but generally give a feasible solution without a guarantee of optimality and therefore depend on the initial guess. Sampling-based planners such as PRM* and RRT* offer asymptotic optimality, but not a global certificate after a finite computation [KF11]. Mixed-integer formulations of [DT14, DIT19] provide global statements for selected problems such as footstep placement and inverse kinematics, although their computational cost and model approximations are inefficient for general multibody dynamics. [PCT14, MDWK19]. investigated contact-implicit methods by including contact forces and mode changes directly in the optimization.

Geometric mechanics and variational integration Rigid-body attitudes naturally belong to Lie groups rather than Euclidean vector spaces. The required foundations of Lie groups, mechanical symmetry, rigid-body mechanics, inertia, and matrix analysis are provided in [Hal00, MR99, BR18, Ric, GQ20, YTT11]. Classical integration methods are summarized by Butcher [But16], whereas geometric integration emphasizes the preservation of mechanical structure over long simulations [Hai94, LR05]. Projection can restore manifold constraints after a numerical step [Hai00], while Lie-group integrators update the configuration directly on the group [CO03]. Retraction maps such as the Cayley transform provide a computational implementation of the implicit Lie-group update, described in [Ise01, CJ12, BLdD23].

Variational integrators derive the discrete equations of motion from a discrete action principle and thereby preserve important geometric properties of the continuous mechanics, which are defined in [MW01]. [Lee08] develops this approach directly on matrix Lie groups and applies it to rigid-body mechanics, optimal control, and the full-body problem. The approach is extended in [LLM07b, LLM07a]. The 3D pendulum provides a compact benchmark for these rotational dynamics [SSC⁺04]. For multibody systems, maximal coordinates assign a pose to every body and impose joints through algebraic constraints, defined in [BM21b, BM21a]. These works form the basis for the 3D-pendulum and maximal-coordinate Acrobot LGVIs derived in this thesis.

Polynomial optimization and sparse semidefinite relaxation Semidefinite programming is a convex optimization framework [VB96]. Moment-SOS methods use it to relax polynomial optimization problems and obtain global lower bounds. [PV99, Las01, Yan24] establish their theoretical basis is given by positivity certificates and the Lasserre hierarchy. However, tightness at a selected relaxation order is not guaranteed, as discussed in [Nie14]. Dense moment matrices grow rapidly with the number of variables. Correlative sparsity instead constructs smaller overlapping matrices and can preserve convergence under suitable conditions given in [WKKM06, Las06].

[TJVG25] combine Lie-group variational integrators with sparse moment relaxation for multibody motion planning and therefore provide the closest methodological foundation for this thesis. [KXS⁺26] show that specialized sparse formulations and first-order GPU solvers can substantially accelerate certifiable trajectory optimization. [KLY25] extend sparsity-rich relaxations to contact-rich planning. These developments demonstrate the potential of the approach, while also showing that relaxation size and solver choice remain decisive. This thesis uses the interior-point solver MOSEK for optimization [MOS26]. Related SDP-based MPC research further illustrates the relevance of repeatedly solving relaxations in closed loop, as shown in [HKD⁺26]. The implementation and experiments presented in this thesis are documented in the open-source repository [Els26].

1.3 Research Gap

The connection between LGVIs and semidefinite relaxation has already been investigated in recent works. However, it does not provide certifiably optimal trajectories for every problem. [TJVG25] combines LGVIs with sparse Moment-SOS relaxations for single- and multibody systems. Still, some relaxations are not tight, requiring the extracted solutions to be refined using nonlinear programming (NLP), in order to obtain feasible trajectories. Similarly, [KXS⁺26, KLY25] develop faster solvers and robotics-specific sparse formulations, but the reported suboptimality gaps remain problem-dependent and range from below (1%) to more than (30%) for some problems (see Table II of [KLY25]). Therefore, semidefinite relaxation can provide useful lower bounds and trajectory candidates, but does not generally guarantee a feasible or certifiably optimal trajectory.

Accordingly, a remaining question is how reliably the complete LGVI-SDP framework can be applied to systems of increasing dynamical complexity. In particular, relaxation tightness and feasible trajectory extraction are problem-dependent, and become more difficult for constrained and underactuated multibody systems. This thesis addresses this gap by comparing the complete pipeline on a single-body benchmark and a constrained underactuated multibody system. It further investigates whether repeated first-control extraction can generate feasible closed-loop trajectories when the complete open-loop relaxation is not tight.

The contribution of this thesis is an empirical assessment showing that SDP can generate feasible trajectories using a closed-loop setup and certifiably optimal trajectories in an open-loop setting for tight relaxations, for rigid-body problems of increasing difficulty. Additionally, the offline SDP-MPC implementation can generate feasible stabilizing trajectories for harder problems, which can serve as reference or training data for controllers and learning-based methods, such as imitation learning or offline reinforcement learning.

1.4 Thesis Structure

Following the introduction, Chapter 2 introduces the required matrix Lie groups and Lie algebras.

Secondly, Chapter 3 develops continuous Lagrangian mechanics on Lie groups, and derives the continuous 3D-pendulum model, which momentum conservation property is verified.

Chapter 4 introduces discrete mechanics and LGVIs, first for a single rigid body and subsequently for constrained multibody systems using maximal coordinates. Thereby, the LGVI dynamics of a 3D pendulum and a two-link Acrobot are derived.

Therefore, Chapter 5 formulates their trajectory-optimization problems, introduces semidefinite and Moment-SOS relaxations, defines sparse clique constructions and certification diagnostics, and presents a closed-loop SDP implementation for the Acrobot.

Consequently, the numerical simulation and optimization results of the 3D-pendulum and Acrobot are evaluated in Chapter 6.

Chapter 7 discusses the previous results, especially the non-certifiable cases and the remaining computational and modeling limitations.

Finally, Chapter 8 summarizes the findings and gives the final outlook, proposing future research directions.

Chapter 2

Preliminaries on Special Embedded Matrix Lie Groups

This chapter summarizes the Lie group notation required for the continuous and discrete rigid-body mechanics developed in Chapters 3 and 4, following the definitions of Lie groups and Lie algebras in [Hal00] and the geometric-mechanics notation in [Lee08].

2.1 Special Matrix Lie Groups

A Lie group is a smooth manifold that is also a group, where multiplication and inversion are smooth maps. A real matrix Lie group is a closed subgroup $G \subset \text{GL}(n, \mathbb{R})$. Its group operation is matrix multiplication, and it is an embedded smooth manifold for which multiplication and inversion are smooth maps [Hal00].

Rotations in n dimensions are represented by the special orthogonal group

$$\text{SO}(n) = \left\{ \mathbf{R} \in \mathbb{R}^{n \times n} \mid \mathbf{R}^\top \mathbf{R} = \mathbf{I}_n, \det(\mathbf{R}) = 1 \right\}. \quad (2.1)$$

The groups $\text{SO}(2)$ and $\text{SO}(3)$ describe planar and spatial rotations [Lee08].

Combined rotational and translational motion is represented by the special Euclidean group

$$\text{SE}(n) = \left\{ \mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \mid \mathbf{R} \in \text{SO}(n), \mathbf{p} \in \mathbb{R}^n \right\}. \quad (2.2)$$

Here, $\mathbf{R}$ denotes the attitude and $\mathbf{p}$ the position [Hal00]. However, in this thesis rotations and translations are represented separately in maximal coordinates, elaborated in Section 4.3. Therefore, the derivations mainly use $\text{SO}(n)$ and its Lie algebra $\text{so}(n)$.

2.2 Special Lie Algebras

Lie algebras and the exponential map [Hal00] defines the Lie algebra $\mathfrak{g}$ of a matrix Lie group G as its tangent space at the identity,

$$\mathfrak{g} = T_e G. \quad (2.3)$$

For matrix Lie groups, the Lie bracket $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ is defined as

$$[\mathbf{a}, \mathbf{b}] = \mathbf{a}\mathbf{b} - \mathbf{b}\mathbf{a}, \quad \mathbf{a}, \mathbf{b} \in \mathfrak{g}, \quad (2.4)$$

which is bilinear and skew-symmetric [Hal00]. The Lie algebras corresponding to $\text{SO}(n)$ and $\text{SE}(n)$ are

$$\mathfrak{so}(n) = \{\boldsymbol{\omega} \in \mathbb{R}^{n \times n} \mid \boldsymbol{\omega}^\top = -\boldsymbol{\omega}\}, \quad \mathfrak{so}(n) = T_e \text{SO}(n) \quad (2.5)$$

$$\mathfrak{se}(n) = \left\{ \begin{bmatrix} \boldsymbol{\omega} & \mathbf{v} \\ \mathbf{0}^\top & 0 \end{bmatrix} \mid \boldsymbol{\omega} \in \mathfrak{so}(n), \mathbf{v} \in \mathbb{R}^n \right\}, \quad \mathfrak{se}(n) = T_e \text{SE}(n). \quad (2.6)$$

For $\text{SO}(2)$, [Lee08] defines the hat map $(\cdot)^\wedge : \mathbb{R} \rightarrow \mathfrak{so}(2)$ as

$$\hat{\omega} = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix}, \quad (2.7)$$

with $\omega \in \mathbb{R}$. Additionally, for $\text{SO}(3)$, the hat map $(\cdot)^\wedge : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ is defined by

$$\hat{\boldsymbol{\omega}} = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}, \quad \hat{\boldsymbol{\omega}}\mathbf{b} = \boldsymbol{\omega} \times \mathbf{b}, \quad (2.8)$$

with $\boldsymbol{\omega}, \mathbf{b} \in \mathbb{R}^3$ and the corresponding inverse map $(\cdot)^\vee : \mathfrak{so}(3) \rightarrow \mathbb{R}^3$.

The matrix exponential of an $n \times n$ matrix $\mathbf{X}$

$$\exp(\mathbf{X}) = \sum_{m=0}^{\infty} \frac{\mathbf{X}^m}{m!} \quad (2.9)$$

defines the exponential map $\exp : \mathfrak{g} \rightarrow G$ as a local diffeomorphism between a neighborhood of $\mathbf{0} \in \mathfrak{g}$ and a neighborhood of $e \in G$ [Hal00, Lee08].

Left and right translations For $\mathbf{g}, \mathbf{h} \in G$, the left and right translation maps are

$$\mathbb{L}_h(\mathbf{g}) = \mathbf{h}\mathbf{g}, \quad R_h(\mathbf{g}) = \mathbf{g}\mathbf{h}. \quad (2.10)$$

Thereby, [Lee08] states that for a differentiable matrix-valued curve $\mathbf{g}(t) \in G$, the left-trivialized or body-fixed velocity $\boldsymbol{\xi} \in \mathfrak{g}$ is given by

$$\boldsymbol{\xi} = \mathbf{g}^{-1}\dot{\mathbf{g}}, \quad \dot{\mathbf{g}} = \mathbf{g}\boldsymbol{\xi}. \quad (2.11)$$

Proposition 1 (Differential of left translation on a matrix Lie group: $T_x \mathbb{L}_h$). *Let $G \subset GL(n, \mathbb{R})$ be a matrix Lie group. For $h \in G$, define $\mathbb{L}_h : G \rightarrow G$ by $\mathbb{L}_h(g) = hg$ [Hal00]. Then for each $x \in G$ the differential*

$$T_x \mathbb{L}_h : T_x G \rightarrow T_{hx} G,$$

of the left translation map $\mathbb{L}_h : G \rightarrow G$ is given by

$$T_x \mathbb{L}_h(v) = hv. \quad (2.12)$$

Proof. By the definition of the tangent space of an embedded submanifold given in [Hal00], there exists a smooth matrix-valued curve $\gamma : (-\epsilon, \epsilon) \rightarrow G$ with $\gamma(0) = x$ and the tangent matrix $\dot{\gamma}(0) = v \in T_x G$. In [Lee08], the differential is defined as

$$T_x \mathbb{L}_h(v) = \left. \frac{d}{dt} \mathbb{L}_h(\gamma(t)) \right|_{t=0} = \left. \frac{d}{dt} (h\gamma(t)) \right|_{t=0} = h\dot{\gamma}(0),$$

with the left translation map $\mathbb{L}_h(g) = hg$. Since h is constant and $\dot{\gamma}(0) = v$, it follows that

$$T_x \mathbb{L}_h(v) = h\dot{\gamma}(0) = hv.$$

□

It can be observed that $\mathbb{L}_h(g) = hg$ and $T_x \mathbb{L}_h(v) = hv$ have the same matrix expression, but act on different objects. The left translation map $\mathbb{L}_h(g)$ multiplies group elements $g \in G$, while its differential is the pushforward of tangent matrices $v \in T_x G \subset \mathbb{R}^{n \times n}$. The differential of a left translation can be applied to the special embedded matrix Lie groups $SO(n)$ and their corresponding Lie algebras $\mathfrak{so}(n)$.

Differential of left translation $T_R \mathbb{L}_H$ on $SO(n)$ The left translation map in $SO(n)$ is defined as $\mathbb{L}_H(R) = HR$ with $H, R \in G$. Applied to eq. (2.12) with $v \in T_R G$, it follows that

$$T_R \mathbb{L}_H(v) = Hv.$$

The tangent matrix $v \in T_R G$ can be represented by $v = T_e \mathbb{L}_R(\omega) = R\omega$ using the left translation map of $\omega \in \mathfrak{g} := \mathfrak{so}(n)$, resulting in

$$T_R \mathbb{L}_H(R\omega) = H(R\omega). \quad (2.13)$$

Adjoint and coadjoint operators [Lee08] introduces the adjoint operator $\text{Ad}_g : \mathfrak{g} \rightarrow \mathfrak{g}$ as

$$\text{Ad}_g \eta = g\eta g^{-1}. \quad (2.14)$$

Its derivative $\text{ad}_\xi : \mathfrak{g} \rightarrow \mathfrak{g}$ is defined as

$$\text{ad}_\xi \eta = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \text{Ad}_{\exp(\epsilon\xi)} \eta = [\xi, \eta] = \xi\eta - \eta\xi, \quad (2.15)$$

using the Lie bracket $[\cdot, \cdot]$ defined in (2.4).

Let $\mathfrak{g}^*$ be the dual space of $\mathfrak{g}$. The notation $\langle \boldsymbol{\mu}, \boldsymbol{\xi} \rangle = \boldsymbol{\mu}(\boldsymbol{\xi})$ denotes the pairing of a covector $\boldsymbol{\mu} \in \mathfrak{g}^*$ with a vector $\boldsymbol{\xi} \in \mathfrak{g}$ [Lee08]. Using the pairing, the coadjoint operator $\text{Ad}_g^* : \mathfrak{g}^* \rightarrow \mathfrak{g}^*$ is defined as

$$\langle \text{Ad}_g^* \boldsymbol{\mu}, \boldsymbol{\xi} \rangle = \langle \boldsymbol{\mu}, \text{Ad}_g \boldsymbol{\xi} \rangle. \quad (2.16)$$

Its derivative $\text{ad}_\eta^* : \mathfrak{g}^* \rightarrow \mathfrak{g}^*$ is given by

$$\langle \text{ad}_\eta^* \boldsymbol{\mu}, \boldsymbol{\xi} \rangle = \langle \boldsymbol{\mu}, \text{ad}_\eta \boldsymbol{\xi} \rangle = \langle \boldsymbol{\mu}, [\boldsymbol{\eta}, \boldsymbol{\xi}] \rangle = \langle \boldsymbol{\mu}, \boldsymbol{\eta} \boldsymbol{\xi} - \boldsymbol{\xi} \boldsymbol{\eta} \rangle. \quad (2.17)$$

Additionally, the pairing can be used to define the inner product on the dual space of the Lie algebra [Lee08]. Using the trace inner product, $\mathfrak{so}(n)$ is identified with its dual. For $n \in \{2, 3\}$, the hat map gives

$$\langle \hat{\boldsymbol{a}}, \hat{\boldsymbol{b}} \rangle = \frac{1}{2} \text{tr}(\hat{\boldsymbol{a}}^\top \hat{\boldsymbol{b}}) = \boldsymbol{a}^\top \boldsymbol{b} = \boldsymbol{a} \cdot \boldsymbol{b}, \quad (2.18)$$

where $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}$ for $n = 2$ and $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^3$ for $n = 3$.

Variations on Matrix Lie Groups For $g \in G$ and $\eta \in \mathfrak{g} = T_e G$, define the local variation curve $\gamma : (-c, c) \rightarrow G$ by

$$\gamma(\epsilon) = g \exp(\epsilon \eta), \quad c > 0, \quad (2.19)$$

where ϵ is the variation parameter. Since $\exp(\epsilon \eta) \in G$, the curve $\gamma(\epsilon)$ remains in G and satisfies $\gamma(0) = g$ [Lee08]. The corresponding infinitesimal variation is

$$\delta g = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \gamma(\epsilon) = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} g \exp(\epsilon \eta) = g \eta = T_e \mathbb{L}_g(\eta) \in T_g G. \quad (2.20)$$

Chapter 3

Continuous Mechanics of Rigid Bodies on Lie Groups

This chapter introduces the continuous mechanics of rigid bodies on Lie groups, which is the foundation for the discrete mechanics formulation in Chapter 4. Therefore, Section 3.1 presents the representation of geometric mechanics on Lie groups by deriving the Euler-Lagrange equations and the Legendre transformation. Moreover, the properties of the Lagrangian flow are discussed. Consequently, Section 3.2 applies the derived framework to the unforced 3D-pendulum as an example of a mechanical system on a Lie group, and verifies the momentum preservation properties of the Lagrangian flow of the model, based on the work of [Lee08].

3.1 Lagrangian Mechanics

Differential geometry provides a framework for describing geometric mechanics, by exploring the geometric structure of a Lagrangian or Hamiltonian system [Lee08]. Thereby, Lagrangian mechanics describes the motion of a mechanical system through a variational principle. Instead of directly discretizing or integrating the equations of motion, the trajectory is obtained from the stationary value of the action integral. This formulation provides the continuous foundation for the discrete variational integrators introduced in Chapter 4. In particular, the variational construction keeps the symplectic and momentum-preserving structure of mechanical systems, which is later used for the geometric trajectory optimization formulation via Semidefinite Programming (SDP) Relaxation in Chapter 5 [Lee08, MW01, TJVG25].

Figure 3.1, adapted from [Lee08], summarizes the general method to derive the Euler-Lagrange equations of a mechanical system on a Lie group. Thereby, the path for which the action of a Lagrangian is stationary over time has to be derived. The variation of the trajectory gives the Euler-Lagrange equations. Alternatively, the Legendre transformation can be used to derive the Hamiltonian equations of motion.

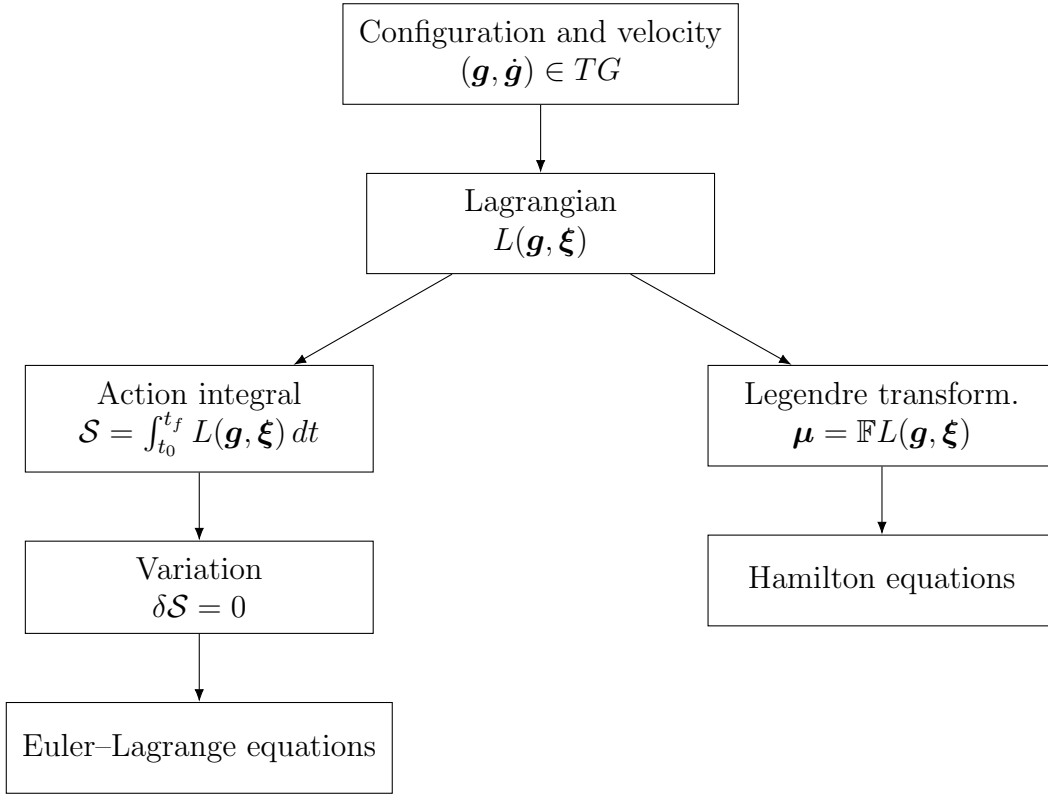


Figure 3.1: General method to derive the Euler-Lagrange equations of a mechanical system on a Lie group (adapted from [Lee08]).

3.1.1 Euler–Lagrange Equations

Following [Lee08], the configuration manifold is a matrix Lie group G with Lie algebra $\mathfrak{g}$. By left trivialization, the tangent bundle TG is identified with $G \times \mathfrak{g}$ in (2.11). A mechanical system on G is described by the Lagrangian

$$L : G \times \mathfrak{g} \rightarrow \mathbb{R}, \quad L(\mathbf{g}, \boldsymbol{\xi}) = T(\mathbf{g}, \boldsymbol{\xi}) - U(\mathbf{g}), \quad (3.1)$$

where T and U denote the kinetic and potential energy. On the interval $[t_0, t_f]$, the action integral is given by

$$\mathcal{S} = \int_{t_0}^{t_f} L(\mathbf{g}, \boldsymbol{\xi}) dt. \quad (3.2)$$

To derive the Euler–Lagrange equations, the variation of the action integral has to be computed. By definition, Hamilton’s principle states that the action is stationary

$$\delta \mathcal{S} = \delta \int_{t_0}^{t_f} L(\mathbf{g}, \boldsymbol{\xi}) dt = \int_{t_0}^{t_f} \delta L(\mathbf{g}, \boldsymbol{\xi}) dt = 0, \quad (3.3)$$

for all variations with fixed endpoints at t_0 and t_f [MW01]. The variation of the Lagrangian is expressed as

$$\delta L(\mathbf{g}, \boldsymbol{\xi}) = \langle D_{\mathbf{g}} L(\mathbf{g}, \boldsymbol{\xi}), \delta \mathbf{g} \rangle + \langle D_{\boldsymbol{\xi}} L(\mathbf{g}, \boldsymbol{\xi}), \delta \boldsymbol{\xi} \rangle, \quad (3.4)$$

with the partial derivatives $D_{\mathbf{g}}L(\mathbf{g}, \boldsymbol{\xi}) \in T_{\mathbf{g}}^*G$ with respect to $\mathbf{g}$ and $D_{\boldsymbol{\xi}}L(\mathbf{g}, \boldsymbol{\xi}) \in \mathfrak{g}^*$ with respect to $\boldsymbol{\xi}$ [MW01]. [Lee08] extends the local group variation introduced in (2.20) to a trajectory defined by

$$\mathbf{g}^\epsilon(t) = \mathbf{g}(t) \exp(\epsilon \boldsymbol{\eta}(t)), \quad \text{for} \quad \boldsymbol{\eta}(t_0) = \boldsymbol{\eta}(t_f) = \mathbf{0}, \quad (3.5)$$

for a differentiable matrix-valued curve $\boldsymbol{\eta} : [t_0, t_f] \rightarrow \mathfrak{g}$ and defines the corresponding infinitesimal variations as

$$\delta \mathbf{g} = \mathbf{g} \boldsymbol{\eta}, \quad \delta \boldsymbol{\xi} = \dot{\boldsymbol{\eta}} + \text{ad}_{\boldsymbol{\xi}} \boldsymbol{\eta}. \quad (3.6)$$

The endpoint conditions in (3.5) keep the initial and terminal configurations fixed and remove the boundary contribution arising from integration by parts.

Substituting (3.4) and (3.6) into (3.3), the variation of the action integral is given by

$$\delta \mathcal{S} = \int_{t_0}^{t_f} \left\langle T_e^* \mathbb{L}_{\mathbf{g}} D_{\mathbf{g}}L(\mathbf{g}, \boldsymbol{\xi}) + \text{ad}_{\boldsymbol{\xi}}^* D_{\boldsymbol{\xi}}L(\mathbf{g}, \boldsymbol{\xi}) - \frac{d}{dt} D_{\boldsymbol{\xi}}L(\mathbf{g}, \boldsymbol{\xi}), \boldsymbol{\eta} \right\rangle dt = 0, \quad (3.7)$$

where $T_e^* \mathbb{L}_{\mathbf{g}} : T_{\mathbf{g}}^*G \rightarrow \mathfrak{g}^*$ denotes the dual of the differential $T_e \mathbb{L}_{\mathbf{g}}$ [Lee08]. Since (3.7) holds for all $\boldsymbol{\eta} \in \mathfrak{g}$ satisfying $\boldsymbol{\eta}(t_0) = \boldsymbol{\eta}(t_f) = \mathbf{0}$, the continuous Euler–Lagrange equations on G are defined as

$$\frac{d}{dt} D_{\boldsymbol{\xi}}L(\mathbf{g}, \boldsymbol{\xi}) - \text{ad}_{\boldsymbol{\xi}}^* D_{\boldsymbol{\xi}}L(\mathbf{g}, \boldsymbol{\xi}) - T_e^* \mathbb{L}_{\mathbf{g}} D_{\mathbf{g}}L(\mathbf{g}, \boldsymbol{\xi}) = 0, \quad (3.8a)$$

$$\dot{\mathbf{g}} = \mathbf{g} \boldsymbol{\xi}. \quad (3.8b)$$

The equation (3.8a) is the dynamical representation of the system with the velocity $\boldsymbol{\xi}$ expressed in the body-fixed frame, while (3.8b) reconstructs the trajectory on the configuration manifold [Lee08].

In order to derive the Euler-Lagrange equations in the following Sections, a general method for calculating the partial derivatives of the Lagrangian with respect to $\boldsymbol{\xi}$ and $\mathbf{g}$ has to be defined.

Proposition 2 (Directional derivative representation of $D_{\boldsymbol{\xi}}L(\mathbf{g}, \boldsymbol{\xi})$). *Let G be a Lie group with Lie algebra $\mathfrak{g}$, and let $L : G \times \mathfrak{g} \rightarrow \mathbb{R}$ be smooth [Hal00]. For any $\delta \boldsymbol{\xi} \in \mathfrak{g}$ the partial derivative $D_{\boldsymbol{\xi}}L(\mathbf{g}, \boldsymbol{\xi}) \in \mathfrak{g}^*$ is characterized by*

$$\langle D_{\boldsymbol{\xi}}L(\mathbf{g}, \boldsymbol{\xi}), \delta \boldsymbol{\xi} \rangle = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} L(\mathbf{g}, \boldsymbol{\xi} + \varepsilon \delta \boldsymbol{\xi}). \quad (3.9)$$

Proof. Since $\mathfrak{g}$ is a vector space, the differential of the map $L(\mathbf{g}, \boldsymbol{\xi})$ at $\boldsymbol{\xi}$ is a linear map $\mathfrak{g} \rightarrow \mathbb{R}$, as $\mathbf{g}$ is fixed. Therefore the partial derivative $D_{\boldsymbol{\xi}}L(\mathbf{g}, \boldsymbol{\xi})$ is an element of the dual space $\mathfrak{g}^*$. Moreover, for any $\delta \boldsymbol{\xi} \in \mathfrak{g}$, the curve $\boldsymbol{\gamma}(\varepsilon) = \boldsymbol{\xi} + \varepsilon \delta \boldsymbol{\xi}$ satisfies

$\gamma(0) = \xi$ and $\dot{\gamma}(0) = \delta\xi$, which realizes $\delta\xi$ as a tangent direction at ξ for any $\varepsilon \in (-c, c)$ [Lee08]. The corresponding directional derivative is

$$\langle D_\xi L(\mathbf{g}, \xi), \delta\xi \rangle = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} L(\mathbf{g}, \gamma(\varepsilon)) = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} L(\mathbf{g}, \xi + \varepsilon \delta\xi),$$

using $\langle D_\xi L(\mathbf{g}, \xi), \delta\xi \rangle = D_\xi L(\mathbf{g}, \xi)(\delta\xi)$. $\square$

Proposition 3 (Directional derivative representation of $D_g L(\mathbf{g}, \xi)$). *Let G be a Lie group with Lie algebra $\mathfrak{g}$, and let $L : G \times \mathfrak{g} \rightarrow \mathbb{R}$ be smooth and η an arbitrary matrix-valued curve in $\mathfrak{g}$ [Hal00]. For any $\delta\mathbf{g} = T_e \mathbb{L}_g(\eta) = \mathbf{g}\eta \in T_g G$ (3.6) the partial derivative $D_g L(\mathbf{g}, \xi) \in T_g^* G$ is characterized by*

$$\langle D_g L(\mathbf{g}, \xi), \delta\mathbf{g} \rangle = \langle T_e^* \mathbb{L}_g D_g L(\mathbf{g}, \xi), \eta \rangle = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} L(\mathbf{g} \exp(\varepsilon \eta), \xi). \quad (3.10)$$

Proof. Since G is a smooth manifold, the differential of the map $L(\mathbf{g}, \xi)$ is a linear map $T_g G \rightarrow \mathbb{R}$, as ξ is fixed. Therefore the partial derivative $D_g L(\mathbf{g}, \xi)$ is an element of the dual space $T_g^* G$. Using the definition of the variation $\mathbf{g}^\varepsilon$ (3.5), the corresponding directional derivative is

$$\langle D_g L(\mathbf{g}, \xi), \delta\mathbf{g} \rangle = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} L(\mathbf{g}^\varepsilon, \xi) = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} L(\mathbf{g} \exp(\varepsilon \eta), \xi),$$

using $\langle D_g L(\mathbf{g}, \xi), \delta\mathbf{g} \rangle = D_g L(\mathbf{g}, \xi)(\delta\mathbf{g})$ [Lee08]. Additionally, the directional derivative can be expressed by the covector $T_e^* \mathbb{L}_g$ and the definition of the differential of the left translation $\mathbf{g}\eta = T_e \mathbb{L}_g(\eta)$ defined in (2.12), which leads to

$$\langle D_g L(\mathbf{g}, \xi), \delta\mathbf{g} \rangle = \langle D_g L(\mathbf{g}, \xi), \mathbf{g}\eta \rangle = \langle D_g L(\mathbf{g}, \xi), T_e \mathbb{L}_g(\eta) \rangle = \langle T_e^* \mathbb{L}_g D_g L(\mathbf{g}, \xi), \eta \rangle.$$

$\square$

3.1.2 Legendre Transformation and Hamiltonian Mechanics

In contrast to the Lagrangian $L : G \times \mathfrak{g} \rightarrow \mathbb{R}$ formulation, the Hamiltonian formulation $H : G \times \mathfrak{g}^* \rightarrow \mathbb{R}$ represents the state on the cotangent bundle T^*G identified with $G \times \mathfrak{g}^*$ by left trivialization. Thereby, the state is written as $(\mathbf{g}, \boldsymbol{\mu})$ with the body-fixed momentum $\boldsymbol{\mu} \in \mathfrak{g}^*$ [MW01]. The Lagrangian and Hamiltonian descriptions are related by the Legendre transformation $\mathbb{F}L : G \times \mathfrak{g} \rightarrow G \times \mathfrak{g}^*$

$$\mathbb{F}L : (\mathbf{g}, \xi) \mapsto (\mathbf{g}, \boldsymbol{\mu}) = (\mathbf{g}, D_\xi L(\mathbf{g}, \xi)), \quad (3.11)$$

Consequently, applying the Legendre transformation (3.11) to the Euler-Lagrange equations (3.8), the Hamiltonian equations of motion are derived as

$$\dot{\boldsymbol{\mu}} - \text{ad}_\xi^* \boldsymbol{\mu} - T_e^* \mathbb{L}_g D_g L(\mathbf{g}, \xi) = 0, \quad (3.12a)$$

$$\dot{\mathbf{g}} = \mathbf{g}\xi. \quad (3.12b)$$

3.1.3 Properties of the Lagrangian Flow

[MW01] defines the Lagrangian flow $\mathcal{F}_L^T : G \times \mathfrak{g} \rightarrow G \times \mathfrak{g}$, which maps an initial state to the corresponding state at time t . Thereby, for a regular Lagrangian, the Lagrangian flow evolves the system in time while preserving its geometric structure. Under a group action, it has symplecticity and momentum-preservation properties [Lee08, MW01].

Symplecticity The Lagrangian one-form Θ_L on $G \times \mathfrak{g}$ is defined by

$$\Theta_L(\mathbf{g}, \boldsymbol{\xi}) \cdot (\delta \mathbf{g}, \delta \boldsymbol{\xi}) = \langle D_{\boldsymbol{\xi}} L(\mathbf{g}, \boldsymbol{\xi}), \mathbf{g}^{-1} \delta \mathbf{g} \rangle. \quad (3.13)$$

With the Lagrangian symplectic two-form $\Omega_L = d\Theta_L$, the flow preserves symplecticity according to

$$(F_L^T)^* \Omega_L = \Omega_L, \quad (3.14)$$

for $T = t_f - t_0$ [Lee08].

Noether's theorem Let a left action of a Lie group H with Lie algebra $\mathfrak{h}$ act smoothly on G by the mapping $\Phi : H \times G \rightarrow G$ such that $\Phi_{\mathbf{h}}(\mathbf{g}) = \Phi(\mathbf{h}, \mathbf{g})$. [Lee08] defines the left trivialization $\phi_L : TG \rightarrow G \times \mathfrak{g}$ by

$$\phi_L(\mathbf{g}, \dot{\mathbf{g}}) = (\mathbf{g}, \mathbf{g}^{-1} \dot{\mathbf{g}}), \quad \phi_L^{-1}(\mathbf{g}, \mathbf{g}^{-1} \dot{\mathbf{g}}) = (\mathbf{g}, \dot{\mathbf{g}}). \quad (3.15)$$

Additionally, the infinitesimal generator $\zeta_{G \times \mathfrak{g}} : G \times \mathfrak{g} \rightarrow T(G \times \mathfrak{g})$ is given by

$$\zeta_{G \times \mathfrak{g}}(\mathbf{g}, \boldsymbol{\xi}) = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \phi_L \circ T_{\mathbf{g}} \Phi_{\exp_H(\epsilon \boldsymbol{\zeta})}(\mathbf{g}) \cdot (\phi_L^{-1}(\mathbf{g}, \boldsymbol{\xi})), \quad (3.16)$$

for $\boldsymbol{\zeta} \in \mathfrak{h}$ [Lee08]. Moreover, the corresponding Lagrangian momentum map $J_L : G \times \mathfrak{g} \rightarrow \mathfrak{h}^*$ is defined by

$$J_L(\mathbf{g}, \boldsymbol{\xi}) \cdot \boldsymbol{\zeta} = \Theta_L \cdot \zeta_{G \times \mathfrak{g}}. \quad (3.17)$$

Theorem 1 (Noether's theorem). *If the Lagrangian is infinitesimally invariant under the lifted action, such that*

$$dL(\mathbf{g}, \boldsymbol{\xi}) \cdot \zeta_{G \times \mathfrak{g}}(\mathbf{g}, \boldsymbol{\xi}) = 0 \quad \text{for all } \boldsymbol{\zeta} \in \mathfrak{h}, \quad (3.18)$$

then the associated momentum map is preserved along the Lagrangian flow,

$$J_L(F_L^t(\mathbf{g}, \boldsymbol{\xi})) = J_L(\mathbf{g}, \boldsymbol{\xi}), \quad (3.19)$$

which is considered as Noether's theorem [MW01].

Noether's theorem connects a continuous symmetry of the Lagrangian with a conserved momentum quantity [Lee08, MW01]. This result is applied in Section 3.2.2 to identify and verify the momentum preserved by the unforced 3D-pendulum. However, [MW01] states that the variational integrator does not generally conserve the continuous energy of the system exactly. Nevertheless, due to the symplectic structure, the energy is bounded and oscillates around the exact energy over a long-time horizon.

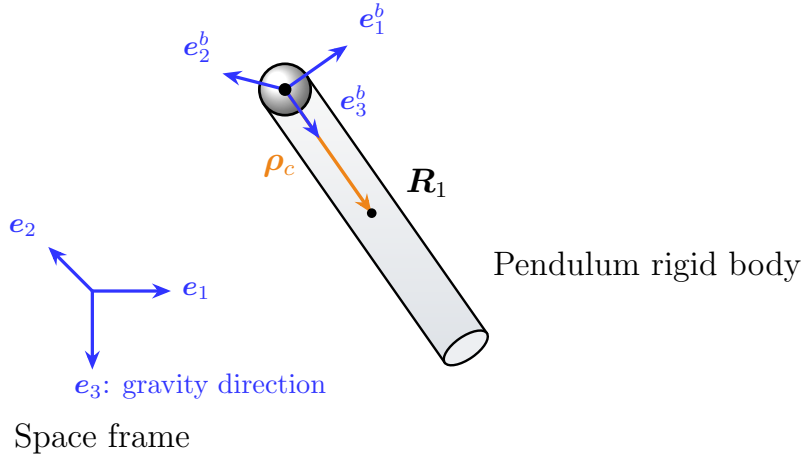


Figure 3.2: 3D-pendulum system attached to a fixed pivot.

3.2 Unforced 3D-Pendulum

In the following Section, the unforced mechanics of a three-degree-of-freedom pendulum given by [Lee08] are presented, using Lagrangian mechanics of Section 3.1. Following, the momentum preservation properties of the Lagrangian flow of the model will be verified.

3.2.1 Derivation of Lie Group Mechanics

Configuration Manifold The 3D-Pendulum model, illustrated in 3.2, consist of a rigid body rotating under the influence of gravity g attached to a fixed frictionless pivot [SSC⁺04]. Moreover, the system has no external control moment, since it is unforced. A space frame and a body-fixed frame attached to the pendulum, both with their origins at the pivot, are defined. The unit vector $\mathbf{e}_3$ represents the gravity direction in the space frame, while $\boldsymbol{\rho}_c$ denotes the vector from the pivot to the center of mass expressed in the body-fixed frame. Following the fixed pivot, there is no translational degree of freedom. Therefore, the configuration manifold of the model is given by the special orthogonal group

$$SO(3) = \{ \mathbf{R} \in \mathbb{R}^{3 \times 3} \mid \mathbf{R}^\top \mathbf{R} = \mathbf{I}_3, \det(\mathbf{R}) = 1 \}.$$

The configuration is completely determined by the rotation matrix $\mathbf{R} \in SO(3)$, which maps the body-fixed frame to the space frame. Additionally, in $SO(3)$ the attitude kinematics equation is defined by

$$\dot{\mathbf{R}} = \mathbf{R}\hat{\boldsymbol{\omega}}, \quad (3.20)$$

with the angular velocity $\boldsymbol{\omega} \in \mathbb{R}^3$, mapped to the Lie algebra $\mathfrak{so}(3)$ by the hat map $\hat{\cdot} : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ [Lee08]. Consequently, $\hat{\boldsymbol{\omega}}$ is a 3×3 skew-symmetric matrix.

Lagrangian To derive the Lagrangian $L : SO(3) \times \mathfrak{so}(3) \rightarrow \mathbb{R}$, the kinetic $T : SO(3) \times \mathfrak{so}(3) \rightarrow \mathbb{R}$ and potential $V : SO(3) \rightarrow \mathbb{R}$ energy of the rigid body are introduced. The rotational inertia of the mass is expressed in the body-frame by the inertia matrix $\mathbf{J} \in \mathbb{R}^{3 \times 3}$. [Lee08] defines

$$T(\boldsymbol{\omega}) = \frac{1}{2} \int_B \|\hat{\boldsymbol{\omega}} \boldsymbol{\rho}\|^2 dm(\boldsymbol{\rho}) = \frac{1}{2} \text{tr}[\hat{\boldsymbol{\omega}} \mathbf{J}_d \hat{\boldsymbol{\omega}}^\top] = \frac{1}{2} \boldsymbol{\omega}^\top \mathbf{J} \boldsymbol{\omega}, \quad (3.21)$$

as a general expression for the kinetic energy of the single body B with the velocity $\boldsymbol{\omega} \times \boldsymbol{\rho} = \hat{\boldsymbol{\omega}} \boldsymbol{\rho}$. Thereby,

$$\mathbf{J}_d = \int_B \hat{\boldsymbol{\rho}} \hat{\boldsymbol{\rho}}^\top dm(\boldsymbol{\rho}) = \frac{1}{2} \text{tr}[\mathbf{J}] \mathbf{I} - \mathbf{J}, \quad (3.22)$$

is a nonstandard inertia matrix, with the relation

$$(\mathbf{J} \boldsymbol{\omega})^\wedge = \hat{\boldsymbol{\omega}} \mathbf{J}_d + \mathbf{J}_d \hat{\boldsymbol{\omega}}, \quad (3.23)$$

for any $\boldsymbol{\omega} \in \mathbb{R}^3$. The potential energy is given by

$$V(\mathbf{R}) = -mg \mathbf{e}_3^\top \mathbf{R} \boldsymbol{\rho}_c, \quad (3.24)$$

where $\mathbf{R} \boldsymbol{\rho}_c$ denotes the vector from the pivot to the center of mass expressed in the space-fixed frame, and $\mathbf{e}_3^\top \mathbf{R} \boldsymbol{\rho}_c$ is its projection onto the gravity direction $\mathbf{e}_3$ [Lee08]. Using (3.21) and (3.24), the Lagrangian of the unforced 3D pendulum is given by

$$L(\mathbf{R}, \hat{\boldsymbol{\omega}}) = T(\hat{\boldsymbol{\omega}}) - V(\mathbf{R}) = \frac{1}{2} \text{tr}[\hat{\boldsymbol{\omega}} \mathbf{J}_d \hat{\boldsymbol{\omega}}^\top] + mg \mathbf{e}_3^\top \mathbf{R} \boldsymbol{\rho}_c. \quad (3.25)$$

Euler–Lagrange Equations In order to define the Euler–Lagrange equations (3.8), the derivatives of (3.25) have to be calculated. Firstly, the directional derivative $D_{\hat{\boldsymbol{\omega}}} L(\mathbf{R}, \hat{\boldsymbol{\omega}})$ is given by applying (3.9), which yields

$$\begin{aligned} \langle D_{\hat{\boldsymbol{\omega}}} L(\mathbf{R}, \hat{\boldsymbol{\omega}}), \delta \hat{\boldsymbol{\omega}} \rangle &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L(\mathbf{R}, \hat{\boldsymbol{\omega}} + \epsilon \delta \hat{\boldsymbol{\omega}}) \\ &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \frac{1}{2} \text{tr}[(\hat{\boldsymbol{\omega}} + \epsilon \delta \hat{\boldsymbol{\omega}}) \mathbf{J}_d (\hat{\boldsymbol{\omega}} + \epsilon \delta \hat{\boldsymbol{\omega}})^\top] \\ &= \frac{1}{2} \text{tr}[\delta \hat{\boldsymbol{\omega}} \mathbf{J}_d \hat{\boldsymbol{\omega}}^\top + \hat{\boldsymbol{\omega}} \mathbf{J}_d (\delta \hat{\boldsymbol{\omega}})^\top]. \end{aligned}$$

By using the antisymmetry identity of a cross product $\hat{\mathbf{x}} = -\hat{\mathbf{x}}^\top$ for any $\mathbf{x} \in \mathbb{R}^3$ and $\text{tr}[\mathbf{A}\mathbf{B}] = \text{tr}[\mathbf{B}\mathbf{A}]$ for any $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{3 \times 3}$, the directional derivative is rewritten as

$$\begin{aligned} \langle D_{\hat{\boldsymbol{\omega}}} L(\mathbf{R}, \hat{\boldsymbol{\omega}}), \delta \hat{\boldsymbol{\omega}} \rangle &= \frac{1}{2} \text{tr}[-\delta \hat{\boldsymbol{\omega}} \mathbf{J}_d \hat{\boldsymbol{\omega}} - \hat{\boldsymbol{\omega}} \mathbf{J}_d \delta \hat{\boldsymbol{\omega}}] \\ &= \frac{1}{2} \text{tr}[-(\mathbf{J}_d \hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}} \mathbf{J}_d) \delta \hat{\boldsymbol{\omega}}] \\ &= \langle \mathbf{J}_d \hat{\boldsymbol{\omega}} + \hat{\boldsymbol{\omega}} \mathbf{J}_d, \delta \hat{\boldsymbol{\omega}} \rangle \\ &= \langle (\mathbf{J} \boldsymbol{\omega})^\wedge, \delta \hat{\boldsymbol{\omega}} \rangle, \end{aligned} \quad (3.26)$$

with the definition of the inner product (2.18) and the skew-symmetric property of (3.23).

Moreover, the derivative $D_{\mathbf{R}}L(\mathbf{R}, \hat{\omega})$, which in this model depends only on the potential energy $V(\mathbf{R})$, is calculated using (3.10) as

$$\begin{aligned}
 \langle D_{\mathbf{R}}L(\mathbf{R}, \hat{\omega}), \delta \mathbf{R} \rangle &= \langle T_e^* \mathbb{L}_{\mathbf{R}} D_{\mathbf{R}}L(\mathbf{R}, \hat{\omega}), \hat{\eta} \rangle \\
 &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L(\mathbf{R} \exp(\epsilon \hat{\eta}), \hat{\omega}) \\
 &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} mg \mathbf{e}_3^\top \mathbf{R} \exp(\epsilon \hat{\eta}) \rho_c \\
 &= mg \mathbf{e}_3^\top \mathbf{R} \hat{\eta} \rho_c \\
 &= mg (\mathbf{R}^\top \mathbf{e}_3)^\top \hat{\eta} \rho_c \\
 &= mg (\rho_c \times \mathbf{R}^\top \mathbf{e}_3) \cdot \eta,
 \end{aligned}$$

since $\mathbf{a}^\top \hat{\mathbf{x}} \mathbf{b} = \mathbf{a} \cdot (\mathbf{x} \times \mathbf{b}) = (\mathbf{b} \times \mathbf{a}) \cdot \mathbf{x}$ for any $\mathbf{a}, \mathbf{b}, \mathbf{x} \in \mathbb{R}^3$. Applying the definition of inner products (2.18), it follows that

$$\langle T_e^* \mathbb{L}_{\mathbf{R}} D_{\mathbf{R}}L(\mathbf{R}, \hat{\omega}), \hat{\eta} \rangle = \langle (mg (\rho_c \times \mathbf{R}^\top \mathbf{e}_3))^\wedge, \hat{\eta} \rangle. \quad (3.27)$$

Since the directional derivative

$$D_{\hat{\omega}}L(\mathbf{R}, \hat{\omega}) \hat{=} \mathbf{J}_d \hat{\omega} + \hat{\omega} \mathbf{J}_d,$$

was derived in (3.26), the coadjoint term in the Euler–Lagrange equations can be written in matrix form by using the identity (2.15). Therefore,

$$\begin{aligned}
 ad_{\hat{\omega}}^* D_{\hat{\omega}}L(\mathbf{R}, \hat{\omega}) &= [\mathbf{J}_d \hat{\omega} + \hat{\omega} \mathbf{J}_d, \hat{\omega}] \\
 &= (\mathbf{J}_d \hat{\omega} + \hat{\omega} \mathbf{J}_d) \hat{\omega} - \hat{\omega} (\mathbf{J}_d \hat{\omega} + \hat{\omega} \mathbf{J}_d) \\
 &= (\mathbf{J} \omega \times \omega)^\wedge
 \end{aligned}$$

with the relation $\hat{\mathbf{x}} \hat{\mathbf{y}} - \hat{\mathbf{y}} \hat{\mathbf{x}} = (\mathbf{x} \times \mathbf{y})^\wedge$ for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^3$ and (3.23). Furthermore, by the antisymmetry of the cross product, it follows that

$$ad_{\hat{\omega}}^* D_{\hat{\omega}}L(\mathbf{R}, \hat{\omega}) = (\mathbf{J} \omega \times \omega)^\wedge = -(\omega \times \mathbf{J} \omega)^\wedge \quad (3.28)$$

Finally, the Euler-Lagrange equations of the unforced 3D-pendulum are expressed in $\mathfrak{so}(3)$ and $SO(3)$ form, by substituting (3.26), (3.27) and (3.28) into (3.8), which results in

$$(\mathbf{J} \dot{\omega})^\wedge + (\omega \times (\mathbf{J} \omega))^\wedge = (mg \rho_c \times \mathbf{R}^\top \mathbf{e}_3)^\wedge, \quad (3.29a)$$

$$\dot{\mathbf{R}} = \mathbf{R} \hat{\omega}. \quad (3.29b)$$

Legendre Transformation In order to obtain Hamilton's equations for the unforced 3D-pendulum model, the Legendre transformation $\mathbb{F}L : (SO(3) \times \mathfrak{so}(3)) \rightarrow (SO(3) \times \mathfrak{so}(3)^*)$ (3.11) has to be applied, which results in the expression for the momentum

$$\hat{\Pi} = D_{\hat{\omega}}L(\mathbf{R}, \hat{\omega}) = (\mathbf{J}\hat{\omega})^\wedge \quad (3.30)$$

where $\hat{\Pi} \in \mathfrak{so}(3)^*$. By substituting (3.30) into (3.29), Hamilton's equations of an unforced 3D-pendulum are

$$\dot{\hat{\Pi}} + (\mathbf{J}^{-1}\hat{\Pi} \times \hat{\Pi})^\wedge = (mg \boldsymbol{\rho}_c \times \mathbf{R}^T \mathbf{e}_3)^\wedge, \quad (3.31a)$$

$$\dot{\mathbf{R}} = \mathbf{R}(\mathbf{J}^{-1}\hat{\Pi})^\wedge, \quad (3.31b)$$

with $\mathbf{J}$ being nonsingular. Finally, the derived Euler-Lagrange equations (3.29) and Hamilton's equations (3.31) correspond to the given 3D-pendulum results of [Lee08].

3.2.2 Verification of Momentum Conservation

The verification of momentum conservation is particularly relevant, in order to ensure that the numerical integration of the system's dynamics preserves the physical properties of the system over long time. This is essential for accurate numerical simulation and for reliable trajectory optimization methods [TJVG25]. Therefore, in the following, the momentum conservation property of the continuous 3D pendulum is verified based on its symmetry and Noether's theorem 1.

[Lee08] identifies a symmetry of the unforced 3D-pendulum's Lagrangian, which is infinitesimally invariant under an action of the group $H = SO(2) \simeq S^1$. Here, a rotation about the gravity direction $\mathbf{e}_3$ does not change the Lagrangian. Thus, $SO(2)$ and S^1 describe the same symmetry group under the left action $\Phi : S^1 \times SO(3) \rightarrow SO(3)$ of H on G ,

$$\Phi(\theta, \mathbf{R}) = \exp_{SO(3)}(\theta \hat{\mathbf{e}}_3) \mathbf{R}, \quad (3.32)$$

where $\exp_{SO(3)}(\theta \hat{\mathbf{e}}_3)$ expresses a finite rotation about the gravity direction $\mathbf{e}_3$ by the angle θ [Lee08]. By Noether's Theorem 1, the infinitesimal invariance of the 3D-pendulum's Lagrangian is verified, if

$$dL(\mathbf{R}, \hat{\omega}) \cdot \zeta_{SO(3) \times \mathfrak{so}(3)}(\mathbf{R}, \hat{\omega}) = 0. \quad (3.33)$$

Consequently, preserving the Lagrangian momentum map along the flow $J_L(\mathcal{F}_L^T(\mathbf{R}, \hat{\omega})) = J_L(\mathbf{R}, \hat{\omega})$.

Therefore, the infinitesimal generator of the lifted group action on $SO(3) \times \mathfrak{so}(3)$ is required [Lee08]. With the attitude kinematics relation (3.20), the left trivialization $\phi_L(\mathbf{R}, \dot{\mathbf{R}}) = \phi_L(\mathbf{R}, \mathbf{R}\hat{\omega}) = (\mathbf{R}, T_e \mathbb{L}_{\mathbf{R}^{-1}} \mathbf{R}\hat{\omega}) = (\mathbf{R}, \hat{\omega})$, and its inverse $\phi_L^{-1}(\mathbf{R}, \hat{\omega}) = (\mathbf{R}, T_e \mathbb{L}_{\mathbf{R}} \hat{\omega}) = (\mathbf{R}, \mathbf{R}\hat{\omega})$ are taken from (3.15), with the differential $T_{\mathbf{R}}$ of (2.13). Following [Lee08], the lifted infinitesimal generator $\zeta_{SO(3) \times \mathfrak{so}(3)} : SO(3) \times$

$\mathfrak{so}(3) \rightarrow T(SO(3) \times \mathfrak{so}(3))$ (3.16) is computed as

$$\begin{aligned}
\zeta_{SO(3) \times \mathfrak{so}(3)}(\mathbf{R}, \hat{\boldsymbol{\omega}}) &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \phi_L \circ T_{\mathbf{R}} \Phi_{\exp_{SO(2)}(\epsilon \zeta)} \circ \phi_L^{-1}(\mathbf{R}, \hat{\boldsymbol{\omega}}) \\
&= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \phi_L \circ T_{\mathbf{R}} \Phi_{\exp_{SO(2)}(\epsilon \zeta)}(\mathbf{R}, \mathbf{R} \hat{\boldsymbol{\omega}}) \\
&= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \phi_L(\exp(\epsilon \zeta \hat{\mathbf{e}}_3) \mathbf{R}, \exp(\epsilon \zeta \hat{\mathbf{e}}_3) \mathbf{R} \hat{\boldsymbol{\omega}}) \\
&= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} (\exp(\epsilon \zeta \hat{\mathbf{e}}_3) \mathbf{R}, \mathbf{R}^{-1} \exp(\epsilon \zeta \hat{\mathbf{e}}_3)^{-1} (\exp(\epsilon \zeta \hat{\mathbf{e}}_3) \mathbf{R} \hat{\boldsymbol{\omega}})) \\
&= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} (\exp(\epsilon \zeta \hat{\mathbf{e}}_3) \mathbf{R}, \hat{\boldsymbol{\omega}}) \\
&= (\zeta \hat{\mathbf{e}}_3 \mathbf{R}, \mathbf{0}).
\end{aligned} \tag{3.34}$$

This result can be substituted into (3.33), using (3.27), which gives

$$\begin{aligned}
dL(\mathbf{R}, \hat{\boldsymbol{\omega}}) \cdot \zeta_{SO(3) \times \mathfrak{so}(3)}(\mathbf{R}, \hat{\boldsymbol{\omega}}) &= dL(\mathbf{R}, \hat{\boldsymbol{\omega}}) \cdot (\zeta \hat{\mathbf{e}}_3 \mathbf{R}, \mathbf{0}) \\
&= \langle D_{\mathbf{R}} L(\mathbf{R}, \hat{\boldsymbol{\omega}}), \zeta \hat{\mathbf{e}}_3 \mathbf{R} \rangle + \langle D_{\hat{\boldsymbol{\omega}}} L(\mathbf{R}, \hat{\boldsymbol{\omega}}), \mathbf{0} \rangle \\
&= \langle T_e^* \mathbb{L}_{\mathbf{R}} D_{\mathbf{R}} L(\mathbf{R}, \hat{\boldsymbol{\omega}}), \hat{\boldsymbol{\eta}} \rangle.
\end{aligned} \tag{3.35}$$

Since the lifted generator has no component in the $\hat{\boldsymbol{\omega}}$ direction, the component directional derivative $D_{\hat{\boldsymbol{\omega}}}$ equals zero in the zero direction. Moreover, by the definition of the lifted infinitesimal generator (3.16), the result of (3.34) redefines the variational relationship as

$$(\delta \mathbf{R}, \delta \hat{\boldsymbol{\omega}}) = (\zeta \hat{\mathbf{e}}_3 \mathbf{R}, \mathbf{0}). \tag{3.36}$$

Consequently, using the previously given infinitesimal variation (3.6), the corresponding relation $\delta \mathbf{R} = \zeta \hat{\mathbf{e}}_3 \mathbf{R} = \mathbf{R} \hat{\boldsymbol{\eta}}$ is defined, from which the variation $\hat{\boldsymbol{\eta}}$ can be expressed as

$$\hat{\boldsymbol{\eta}} = \zeta \mathbf{R}^{-1} \hat{\mathbf{e}}_3 \mathbf{R}. \tag{3.37}$$

This can be substituted into (3.35), yielding

$$\langle T_e^* \mathbb{L}_{\mathbf{R}} D_{\mathbf{R}} L(\mathbf{R}, \hat{\boldsymbol{\omega}}), \zeta \mathbf{R}^{-1} \hat{\mathbf{e}}_3 \mathbf{R} \rangle = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L(\mathbf{R} \exp(\epsilon \zeta \mathbf{R}^{-1} \hat{\mathbf{e}}_3 \mathbf{R}), \hat{\boldsymbol{\omega}}).$$

Since only the potential term of the 3D-pendulum $V(\mathbf{R})$ (3.24) depends on $\mathbf{R}$ and

(3.37) is skew symmetric, it can be shown that

$$\begin{aligned}
\left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L(\mathbf{R} \exp(\epsilon \zeta \mathbf{R}^{-1} \hat{\mathbf{e}}_3 \mathbf{R}), \hat{\boldsymbol{\omega}}) &= mg \mathbf{e}_3^T \mathbf{R} (\zeta \mathbf{R}^{-1} \hat{\mathbf{e}}_3 \mathbf{R}) \boldsymbol{\rho}_c \\
&= mg (\boldsymbol{\rho}_c^\top (\zeta \mathbf{R}^{-1} \hat{\mathbf{e}}_3 \mathbf{R})^\top \mathbf{R}^\top \mathbf{e}_3)^\top \\
&= -mg \boldsymbol{\rho}_c^\top (\zeta \mathbf{R}^{-1} \hat{\mathbf{e}}_3 \mathbf{R}) \mathbf{R}^\top \mathbf{e}_3 \\
&= -mg \boldsymbol{\rho}_c^\top \zeta \mathbf{R}^{-1} \hat{\mathbf{e}}_3 \mathbf{e}_3 \\
&= -mg \boldsymbol{\rho}_c^\top \zeta \mathbf{R}^{-1} (\mathbf{e}_3 \times \mathbf{e}_3) \\
&= 0,
\end{aligned} \tag{3.38}$$

because $\mathbf{e}_3 \times \mathbf{e}_3 = \mathbf{0}$ and $\mathbf{R}^{-1} = \mathbf{R}^\top$ for any $\mathbf{R} \in SO(3)$. Therefore, the condition of (3.33) is verified and the Lagrangian is infinitesimally invariant under the symmetry action. Consequently, by Noether's Theorem 1, the corresponding momentum map $J_L(\mathbf{R}, \boldsymbol{\omega})$ of the continuous 3D-pendulum is preserved along the Lagrangian flow

$$J_L(\mathcal{F}_L^T(\mathbf{R}, \hat{\boldsymbol{\omega}})) = J_L(\mathbf{R}, \hat{\boldsymbol{\omega}}).$$

Chapter 4

Discrete Mechanics for Rigid Bodies on Lie Groups

This chapter introduces the discrete mechanics of rigid bodies on Lie groups for trajectory optimization computations. Thereby, it is the discrete-time version of the continuous mechanics in Chapter 3. Instead of applying standard numerical integration methods, for example, the midpoint or trapezoidal rule, directly to the continuous equations of motion, variational integrators discretize Hamilton's principle. Therefore, the resulting discrete flow preserves the variational and geometric structure of the mechanical system [MW01, LR05]. In correspondence with [Lee08], the discrete time nodes are defined by

$$t_k = t_0 + kh, \quad k = 0, \dots, N, \quad \text{and} \quad Nh = t_f - t_0,$$

for a fixed time step $h > 0$ and N time steps.

Section 4.1 introduces the forced discrete Euler-Lagrange equations, including the discrete Legendre transformation, and the geometric properties of the discrete Lagrangian flow. Additionally, a computational approach is given for solving the implicit discrete equations. Subsequently, in Section 4.2, the formulation is applied to a single rigid body system and extended to multi-body systems in maximal coordinates in Section 4.3. These discrete models are used for the geometric trajectory optimization formulation and computation in Chapter 5.

4.1 Discrete Lagrangian Mechanics

Following the continuous Lagrangian mechanics in Chapter 3, discrete Lagrangian mechanics is based on the discrete variational principle. The continuous trajectory is replaced by a sequence of configurations on the Lie group, and the action integral is approximated by an action sum. For a regular discrete Lagrangian, the resulting Lie group variational integrator defines a symplectic discrete flow that remains on the configuration manifold. If the discrete Lagrangian is invariant under a lifted

Lie-group action, the associated discrete momentum map is preserved exactly at every time step manifold [MW01, Lee08].

Figure 4.1 summarizes the general method for deriving the discrete Euler–Lagrange equations of a mechanical system on a Lie group. Thereby, the discrete Euler–Lagrange equations are obtained from the variation of the action sum. Alternatively, the discrete Legendre transformation can be used to derive the discrete Hamiltonian equations of motion.

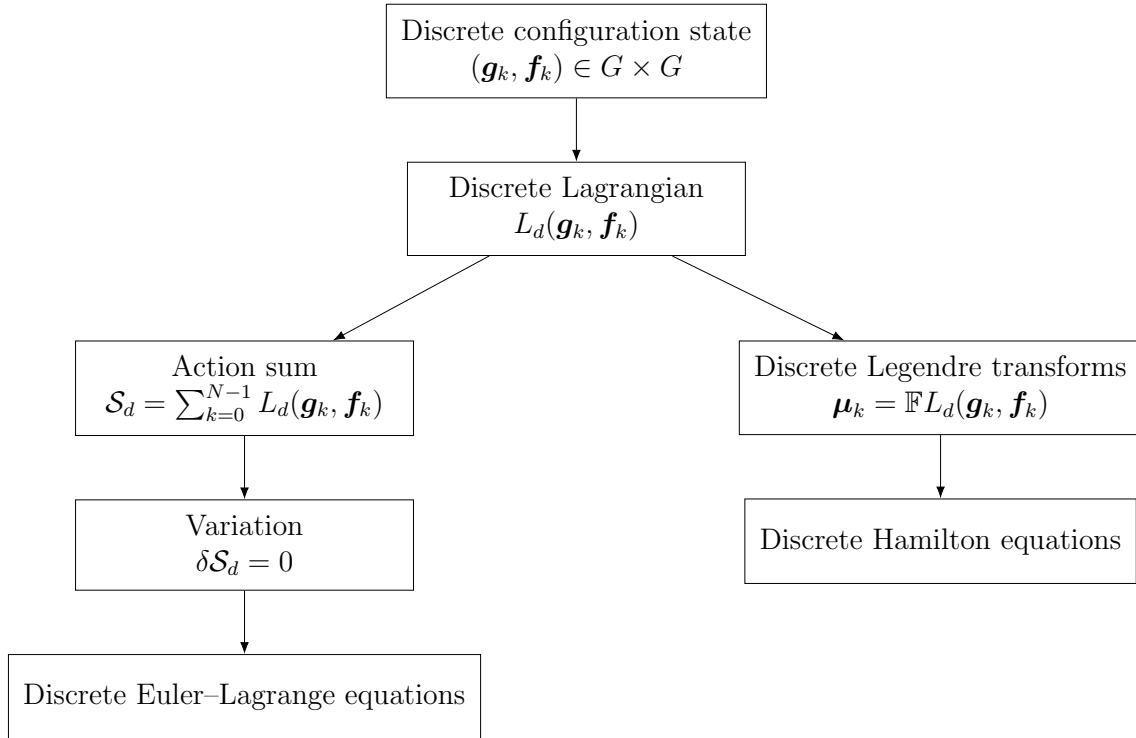


Figure 4.1: General method to derive the discrete Euler–Lagrange equations of a mechanical system on a Lie group. Adapted from [Lee08].

4.1.1 Discrete Euler–Lagrange Equations

Following [Lee08, MW01], G is a matrix Lie group with Lie algebra $\mathfrak{g}$. Thereby, a discrete trajectory $\mathbf{g}_d$ is defined by a sequence of configurations $\{\mathbf{g}_k\}_{k=0}^N$, where $\mathbf{g}_k = \mathbf{g}_d(t_k)$. The discrete state space is defined on $G \times G$. By defining the relative group element $\mathbf{f}_k \in G$, the discrete-time kinematic equation is given by

$$\mathbf{g}_{k+1} = \mathbf{g}_k \mathbf{f}_k. \quad (4.1)$$

Consequently, the multiplicative update guarantees that $\mathbf{g}_{k+1} \in G$.

The discrete Lagrangian $L_d : G \times G \rightarrow \mathbb{R}$ approximates the action integral of the continuous Lagrangian over the interval $[t_k, t_{k+1}]$. [MW01] states that, for sufficiently

small h and a regular continuous Lagrangian, the exact discrete Lagrangian is defined by

$$L_d^{\text{exact}}(\mathbf{g}_k, \mathbf{f}_k) = \int_0^h L(\tilde{\mathbf{g}}(t), \tilde{\mathbf{g}}(t)^{-1} \dot{\tilde{\mathbf{g}}}(t)) dt, \quad (4.2)$$

with the boundary conditions $\tilde{\mathbf{g}}(0) = \mathbf{g}_k, \tilde{\mathbf{g}}(h) = \mathbf{g}_k \mathbf{f}_k$. Moreover, $\tilde{\mathbf{g}} : [0, h] \rightarrow G$ satisfies the continuous Euler-Lagrange equations (3.8) [Lee08]. However, in this thesis, L_d is calculated using the trapezoidal rule (4.34) to approximate (4.2).

The discrete action sum is given by

$$\mathcal{S}_d = \sum_{k=0}^{N-1} L_d(\mathbf{g}_k, \mathbf{f}_k). \quad (4.3)$$

Similarly to the continuous action integral (3.3) and the variation of the continuous Lagrangian (3.4), the discrete Hamilton principle states that the action sum is stationary

$$\delta \mathcal{S}_d = \sum_{k=0}^{N-1} \delta L_d(\mathbf{g}_k, \mathbf{f}_k) = \sum_{k=0}^{N-1} \langle D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \delta \mathbf{g}_k \rangle + \langle D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \delta \mathbf{f}_k \rangle = 0, \quad (4.4)$$

for all possible variations of $\mathbf{g}_d$ with fixed endpoints at $k = 0$ and $k = N$ [MW01].

Analogously to the continuous variation in (3.5), the variation of the discrete configuration sequence is defined by

$$\mathbf{g}_k^\epsilon = \mathbf{g}_k \exp(\epsilon \boldsymbol{\eta}_k) \quad (4.5)$$

with the discrete matrix-valued curve $\{\boldsymbol{\eta}_k\}_{k=0}^N$ and $\boldsymbol{\eta}_0 = \boldsymbol{\eta}_N = \mathbf{0}$ [Lee08]. Using the infinitesimal variation introduced in (2.20), it follows that

$$\delta \mathbf{g}_k = \mathbf{g}_k \boldsymbol{\eta}_k = T_e \mathbb{L}_{\mathbf{g}_k}(\boldsymbol{\eta}_k). \quad (4.6)$$

Moreover, [Lee08] derives the infinitesimal variation of $\mathbf{f}_k$ as

$$\delta \mathbf{f}_k = T_e \mathbb{L}_{\mathbf{f}_k} \cdot \left(-\text{Ad}_{\mathbf{f}_k^{-1}} \boldsymbol{\eta}_k + \boldsymbol{\eta}_{k+1} \right) = \mathbf{f}_k \mathbf{X}_k, \quad (4.7)$$

with $\mathbf{X}_k = -\text{Ad}_{\mathbf{f}_k^{-1}} \boldsymbol{\eta}_k + \boldsymbol{\eta}_{k+1} \in \mathfrak{g}$. After substituting (4.6) and (4.7) into (4.4), the variation of the action sum becomes

$$\begin{aligned} \delta \mathcal{S}_d = \sum_{k=1}^{N-1} \left\langle T_e^* \mathbb{L}_{\mathbf{f}_{k-1}} \cdot D_{\mathbf{f}_{k-1}} L_d(\mathbf{g}_{k-1}, \mathbf{f}_{k-1}) + T_e^* \mathbb{L}_{\mathbf{g}_k} \cdot D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k) \right. \\ \left. - \text{Ad}_{\mathbf{f}_k^{-1}}^* \cdot (T_e^* \mathbb{L}_{\mathbf{f}_k} \cdot D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k)), \boldsymbol{\eta}_k \right\rangle = 0, \end{aligned} \quad (4.8)$$

shifting the summation index, and using $\boldsymbol{\eta}_0 = \boldsymbol{\eta}_N = \mathbf{0}$ [Lee08].

Since (4.8) holds for all sequences $\{\boldsymbol{\eta}_k\}_{k=1}^{N-1} \subset \mathfrak{g}$, the discrete Euler-Lagrange equations on G are defined as

$$\begin{aligned} T_e^* \mathbb{L}_{\mathbf{f}_{k-1}} \cdot D_{\mathbf{f}_{k-1}} L_d(\mathbf{g}_{k-1}, \mathbf{f}_{k-1}) - \text{Ad}_{\mathbf{f}_k}^* \cdot (T_e^* \mathbb{L}_{\mathbf{f}_k} \cdot D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k)) \\ + T_e^* \mathbb{L}_{\mathbf{g}_k} \cdot D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k) = 0, \end{aligned} \quad (4.9a)$$

$$\mathbf{g}_{k+1} = \mathbf{g}_k \mathbf{f}_k, \quad (4.9b)$$

[Lee08]. Equation (4.9a) is the dynamical representation of the system in the body-fixed frame, while (4.9b) updates the configuration on the Lie group.

In order to derive the discrete Euler-Lagrange equations in the following sections, the definitions of the partial derivatives $D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k)$ and $D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k)$ are required.

Proposition 4 (Directional derivative representation of $D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k)$). *Let G be a Lie group with Lie algebra $\mathfrak{g}$, and let $L_d : G \times G \rightarrow \mathbb{R}$ be smooth [Hal00]. For any $\delta \mathbf{g}_k = T_e \mathbb{L}_{\mathbf{g}_k}(\boldsymbol{\eta}_k) = \mathbf{g}_k \boldsymbol{\eta}_k \in T_{\mathbf{g}_k} G$ (4.6), the partial derivative $D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k) \in T_{\mathbf{g}_k}^* G$ is characterized by*

$$\langle D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \delta \mathbf{g}_k \rangle = \langle T_e^* \mathbb{L}_{\mathbf{g}_k} \cdot D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \boldsymbol{\eta}_k \rangle = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L_d(\mathbf{g}_k \exp(\epsilon \boldsymbol{\eta}_k), \mathbf{f}_k). \quad (4.10)$$

Proof. By definition of the infinitesimal variation $\mathbf{g}_k^\epsilon$ (4.6) and the definition of the infinitesimal variation on matrix Lie groups (2.20), the corresponding directional derivative is

$$\langle D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \delta \mathbf{g}_k \rangle = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L_d(\mathbf{g}_k \exp(\epsilon \boldsymbol{\eta}_k), \mathbf{f}_k).$$

using $\langle D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \delta \mathbf{g}_k \rangle = D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k)(\delta \mathbf{g}_k)$. Additionally, the directional derivative can be expressed by the covector $T_e^* \mathbb{L}_{\mathbf{g}_k}$ and the definition of the differential of the left translation $\mathbf{g}_k \boldsymbol{\eta}_k = T_e \mathbb{L}_{\mathbf{g}_k}(\boldsymbol{\eta}_k)$ defined in (2.12), which leads to

$$\begin{aligned} \langle D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \delta \mathbf{g}_k \rangle &= \langle D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \mathbf{g}_k \boldsymbol{\eta}_k \rangle \\ &= \langle D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), T_e \mathbb{L}_{\mathbf{g}_k}(\boldsymbol{\eta}_k) \rangle \\ &= \langle T_e^* \mathbb{L}_{\mathbf{g}_k} \cdot D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \boldsymbol{\eta}_k \rangle. \end{aligned}$$

□

Proposition 5 (Directional derivative representation of $D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k)$). *Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$, and let $L_d : G \times G \rightarrow \mathbb{R}$ be smooth [Hal00]. For any $\delta \mathbf{f}_k = T_e \mathbb{L}_{\mathbf{f}_k}(\mathbf{X}_k) = \mathbf{f}_k \mathbf{X}_k \in T_{\mathbf{f}_k} G$ (4.7), the partial derivative $D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k) \in T_{\mathbf{f}_k}^* G$ is characterized by*

$$\begin{aligned} \langle D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \delta \mathbf{f}_k \rangle &= \langle T_e^* \mathbb{L}_{\mathbf{f}_k} \cdot D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \mathbf{X}_k \rangle \\ &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L_d(\mathbf{g}_k, \mathbf{f}_k \exp(\epsilon \mathbf{X}_k)). \end{aligned} \quad (4.11)$$

Proof. By definition of the infinitesimal variation $\mathbf{f}_k^\epsilon$ (4.7) and the definition of the infinitesimal variation on matrix Lie groups (2.20), the corresponding directional derivative is

$$\langle D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \delta \mathbf{f}_k \rangle = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L_d(\mathbf{g}_k, \mathbf{f}_k \exp(\epsilon \mathbf{X}_k)).$$

using $\langle D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \delta \mathbf{f}_k \rangle = D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k)(\delta \mathbf{f}_k)$. Additionally, the directional derivative can be expressed by the covector $T_e^* \mathbb{L}_{\mathbf{f}_k}$ and the definition of the differential of the left translation $\mathbf{f}_k \mathbf{X}_k = T_e \mathbb{L}_{\mathbf{f}_k}(\mathbf{X}_k)$ defined in (2.12), which leads to

$$\begin{aligned} \langle D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \delta \mathbf{f}_k \rangle &= \langle D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \mathbf{f}_k \mathbf{X}_k \rangle \\ &= \langle D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), T_e \mathbb{L}_{\mathbf{f}_k}(\mathbf{X}_k) \rangle \\ &= \langle T_e^* \mathbb{L}_{\mathbf{f}_k} \cdot D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \mathbf{X}_k \rangle. \end{aligned}$$

□

4.1.2 Discrete Legendre Transformation and Hamiltonian Mechanics

Correspondingly to the continuous Legendre transformation in Section 3.1.2, the same discrete dynamics can be represented on the left-trivialized cotangent bundle T^*G identified with $G \times \mathfrak{g}^*$. Thereby, [Lee08] defines the discrete Legendre transformations $\mathbb{F}^+ L_d, \mathbb{F}^- L_d : G \times G \rightarrow G \times \mathfrak{g}^*$ as

$$\mathbb{F}^+ L_d(\mathbf{g}_k, \mathbf{f}_k) = (\mathbf{g}_k \mathbf{f}_k, \boldsymbol{\mu}_{k+1}), \quad (4.12a)$$

$$\mathbb{F}^- L_d(\mathbf{g}_k, \mathbf{f}_k) = (\mathbf{g}_k, \boldsymbol{\mu}_k), \quad (4.12b)$$

where the discrete momenta $\boldsymbol{\mu}_k, \boldsymbol{\mu}_{k+1} \in \mathfrak{g}^*$ are

$$\boldsymbol{\mu}_k = -T_e^* \mathbb{L}_{\mathbf{g}_k} \cdot D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k) + \text{Ad}_{\mathbf{f}_k^{-1}}^* \cdot (T_e^* \mathbb{L}_{\mathbf{f}_k} \cdot D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k)), \quad (4.13a)$$

$$\boldsymbol{\mu}_{k+1} = T_e^* \mathbb{L}_{\mathbf{f}_k} \cdot D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k). \quad (4.13b)$$

[MW01] describes that the discrete Euler-Lagrange equation (4.9a) is equivalent to the momentum matching condition

$$\mathbb{F}^+ L_d(\mathbf{g}_{k-1}, \mathbf{f}_{k-1}) = \mathbb{F}^- L_d(\mathbf{g}_k, \mathbf{f}_k). \quad (4.14)$$

Thus, the momentum obtained from the preceding interval is equal to the momentum obtained from the following interval at the common node $\mathbf{g}_k$ [MW01].

Using (4.13), the corresponding discrete Hamilton equations are given by

$$\text{Ad}_{\mathbf{f}_k^{-1}}^* \cdot (T_e^* \mathbb{L}_{\mathbf{f}_k} \cdot D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k)) = \boldsymbol{\mu}_k + T_e^* \mathbb{L}_{\mathbf{g}_k} \cdot D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k), \quad (4.15a)$$

$$\mathbf{g}_{k+1} = \mathbf{g}_k \mathbf{f}_k, \quad (4.15b)$$

$$\boldsymbol{\mu}_{k+1} = \text{Ad}_{\mathbf{f}_k}^* \cdot (\boldsymbol{\mu}_k + T_e^* \mathbb{L}_{\mathbf{g}_k} \cdot D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k)). \quad (4.15c)$$

First, the implicit equation (4.15a) is solved for $\mathbf{f}_k$, given the state $(\mathbf{g}_k, \boldsymbol{\mu}_k)$. Consequently, the next configuration $\mathbf{g}_{k+1}$ and momentum $\boldsymbol{\mu}_{k+1}$ are then obtained from (4.15b) and (4.15c). Consequently, (4.15) defines the discrete flow $(\mathbf{g}_k, \boldsymbol{\mu}_k) \mapsto (\mathbf{g}_{k+1}, \boldsymbol{\mu}_{k+1})$ [Lee08].

4.1.3 Properties of the Discrete Lagrangian Flow

Similarly to the continuous case in Section 3.1.3, the discrete-time Lagrangian flow associated with a regular discrete Lagrangian is symplectic. Moreover, if the discrete Lagrangian is invariant under the lifted action of a Lie group, the corresponding discrete momentum map is preserved [Lee08].

Symplecticity In contrast to the continuous case (3.13), in the discrete case, there exist two Lagrangian one-forms $\Theta_{L_d}^+$, $\Theta_{L_d}^-$ on $G \times G$ through boundary terms [MW01], which relation is given by

$$dL_d = \Theta_{L_d}^+ - \Theta_{L_d}^-.$$

Moreover, [MW01] shows that $d\Theta_{L_d}^+ = d\Theta_{L_d}^-$, as the second-order derivative d^2L_d equals zero. Therefore, the discrete Lagrangian symplectic form is defined as $\Omega_{L_d} = d\Theta_{L_d}^+ = d\Theta_{L_d}^-$, which is preserved by the discrete Lagrangian flow $\mathcal{F}_{L_d} : G \times G \rightarrow G \times G$. Consequently,

$$(\mathcal{F}_{L_d}^{N-1})^* \Omega_{L_d} = \Omega_{L_d}, \quad (4.16)$$

for $N - 1$ discrete steps [MW01].

Discrete Noether's theorem Considering the defined action of a Lie group H on G , $\Phi : H \times G \rightarrow G$ in Section 3.1.3, there exist two Lagrangian momentum maps $J_{L_d}^+$ and $J_{L_d}^- : G \times G \rightarrow \mathfrak{h}^*$.

Theorem 2 (Discrete Noether's theorem). *If the discrete Lagrangian $L_d : G \times G \rightarrow \mathbb{R}$ is invariant under the lifted action of $\Phi : H \times G \rightarrow G$, then the two discrete Lagrangian momentum maps are equal $J_{L_d}^+ = J_{L_d}^- = J_{L_d}$, and the associated discrete momentum map is preserved by the discrete Lagrangian flow,*

$$J_{L_d}(\mathcal{F}_{L_d}^{N-1}(\mathbf{g}_0, \mathbf{f}_0)) = J_{L_d}(\mathbf{g}_0, \mathbf{f}_0), \quad (4.17)$$

which is called discrete Noether's theorem [MW01].

In contrast to the continuous Lagrangian mechanics of Section 3.1, the mathematical validation of momentum preservation for discrete variational integrators of single and multi-body systems is not considered in this thesis. However, their long-time energy behaviour will be evaluated through numerical simulations in Section 6.1.

4.1.4 Forced Discrete Euler-Lagrange Equations

Currently, the discrete Euler-Lagrange equations (4.9) describe unforced mechanical systems. However, in most cases the system is subjected to external forces or control inputs. The Lagrange-d'Alembert principle can be applied to derive the forced Euler-Lagrange equations for continuous mechanics, given by

$$\delta \int_{t_0}^{t_f} L(\mathbf{g}, \boldsymbol{\xi}) dt + \int_{t_0}^{t_f} \mathbf{u}(t) \cdot \boldsymbol{\eta} dt = 0, \quad (4.18)$$

with generalized forces $\mathbf{u}(t) : [t_0, t_f] \rightarrow \mathfrak{g}^*$ acting on the system for any $\boldsymbol{\eta} = \mathbf{g}^{-1} \delta \mathbf{g} \in \mathfrak{g}$ in continuous time [Lee08].

In discrete time the force in the Lagrange-d'Alembert principle is approximated by

$$\int_{t_k}^{t_{k+1}} \mathbf{u}(t) \cdot \boldsymbol{\eta} dt \approx \mathbf{u}_{d_k}^- \cdot \boldsymbol{\eta}_k + \mathbf{u}_{d_k}^+ \cdot \boldsymbol{\eta}_{k+1}, \quad (4.19)$$

using discrete generalized forces $\mathbf{u}_{d_k}^+, \mathbf{u}_{d_k}^- \in \mathfrak{g}^*$ acting on the system at the discrete time step k [Lee08]. Therefore, corresponding to the action sum of the discrete Lagrangian (4.3), the discrete Lagrange-d'Alembert principle is given by

$$\delta \sum_{k=0}^{N-1} L_d(\mathbf{g}_k, \mathbf{f}_k) + \sum_{k=0}^{N-1} (\mathbf{u}_{d_k}^- \cdot \boldsymbol{\eta}_k + \mathbf{u}_{d_k}^+ \cdot \boldsymbol{\eta}_{k+1}) = 0. \quad (4.20)$$

[Lee08] states that (4.20) is equivalent to the discrete Hamilton's principle (4.4) with the additional forcing term. In consequence, the corresponding forced version of the unforced discrete Euler-Lagrange equations (4.9) is given by

$$\begin{aligned} T_e^* \mathbb{L}_{\mathbf{f}_{k-1}} \cdot D_{\mathbf{f}_{k-1}} L_d(\mathbf{g}_{k-1}, \mathbf{f}_{k-1}) - \text{Ad}_{\mathbf{f}_k}^* \cdot (T_e^* \mathbb{L}_{\mathbf{f}_k} \cdot D_{\mathbf{f}_k} L_d(\mathbf{g}_k, \mathbf{f}_k)) \\ + T_e^* \mathbb{L}_{\mathbf{g}_k} \cdot D_{\mathbf{g}_k} L_d(\mathbf{g}_k, \mathbf{f}_k) + \mathbf{u}_{d_k}^- + \mathbf{u}_{d_{k-1}}^+ = \mathbf{0}, \end{aligned} \quad (4.21a)$$

$$\mathbf{g}_{k+1} = \mathbf{g}_k \mathbf{f}_k. \quad (4.21b)$$

According to [Lee08], for the discrete Lagrangian of the form

$$L_d(\mathbf{g}_k, \mathbf{f}_k) = T_d(\mathbf{f}_k) - (1 - c)hV(\mathbf{g}_k) - chV(\mathbf{g}_k \mathbf{f}_k), \quad (4.22)$$

with $T_d(\mathbf{f}_k)$ being the discrete kinetic energy, $V(\mathbf{g}_k)$ the potential energy and $c \in [0, 1]$ a constant parameter, the discrete forces can be generalized as

$$\mathbf{u}_{d_k}^- = (1 - c)h\mathbf{u}_k, \quad \mathbf{u}_{d_k}^+ = ch\mathbf{u}_{k+1}, \quad (4.23)$$

at the discrete time nodes t_k and t_{k-1} .

4.1.5 Computational Implementation

In contrast to the continuous mechanics of Chapter 3, numerical simulations can be performed for discrete-time mechanical model representations. Therefore, a simulation framework has to be defined. To determine the relative attitude update $\mathbf{F}_k \in SO(3)$, the implicit Euler-Lagrange (4.9) and Hamilton equations (4.15) have to be solved at every time step k . Therefore, an efficient computational algorithm is required for practical numerical simulations, especially for complex multibody-mechanical models. [Lee08] proposes an approach of expressing the group element $\mathbf{F}_k \in G$ on the linear vector space of the Lie algebra, by using the exponential map. Thereby, the implicit equation is solved numerically by a Newton iteration [MW01] in $\mathbb{R}^3$.

Firstly, the implicit discrete-time Euler-Lagrange equations or the discrete-time Hamilton equations have to be converted into an equivalent vector equation in $\mathbb{R}^3$, for example, using the Rodrigues formula

$$\mathbf{F} = \exp(\hat{\mathbf{f}}) = \mathbf{I}_{3 \times 3} + \frac{\sin\|\mathbf{f}\|}{\|\mathbf{f}\|} \hat{\mathbf{f}} + \frac{1 - \cos\|\mathbf{f}\|}{\|\mathbf{f}\|^2} \hat{\mathbf{f}}^2, \quad (4.24)$$

for $\mathbf{f} \in \mathbb{R}^3$ [MR99]. However, it requires the evaluation of expensive sin and cos terms, which increase the computational time and costs. In order to improve the computational efficiency, the exponential map (4.24) is replaced by the Cayley transformation [Ise01].

Cayley Transformation The Cayley transformation is a local diffeomorphism between $\mathbb{R}^3$ and $SO(3)$, which is defined as

$$\mathbf{F} = \text{Cay}(\mathbf{f}) = (\mathbf{I} + \hat{\mathbf{f}})(\mathbf{I} - \hat{\mathbf{f}})^{-1} = \frac{1}{1 + \mathbf{f}^\top \mathbf{f}} \left((1 - \mathbf{f}^\top \mathbf{f})\mathbf{I} + 2\hat{\mathbf{f}} + 2\mathbf{f}\mathbf{f}^\top \right), \quad (4.25)$$

representing a rotation θ given by $\|\mathbf{f}\| = \tan\left(\frac{\theta}{2}\right)$ [BLdD23]. Therefore, the Cayley transformation is subject to the constraint $\theta < \pi$, which requires the step size h to be chosen sufficiently small. After substituting the Cayley transformation (4.25) into the implicit equations (4.9a) or (4.15a), a equivalent vector equation is obtained

$$\mathbf{0} = \mathbf{A}(\mathbf{f}), \quad (4.26)$$

where $\mathbf{A} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ [CJ12].

Newton Iteration For solving the implicit vector equation (4.26), Newton's method is used [MW01]. The iteration rule is given by

$$\mathbf{f}^{(i+1)} = \mathbf{f}^{(i)} - \nabla \mathbf{A}(\mathbf{f}^{(i)})^{-1} \mathbf{A}(\mathbf{f}^{(i)}), \quad (4.27)$$

where convergence to the solution is reached until $\|\mathbf{f}^{(i+1)} - \mathbf{f}^{(i)}\| < \varepsilon_N$ for a defined ε_N . An initial guess $\mathbf{f}^0$ can be obtained by linearizing the vector equation (4.26) [Lee08]. Finally, the rotation matrix $\mathbf{F}_k \in SO(3)$ at k is calculated using the Cayley transformation (4.25).

4.2 Single Rigid Body Systems

4.2.1 3D-Pendulum

In the following Section, the unforced mechanics of a three-degree-of-freedom pendulum given by [Lee08] are presented, using discrete Lagrangian mechanics of section 4.1. This corresponds to the equivalent discrete case of the continuous unforced 3D-pendulum of Section 3.2.

Configuration Manifold The discrete configuration manifold of the unforced 3D-pendulum uses the same space and body-fixed frame definitions as the continuous model. The gravity direction in the space frame is represented by the unit vector $\mathbf{e}_3$, and the vector from the pivot to the center of mass is $\boldsymbol{\rho}_c$ in the body frame. As in the continuous formulation of Section 3.2, the discrete configuration manifold is $SO(3)$. Therefore, the attitude of the 3D-pendulum is described by a rotation matrix $\mathbf{R}_k \in SO(3)$, which maps the body-fixed frame to the reference frame at the discrete time step k . [Lee08] defines the discrete-time attitude kinematics equation as

$$\mathbf{R}_{k+1} = \mathbf{R}_k \mathbf{F}_k, \quad (4.28)$$

with the rotation matrix $\mathbf{F}_k$, representing the relative attitude update between two integration steps. Additionally, [Lee08] introduces adjoint operator relations for $\mathbf{F} \in SO(3)$ and the skew-symmetric matrix $\hat{\boldsymbol{\eta}} \in \mathfrak{so}(3)$, which are given as

$$\text{Ad}_{\mathbf{F}} \hat{\boldsymbol{\eta}} = \mathbf{F} \hat{\boldsymbol{\eta}} \mathbf{F}^\top = (\mathbf{F} \boldsymbol{\eta})^\wedge, \quad (4.29)$$

$$\text{Ad}_{\mathbf{F}}^* \hat{\boldsymbol{\eta}} = \mathbf{F}^\top \hat{\boldsymbol{\eta}} \mathbf{F} = (\mathbf{F}^\top \boldsymbol{\eta})^\wedge. \quad (4.30)$$

Discrete Lagrangian As defined in (4.2), the discrete Lagrangian $L_d(\mathbf{R}_k, \mathbf{F}_k) : SO(3) \times SO(3) \rightarrow \mathbb{R}$ is an approximation of the action integral of the continuous Lagrangian over one time step h . In order to discretize the given continuous Lagrangian of (3.25)

$$L(\mathbf{R}, \hat{\boldsymbol{\omega}}) = T(\mathbf{R}, \hat{\boldsymbol{\omega}}) - V(\mathbf{R}) = \frac{1}{2} \text{tr}[\hat{\boldsymbol{\omega}} \mathbf{J}_d \hat{\boldsymbol{\omega}}^\top] + mg \mathbf{e}_3^\top \mathbf{R} \boldsymbol{\rho}_c, \quad (4.31)$$

the time dependent angular velocity $\hat{\boldsymbol{\omega}}(t)$ and rotation matrix $\mathbf{R}(t)$ have to be represented on the discrete time interval $[t_k, t_{k+1}]$ [MW01]. Given the definition of the exponential (2.9) for an $n \times n$ matrix $\mathbf{X}$, its first-order expansion can be written as

$$\exp(\mathbf{X}) = \mathbf{I} + \mathbf{X} + O(\mathbf{X}^2). \quad (4.32)$$

Therefore, for small h , the exponential map $\mathbf{F}_k = \exp(h\hat{\boldsymbol{\omega}}_k)$ can be approximated using (4.32), by

$$\mathbf{F}_k = \exp(h\hat{\boldsymbol{\omega}}_k) \approx \mathbf{I} + h\hat{\boldsymbol{\omega}}_k.$$

This gives the general discrete approximation of the angular velocity

$$\hat{\boldsymbol{\omega}}_k \approx \frac{1}{h}(\mathbf{F}_k - \mathbf{I}), \quad (4.33)$$

[Lee08]. Secondly, quadrature rules, for example, the midpoint rule and the trapezoidal rule, can be applied to create an integrator $F_{L_d} : (\mathbf{R}_0, \mathbf{F}_0) \rightarrow (\mathbf{R}_1, \mathbf{F}_1)$ [MW01]. In order to align with [Lee08], the trapezoidal rule is applied to approximate an integral, as given by

$$\int_{t_k}^{t_{k+1}} f(t) dt \approx \frac{h}{2} (f(t_k) + f(t_{k+1})), \quad (4.34)$$

for $h = t_{k+1} - t_k$ [But16]. Applying (4.34) to the Lagrangian (4.31), it follows

$$\begin{aligned} \int_{t_k}^{t_{k+1}} V(\mathbf{R}) dt &= \int_{t_k}^{t_{k+1}} -mg \mathbf{e}_3^\top \mathbf{R} \boldsymbol{\rho}_c dt \approx -\frac{h}{2} mg \mathbf{e}_3^\top \mathbf{R}_k \boldsymbol{\rho}_c - \frac{h}{2} mg \mathbf{e}_3^\top \mathbf{R}_{k+1} \boldsymbol{\rho}_c \\ &= -\frac{h}{2} mg \mathbf{e}_3^\top \mathbf{R}_k \boldsymbol{\rho}_c - \frac{h}{2} mg \mathbf{e}_3^\top \mathbf{R}_k \mathbf{F}_k \boldsymbol{\rho}_c, \\ &= \frac{h}{2} V(\mathbf{R}_k) + \frac{h}{2} V(\mathbf{R}_k \mathbf{F}_k), \end{aligned} \quad (4.35)$$

using (4.28), as only the potential energy $V(\mathbf{R})$ of the continuous Lagrangian depends on $\mathbf{R}$. Substituting (4.33) and (4.35) into (4.31), the discrete Lagrangian is calculated as

$$\begin{aligned} L_d(\mathbf{R}_k, \mathbf{F}_k) &= \frac{1}{2h} \text{tr}[(\mathbf{F}_k - \mathbf{I}) \mathbf{J}_d (\mathbf{F}_k - \mathbf{I})^\top] - \frac{h}{2} (V(\mathbf{R}_k) + V(\mathbf{R}_k \mathbf{F}_k)) \\ &= \frac{1}{2h} \text{tr}[\mathbf{F}_k \mathbf{J}_d \mathbf{F}_k^\top - \mathbf{J}_d \mathbf{F}_k^\top - \mathbf{F}_k \mathbf{J}_d + \mathbf{J}_d] - \frac{h}{2} (V(\mathbf{R}_k) + V(\mathbf{R}_k \mathbf{F}_k)) \\ &= \frac{1}{2h} \text{tr}[2\mathbf{J}_d - 2\mathbf{F}_k \mathbf{J}_d] - \frac{h}{2} (V(\mathbf{R}_k) + V(\mathbf{R}_k \mathbf{F}_k)) \\ &= \frac{1}{h} \text{tr}[(\mathbf{I} - \mathbf{F}_k) \mathbf{J}_d] - \frac{h}{2} (V(\mathbf{R}_k) + V(\mathbf{R}_k \mathbf{F}_k)), \end{aligned} \quad (4.36)$$

using $\text{tr}[\mathbf{F}_k \mathbf{J}_d \mathbf{F}_k^\top] = \text{tr}[\mathbf{J}_d \mathbf{F}_k^\top \mathbf{F}_k] = \text{tr}[\mathbf{J}_d]$ and $\text{tr}[\mathbf{J}_d \mathbf{F}_k^\top] = \text{tr}[\mathbf{F}_k \mathbf{J}_d]$, since $\mathbf{J}_d$ is defined as a diagonal matrix $\mathbf{J}_d = \mathbf{J}_d^\top$ [Lee08].

Discrete Euler-Lagrange Equations In order to define the Euler-Lagrange Equation (4.9), the derivatives of (4.36) have to be calculated. Firstly, the directional derivative $\langle D_{\mathbf{F}_k} L_d(\mathbf{R}_k, \mathbf{F}_k), \delta \mathbf{F}_k \rangle = \left\langle T_e^* \mathbb{L}_{\mathbf{F}_k} D_{\mathbf{F}_k} L_d(\mathbf{R}_k, \mathbf{F}_k), \hat{\mathbf{X}}_k \right\rangle$ is given by applying Proposition (5), which yields

$$\begin{aligned} \left\langle T_e^* \mathbb{L}_{\mathbf{F}_k} D_{\mathbf{F}_k} L_d(\mathbf{R}_k, \mathbf{F}_k), \hat{\mathbf{X}}_k \right\rangle &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L_d(\mathbf{R}_k, \mathbf{F}_k \exp(\epsilon \hat{\mathbf{X}}_k)) \\ &= -\frac{1}{h} \text{tr}[\mathbf{F}_k \hat{\mathbf{X}}_k \mathbf{J}_d] + \frac{h}{2} mg \mathbf{e}_3^\top \mathbf{R}_k \mathbf{F}_k \hat{\mathbf{X}}_k \boldsymbol{\rho}_c \\ &= -\frac{1}{h} \text{tr}[\hat{\mathbf{X}}_k \mathbf{J}_d \mathbf{F}_k] + \frac{h}{2} mg \left((\mathbf{R}_k \mathbf{F}_k)^\top \mathbf{e}_3 \right)^\top \hat{\mathbf{X}}_k \boldsymbol{\rho}_c, \end{aligned}$$

using the trace identity $\text{tr}[\mathbf{ABC}] = \text{tr}[\mathbf{BCA}]$ for any $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{R}^{n \times n}$. Additionally, [Lee08] introduces the relation $\text{tr}[\hat{\mathbf{s}}\mathbf{B}] = -\langle \mathbf{B} - \mathbf{B}^\top, \hat{\mathbf{s}} \rangle$ for any $\mathbf{s} \in \mathbb{R}^3, \mathbf{B} \in \mathbb{R}^{3 \times 3}$. Applying it together with $\mathbf{a}^\top \hat{\mathbf{x}}\mathbf{b} = (\mathbf{b} \times \mathbf{a}) \cdot \mathbf{x}$ for any $\mathbf{a}, \mathbf{b}, \mathbf{x} \in \mathbb{R}^3$, the inner product definition (2.18), and the linearity of the pairing $\langle \mathbf{A}, \mathbf{B} \rangle + \langle \mathbf{C}, \mathbf{B} \rangle = \langle \mathbf{A} + \mathbf{C}, \mathbf{B} \rangle$ for any $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{R}^{n \times n}$, the final directional derivative of the discrete Lagrangian with respect to $\mathbf{F}_k$ is calculated as

$$\begin{aligned} \langle T_e^* \mathbb{L}_{\mathbf{F}_k} D_{\mathbf{F}_k} L_d(\mathbf{R}_k, \mathbf{F}_k), \hat{\mathbf{X}}_k \rangle &= \left\langle \frac{1}{h} (\mathbf{J}_d \mathbf{F}_k - \mathbf{F}_k^\top \mathbf{J}_d), \hat{\mathbf{X}}_k \right\rangle + \left\langle \frac{h}{2} mg (\boldsymbol{\rho}_c \times (\mathbf{R}_k \mathbf{F}_k)^\top \mathbf{e}_3)^\wedge, \hat{\mathbf{X}}_k \right\rangle \\ &= \left\langle \frac{1}{h} (\mathbf{J}_d \mathbf{F}_k - \mathbf{F}_k^\top \mathbf{J}_d) + \left(\frac{h}{2} mg \boldsymbol{\rho}_c \times \mathbf{R}_{k+1}^\top \mathbf{e}_3 \right)^\wedge, \hat{\mathbf{X}}_k \right\rangle. \end{aligned}$$

Therefore,

$$T_e^* \mathbb{L}_{\mathbf{F}_k} D_{\mathbf{F}_k} L_d(\mathbf{R}_k, \mathbf{F}_k) \hat{=} \frac{1}{h} (\mathbf{J}_d \mathbf{F}_k - \mathbf{F}_k^\top \mathbf{J}_d) + \frac{h}{2} \hat{\boldsymbol{\tau}}_{g,k+1}, \quad (4.37)$$

with the definition of

$$\boldsymbol{\tau}_{g,k} = mg \boldsymbol{\rho}_c \times \mathbf{R}_k^\top \mathbf{e}_3, \quad (4.38)$$

being the moment due to the potential energy in the body frame [Lee08].

Moreover, the directional derivative

$$\langle D_{\mathbf{R}_{k+1}} L_d(\mathbf{R}_{k+1}, \mathbf{F}_{k+1}), \delta \mathbf{R}_{k+1} \rangle = \langle T_e^* \mathbb{L}_{\mathbf{R}_{k+1}} D_{\mathbf{R}_{k+1}} L_d(\mathbf{R}_{k+1}, \mathbf{F}_{k+1}), \hat{\boldsymbol{\eta}}_{k+1} \rangle,$$

which in this model depends only on the potential energy $V(\mathbf{R}_{k+1}) + V(\mathbf{R}_{k+1} \mathbf{F}_{k+1})$, is calculated using Proposition (4) as

$$\begin{aligned} \langle T_e^* \mathbb{L}_{\mathbf{R}_{k+1}} D_{\mathbf{R}_{k+1}} L_d(\mathbf{R}_{k+1}, \mathbf{F}_{k+1}), \hat{\boldsymbol{\eta}}_{k+1} \rangle &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L_d(\mathbf{R}_{k+1} \exp(\epsilon \hat{\boldsymbol{\eta}}_{k+1}), \mathbf{F}_{k+1}) \\ &= \frac{h}{2} mg \mathbf{e}_3^\top \mathbf{R}_{k+1} \hat{\boldsymbol{\eta}}_{k+1} \boldsymbol{\rho}_c + \frac{h}{2} mg \mathbf{e}_3^\top \mathbf{R}_{k+1} \hat{\boldsymbol{\eta}}_{k+1} \mathbf{F}_{k+1} \boldsymbol{\rho}_c \\ &= \frac{h}{2} mg \left[(\boldsymbol{\rho}_c \times \mathbf{R}_{k+1}^\top \mathbf{e}_3) + (\mathbf{F}_{k+1} \boldsymbol{\rho}_c \times \mathbf{R}_{k+1}^\top \mathbf{e}_3) \right] \cdot \boldsymbol{\eta}_{k+1}. \end{aligned}$$

Applying the definition of inner products (2.18), it follows

$$\begin{aligned} \langle T_e^* \mathbb{L}_{\mathbf{R}_{k+1}} D_{\mathbf{R}_{k+1}} L_d(\mathbf{R}_{k+1}, \mathbf{F}_{k+1}), \hat{\boldsymbol{\eta}}_{k+1} \rangle &= \left\langle \frac{h}{2} mg \left[(\boldsymbol{\rho}_c \times \mathbf{R}_{k+1}^\top \mathbf{e}_3) + (\mathbf{F}_{k+1} \boldsymbol{\rho}_c \times \mathbf{R}_{k+1}^\top \mathbf{e}_3) \right]^\wedge, \hat{\boldsymbol{\eta}}_{k+1} \right\rangle \\ &= \left\langle \frac{h}{2} (\hat{\boldsymbol{\tau}}_{g,k+1} + (\mathbf{F}_{k+1} \boldsymbol{\tau}_{g,k+2})^\wedge), \hat{\boldsymbol{\eta}}_{k+1} \right\rangle, \end{aligned}$$

which implies that the derivative of (4.31) with respect to $\mathbf{R}_{k+1}$ is given by

$$T_e^* \mathbb{L}_{\mathbf{R}_{k+1}} D_{\mathbf{R}_{k+1}} L_d(\mathbf{R}_{k+1}, \mathbf{F}_{k+1}) = \frac{h}{2} [\hat{\boldsymbol{\tau}}_{g,k+1} + (\mathbf{F}_{k+1} \boldsymbol{\tau}_{g,k+2})^\wedge]. \quad (4.39)$$

Since the directional derivative $T_e^* \mathbb{L}_{\mathbf{F}_k} D_{\mathbf{F}_k} L_d(\mathbf{R}_k, \mathbf{F}_k)$ was already derived in (4.37), the coadjoint term in the discrete Euler-Lagrange equation (4.9) can be written in matrix form as

$$\begin{aligned} \text{Ad}_{\mathbf{F}_{k+1}}^* (T_e^* \mathbb{L}_{\mathbf{F}_{k+1}} D_{\mathbf{F}_{k+1}} L_d(\mathbf{R}_{k+1}, \mathbf{F}_{k+1})) &= \mathbf{F}_{k+1} \left[\frac{1}{h} \left(\mathbf{J}_d \mathbf{F}_{k+1} - \mathbf{F}_{k+1}^\top \mathbf{J}_d \right) + \frac{h}{2} \hat{\boldsymbol{\tau}}_{g,k+2} \right] \mathbf{F}_{k+1}^\top \\ &= \frac{1}{h} \left(\mathbf{F}_{k+1} \mathbf{J}_d - \mathbf{J}_d \mathbf{F}_{k+1}^\top \right) + \frac{h}{2} \mathbf{F}_{k+1} \hat{\boldsymbol{\tau}}_{g,k+2} \mathbf{F}_{k+1}^\top \\ &= \frac{1}{h} \left(\mathbf{F}_{k+1} \mathbf{J}_d - \mathbf{J}_d \mathbf{F}_{k+1}^\top \right) + \frac{h}{2} (\mathbf{F}_{k+1} \boldsymbol{\tau}_{g,k+2})^\wedge, \end{aligned} \quad (4.40)$$

by using the identity (4.30).

Finally, the discrete Euler-Lagrange equations of the unforced 3D-pendulum are expressed in $\mathfrak{so}(3)$ $SO(3)$ form, substituting (4.37), (4.39) and (4.40) into (4.9), which results in

$$\frac{1}{h} (\mathbf{F}_{k+1} \mathbf{J}_d - \mathbf{J}_d \mathbf{F}_{k+1}^\top - \mathbf{J}_d \mathbf{F}_k + \mathbf{F}_k^\top \mathbf{J}_d) = h \hat{\boldsymbol{\tau}}_{g,k+1}, \quad (4.41a)$$

$$\mathbf{R}_{k+1} = \mathbf{R}_k \mathbf{F}_k. \quad (4.41b)$$

[Lee08] describes the procedure of obtaining the discrete-time flow map $(\mathbf{R}_k, \mathbf{R}_{k+1}) \rightarrow (\mathbf{R}_{k+1}, \mathbf{R}_{k+2})$ by solving (4.41a) for $\mathbf{F}_{k+1}$ for given $(\mathbf{R}_k, \mathbf{R}_{k+1})$ and $\mathbf{F}_k = \mathbf{R}_k^\top \mathbf{R}_{k+1}$. Secondly, $\mathbf{R}_{k+2}$ is derived from (4.41b).

Discrete Legendre Transformation In order to obtain the discrete Hamilton equations for the unforced 3D-pendulum model, the discrete Legendre transformation $\mathbb{F}^+ L_d, \mathbb{F}^- L_d : (SO(3) \times SO(3)) \rightarrow (SO(3) \times \mathfrak{so}(3)^*)$ with $\mathbb{F}^+ L_d(\mathbf{R}_{k-1}, \mathbf{F}_{k-1}) = \mathbb{F}^- L_d(\mathbf{R}_k, \mathbf{F}_k)$ (4.12) have to be applied. This results in the expression for the discrete momentum

$$\begin{aligned} \hat{\boldsymbol{\Pi}}_k &= -\frac{h}{2} (\hat{\boldsymbol{\tau}}_{g,k} + (\mathbf{F}_k \boldsymbol{\tau}_{g,k+1})^\wedge) + \frac{1}{h} (\mathbf{F}_k \mathbf{J}_d - \mathbf{J}_d \mathbf{F}_k^\top) + \frac{h}{2} (\mathbf{F}_k \boldsymbol{\tau}_{g,k+1})^\wedge \\ &= \frac{1}{h} (\mathbf{F}_k \mathbf{J}_d - \mathbf{J}_d \mathbf{F}_k^\top) - \frac{h}{2} \hat{\boldsymbol{\tau}}_{g,k}, \end{aligned} \quad (4.42a)$$

$$\hat{\boldsymbol{\Pi}}_{k+1} = \frac{1}{h} (\mathbf{J}_d \mathbf{F}_k - \mathbf{F}_k^\top \mathbf{J}_d) + \frac{h}{2} \hat{\boldsymbol{\tau}}_{g,k+1}, \quad (4.42b)$$

where $\hat{\boldsymbol{\Pi}} \in \mathfrak{so}(3)^*$. By substituting (4.37), (4.39) and (4.40) into (4.15), the discrete Hamilton equations of an unforced 3D-pendulum are calculated as

$$\frac{1}{h} (\mathbf{F}_k \mathbf{J}_d - \mathbf{J}_d \mathbf{F}_k^\top) = \hat{\boldsymbol{\Pi}}_k + \frac{h}{2} \hat{\boldsymbol{\tau}}_{g,k}, \quad (4.43a)$$

$$\mathbf{R}_{k+1} = \mathbf{R}_k \mathbf{F}_k, \quad (4.43b)$$

$$\hat{\boldsymbol{\Pi}}_{k+1} = (\mathbf{F}_k^\top \hat{\boldsymbol{\Pi}}_k)^\wedge + \frac{h}{2} (\mathbf{F}_k^\top \boldsymbol{\tau}_{g,k})^\wedge + \frac{h}{2} \hat{\boldsymbol{\tau}}_{g,k+1}. \quad (4.43c)$$

Finally, the discrete map $(\mathbf{R}_k, \mathbf{\Pi}_k) \rightarrow (\mathbf{R}_{k+1}, \mathbf{\Pi}_{k+1})$ is obtained by implicitly solving the equation (4.43a) for $\mathbf{F}_k$ with given $(\mathbf{R}_k, \mathbf{\Pi}_k)$. In consequence, $\mathbf{R}_{k+1}$ is derived from (4.43b), and $\mathbf{\Pi}_{k+1}$ from (4.43c).

Controlled 3D Pendulum The derived Euler-Lagrange and Hamilton equations describe the free motion of the 3D pendulum under the influence of gravity. In order to formulate a control problem, an external control moment $\mathbf{u}_k$ is applied to the system. [Lee08] identifies

$$\mathbf{u}_k = \mathbf{R}_k^\top \mathbf{e}_3 \times \mathbf{u}_{p_k}, \quad (4.44)$$

in the body-fixed frame for a control parameter $\mathbf{u}_{p_k} \in \mathbb{R}^3$. Thereby, the external control moment has no component along the gravity direction, since the vector $\mathbf{R}_k^\top \mathbf{e}_3$ is aligned with the gravity vector. Thus, the control input preserves the angular momentum about the gravity direction.

For the discrete Lagrangian in (4.31), the potential energy is evaluated at both endpoints of the time interval. Comparing its coefficients with the generalized mechanical discrete Lagrangian (4.22) gives $c = \frac{1}{2}$, which corresponds to the trapezoidal approximation of the potential contribution. Consequently, using the representation of the generalized forces in (4.23), the discrete generalized forces are chosen as

$$\mathbf{u}_{d_k}^- = \frac{h}{2} \mathbf{u}_k, \quad \mathbf{u}_{d_k}^+ = \frac{h}{2} \mathbf{u}_{k+1}. \quad (4.45)$$

Following, the controlled 3D-Pendulum Euler-Lagrange equations are derived as

$$\frac{1}{h} (\mathbf{F}_{k+1} \mathbf{J}_d - \mathbf{J}_d \mathbf{F}_{k+1}^\top - \mathbf{J}_d \mathbf{F}_k + \mathbf{F}_k^\top \mathbf{J}_d) = h (\boldsymbol{\tau}_{g,k+1} + \mathbf{u}_{k+1})^\wedge, \quad (4.46a)$$

$$\mathbf{R}_{k+1} = \mathbf{R}_k \mathbf{F}_k. \quad (4.46b)$$

by substituting the generalized forces (4.45) into the previously derived discrete Euler-Lagrange equations of the uncontrolled 3D-pendulum (4.41), considering the Lagrange-d'Alembert principle for forced discrete mechanics (4.21). Correspondingly, the controlled 3D-Pendulum Hamilton equations are given as

$$\hat{\mathbf{\Pi}}_k = \frac{1}{h} (\mathbf{F}_k \mathbf{J}_d - \mathbf{J}_d \mathbf{F}_k^\top) - \frac{h}{2} \hat{\boldsymbol{\tau}}_{g,k} - \frac{h}{2} \hat{\mathbf{u}}_k, \quad (4.47a)$$

$$\mathbf{R}_{k+1} = \mathbf{R}_k \mathbf{F}_k, \quad (4.47b)$$

$$\hat{\mathbf{\Pi}}_{k+1} = (\mathbf{F}_k^\top \mathbf{\Pi}_k)^\wedge + \frac{h}{2} (\mathbf{F}_k^\top (\boldsymbol{\tau}_{g,k} + \mathbf{u}_k))^\wedge + \frac{h}{2} (\boldsymbol{\tau}_{g,k+1} + \mathbf{u}_{k+1})^\wedge, \quad (4.47c)$$

[Lee08]. It can be observed that, for $\mathbf{u}_k = \mathbf{0}$, the forced discrete Euler-Lagrange (4.46) and Hamilton (4.47) equations reduce to the corresponding unforced equations (4.41), (4.43) derived previously.

4.3 Multi-body Systems

The previous sections considered the discrete mechanics of a single rigid body whose configuration evolves directly on a Lie group. For multibody systems, the individual rigid bodies are connected through mechanical joints. These joints impose holonomic constraints on the configuration and therefore have to be included in the variational principle. In this work, constrained multibody systems are represented in maximal coordinates, following the formulation used for variational integrators in maximal coordinates in [BM21b] and for constrained Lie group variational dynamics in [TJVG25].

In minimal coordinates, the configuration is described directly by the independent degrees of freedom of the system. This representation is compact, but the resulting kinematics and dynamics may contain nonlinear trigonometric expressions and dense coupling terms. This is disadvantageous for the polynomial optimization formulation considered later in this thesis. In contrast, maximal coordinates assign each rigid body its own position and orientation variables. The reduction to the physical degrees of freedom is then achieved by imposing algebraic joint constraints with Lagrange multipliers. Therefore, the constraints are not artificial state constraints, but hard mechanical constraints induced by the joints of the system [BM21b].

For a multi-body system with n_b rigid bodies in dimension $d \in \{2, 3\}$, the maximal-coordinate configuration manifold is given by

$$Q_{\max} = \prod_{i=1}^{n_b} (\mathbb{R}^d \times SO(d)). \quad (4.48)$$

Each rigid body i is described by its center of mass position $\mathbf{x}_i \in \mathbb{R}^d$ and its orientation $\mathbf{R}_i \in SO(d)$. Consequently, the full configuration in maximal-coordinates is written as

$$q = (\mathbf{x}_1, \mathbf{R}_1, \dots, \mathbf{x}_{n_b}, \mathbf{R}_{n_b}) \in Q_{\max}. \quad (4.49)$$

The joint structure is encoded by holonomic constraints

$$\phi(q) = \mathbf{0}, \quad \phi : Q_{\max} \rightarrow \mathbb{R}^{n_c}, \quad (4.50)$$

where n_c denotes the number of scalar constraint equations.

A typical holonomic joint constraint can be written by equating two points attached to different rigid bodies. This is illustrated in Fig. 4.2. Let $\boldsymbol{\rho}_i$ and $\boldsymbol{\rho}_j$ denote constant vectors from the corresponding center of mass to the joint location in each body-fixed frame. Then a link-connecting constraint is given by

$$\phi_{ij}(q) = \mathbf{x}_i + \mathbf{R}_i \boldsymbol{\rho}_i - \mathbf{x}_j - \mathbf{R}_j \boldsymbol{\rho}_j = \mathbf{0}. \quad (4.51)$$

To include the holonomic constraints in the discrete variational principle, the discrete action sum is augmented by Lagrange multipliers $\boldsymbol{\lambda}_k \in \mathbb{R}^{n_c}$. Using a trapezoidal

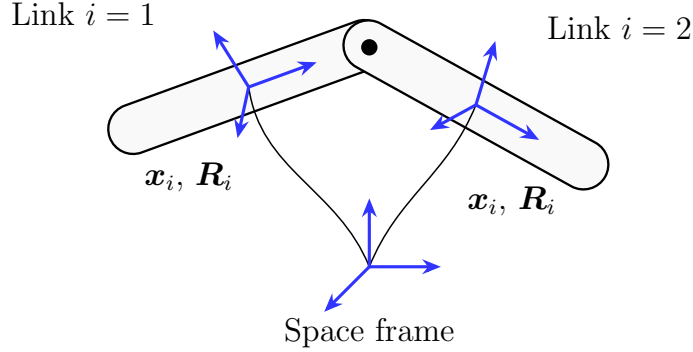


Figure 4.2: Two rigid links connected by a joint in maximal coordinates. Adapted from [BM21a].

approximation (4.34) for the constraint contribution gives the discrete action sum

$$S_d = \sum_{k=0}^{N-1} \left[L_d(q_k, q_{k+1}) + \frac{h}{2} (\boldsymbol{\lambda}_k^\top \boldsymbol{\phi}(q_k) + \boldsymbol{\lambda}_{k+1}^\top \boldsymbol{\phi}(q_{k+1})) \right]. \quad (4.52)$$

4.3.1 Acrobot

Configuration Manifold The Acrobot model, illustrated in Fig. 4.3, consists of two planar rigid links connected by a revolute joint. The first link is attached to the inertial frame by a fixed revolute joint, while the second link is connected to the first link at the elbow. In contrast to the 3D-pendulum, the system is described in maximal coordinates. Therefore, each link is assigned its own center of mass position and orientation.

For each link $i \in \{1, 2\}$, the center of mass position is denoted by $\mathbf{x}_i \in \mathbb{R}^2$, and the orientation of the body-fixed frame with respect to the inertial frame is described by the rotation matrix $\mathbf{R}_i \in SO(2)$. The special orthogonal group in two dimensions is defined as

$$SO(2) = \{ \mathbf{R} \in \mathbb{R}^{2 \times 2} \mid \mathbf{R}^\top \mathbf{R} = \mathbf{I}_2, \det(\mathbf{R}) = 1 \}. \quad (4.53)$$

Every element $\mathbf{R}_i \in SO(2)$ can be written in terms of an angle $\theta_i \in \mathbb{R}$ as

$$\mathbf{R}_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \\ \sin \theta_i & \cos \theta_i \end{bmatrix}. \quad (4.54)$$

Thus, the unconstrained maximal-coordinate configuration manifold is given by

$$Q_{\max} = (\mathbb{R}^2 \times SO(2))^2. \quad (4.55)$$

Consequently, the constrained configuration of the Acrobot is written as

$$Q_c = \{ q \in Q_{\max} \mid \boldsymbol{\phi}(q) = \mathbf{0} \}, \quad (4.56)$$

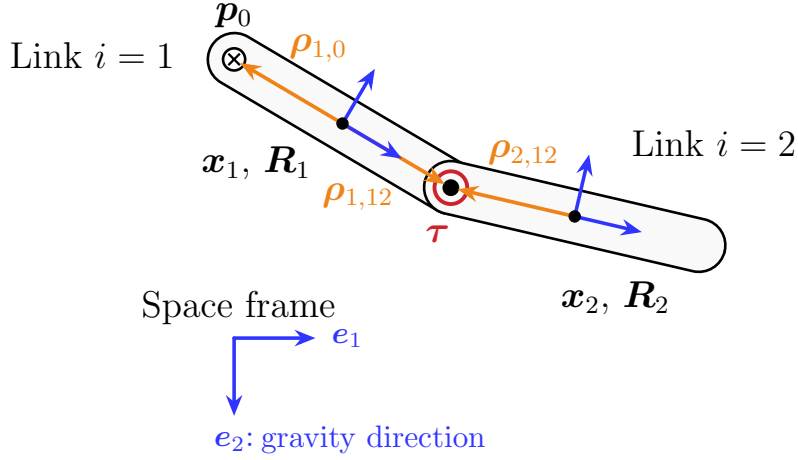


Figure 4.3: Acrobot configuration manifold in maximal coordinates.

with $q = (\mathbf{x}_1, \mathbf{R}_1, \mathbf{x}_2, \mathbf{R}_2)$. Here, $\mathbf{R}_i$ maps vectors from the i -th body-fixed frame to the inertial frame.

Let $\rho_{1,0} \in \mathbb{R}^2$ denote the vector from the center of mass of link 1 to the base joint, expressed in the first body-fixed frame. Furthermore, let $\rho_{1,12} \in \mathbb{R}^2$ and $\rho_{2,12} \in \mathbb{R}^2$ denote the vectors from the corresponding centers of mass to the elbow joint, expressed in the body-fixed frames of link 1 and link 2. The fixed base joint is imposed by

$$\phi_0(q) = \mathbf{x}_1 + \mathbf{R}_1 \rho_{1,0} - \mathbf{p}_0 = \mathbf{0}, \quad (4.57)$$

where $\mathbf{p}_0 \in \mathbb{R}^2$ is the fixed base position in the inertial frame. The elbow joint is imposed by requiring the two joint points on link 1 and link 2 to coincide, yielding

$$\phi_{12}(q) = \mathbf{x}_1 + \mathbf{R}_1 \rho_{1,12} - \mathbf{x}_2 - \mathbf{R}_2 \rho_{2,12} = \mathbf{0}. \quad (4.58)$$

Consequently, the full holonomic constraint vector is defined as

$$\phi(q) = \begin{bmatrix} \phi_0(q) \\ \phi_{12}(q) \end{bmatrix} = \mathbf{0}. \quad (4.59)$$

In the discrete setting, the attitude of each link is updated by a relative rotation $\mathbf{F}_{i,k} \in SO(2)$, such that

$$\mathbf{R}_{i,k+1} = \mathbf{R}_{i,k} \mathbf{F}_{i,k}, \quad i \in \{1, 2\}. \quad (4.60)$$

The discrete configuration at time step k is therefore

$$q_k = (\mathbf{x}_{1,k}, \mathbf{R}_{1,k}, \mathbf{x}_{2,k}, \mathbf{R}_{2,k}) \in Q_c. \quad (4.61)$$

Continuous Lagrangian To derive the continuous Lagrangian of the Acrobot in maximal coordinates, the kinetic and potential energy of both rigid links are introduced. For each link $i \in \{1, 2\}$, the translational velocity of the center of mass is denoted by $\dot{\mathbf{x}}_i = \mathbf{v}_i \in \mathbb{R}^2$. Let $\mathbf{M}_i = m_i \mathbf{I}_2$ be the mass matrix of link i . The defined non-standard inertia matrix in (3.22) is derived for $SO(3)$. In order to apply the non-standard inertia to the planar $SO(2)$ case, the non-standard inertia matrix on $SO(2)$ is derived by rewriting the planar rotational kinetic term of the discrete Lagrangian in [LLM07a]. As a result, the relation of the non-standard inertia to the standard inertia matrix on $SO(2)$ is given by

$$\mathbf{J}_{d,i} = \frac{1}{2} J_i^{\text{com}} \mathbf{I}_2, \quad (4.62)$$

where J_i^{com} is the rotational inertia of a uniform rod about its center of mass. The kinetic energy with its translational T_T and rotational T_R components is written as

$$T = \underbrace{\sum_{i=1}^2 \frac{1}{2} \mathbf{v}_i^\top \mathbf{M}_i \mathbf{v}_i}_{T_T} + \underbrace{\sum_{i=1}^2 \frac{1}{2} \text{tr} [\hat{\boldsymbol{\omega}}_i \mathbf{J}_{d,i} \hat{\boldsymbol{\omega}}_i^\top]}_{T_R}. \quad (4.63)$$

The gravitational potential energy is expressed directly by the center of mass positions. In contrast to the 3D-pendulum, [BM21b] assumes zero potential for the rotational setting, which gives

$$V_R(\mathbf{R}_1, \mathbf{R}_2) = 0. \quad (4.64)$$

Therefore, the gravitational potential energy is only given by the translational term

$$V = V_T(\mathbf{x}_1, \mathbf{x}_2) = - \sum_{i=1}^2 g \mathbf{e}_2^\top \mathbf{M}_i \mathbf{x}_i, \quad (4.65)$$

where $\mathbf{e}_2 \in \mathbb{R}^2$ denotes the unit vector in the gravity direction in the inertial frame. Consequently, the continuous Lagrangian $L : TQ_{\text{max}} \rightarrow \mathbb{R}$ is defined as

$$L(q, \dot{q}) = \sum_{i=1}^2 \left(\frac{1}{2} \mathbf{v}_i^\top \mathbf{M}_i \mathbf{v}_i + \frac{1}{2} \text{tr} [\hat{\boldsymbol{\omega}}_i \mathbf{J}_{d,i} \hat{\boldsymbol{\omega}}_i^\top] \right) + \sum_{i=1}^2 g \mathbf{e}_2^\top \mathbf{M}_i \mathbf{x}_i. \quad (4.66)$$

Discrete Lagrangian In general, the discrete Lagrangian is written as

$$L_d(q_k, q_{k+1}) = T_{d,T}(q_k, q_{k+1}) + T_{d,R}(q_k, q_{k+1}) - V_{d,T}(q_k, q_{k+1}).$$

Following the structure of maximal-coordinate variational integrators in [BM21b], the translational and rotational parts are discretized separately.

For the translational part, the center of mass velocity of each link is

$$\mathbf{v}_{i,k} = \frac{\mathbf{x}_{i,k+1} - \mathbf{x}_{i,k}}{h}, \quad i \in \{1, 2\}. \quad (4.67)$$

[BM21b] defines the discrete translational kinetic energy contribution to be

$$T_{d,T}(q_k, q_{k+1}) = \sum_{i=1}^2 \frac{1}{2h} (\mathbf{x}_{i,k+1} - \mathbf{x}_{i,k})^\top \mathbf{M}_i (\mathbf{x}_{i,k+1} - \mathbf{x}_{i,k}). \quad (4.68)$$

Substituting the angular velocity approximation (4.33) into the rotational kinetic energy T_R of (4.63), the discrete rotational kinetic energy contribution is derived as

$$\begin{aligned} T_{d,R}(q_k, q_{k+1}) &= \sum_{i=1}^2 \frac{1}{2h} \text{tr} \left[(\mathbf{F}_{i,k} - \mathbf{I}_2) \mathbf{J}_{d,i} (\mathbf{F}_{i,k} - \mathbf{I}_2)^\top \right] \\ &= \sum_{i=1}^2 \frac{1}{h} \text{tr} [(\mathbf{I}_2 - \mathbf{F}_{i,k}) \mathbf{J}_{d,i}]. \end{aligned} \quad (4.69)$$

This calculation corresponds to the discrete rotational kinetic energy of the 3D-pendulum in Section 4.2.1.

Finally, using the trapezoidal rule for the discretization of the potential energy integral, the discrete translational potential contribution $V_{d,T}$ is given by

$$V_{d,T}(q_k, q_{k+1}) = -\frac{h}{2} \sum_{i=1}^2 m_i g \mathbf{e}_2^\top (\mathbf{x}_{i,k} + \mathbf{x}_{i,k+1}). \quad (4.70)$$

Combining the translational (4.68), rotational (4.69), and potential (4.70) contributions gives the discrete Lagrangian

$$\begin{aligned} L_d(q_k, q_{k+1}) &= L_d(\mathbf{x}_{1,k}, \mathbf{x}_{1,k+1}, \mathbf{x}_{2,k}, \mathbf{x}_{2,k+1}, \mathbf{F}_{1,k}, \mathbf{F}_{2,k}) \\ &= \sum_{i=1}^2 \frac{1}{2h} (\mathbf{x}_{i,k+1} - \mathbf{x}_{i,k})^\top \mathbf{M}_i (\mathbf{x}_{i,k+1} - \mathbf{x}_{i,k}) \\ &\quad + \sum_{i=1}^2 \frac{1}{h} \text{tr} [(\mathbf{I}_2 - \mathbf{F}_{i,k}) \mathbf{J}_{d,i}] \\ &\quad + \frac{h}{2} \sum_{i=1}^2 g \mathbf{e}_2^\top \mathbf{M}_i (\mathbf{x}_{i,k} + \mathbf{x}_{i,k+1}). \end{aligned} \quad (4.71)$$

Discrete Euler-Lagrange Equations The discrete equations of motion are obtained from the stationary condition of the constrained discrete action. The holonomic constraints introduced in (4.59) are included by extending the action integral with Lagrange multipliers. Let

$$\boldsymbol{\lambda} = \begin{bmatrix} \lambda_0 \\ \lambda_{12} \end{bmatrix} \in \mathbb{R}^4, \quad (4.72)$$

where $\boldsymbol{\lambda}_0 \in \mathbb{R}^2$ enforces the base constraint and $\boldsymbol{\lambda}_{12} \in \mathbb{R}^2$ enforces the elbow constraint. Using the constrained action sum defined in (4.52), the variation is written as

$$\delta S_d = \delta \sum_{k=0}^{N-1} \left[L_d(q_k, q_{k+1}) + \frac{h}{2} (\boldsymbol{\lambda}_k^\top \boldsymbol{\phi}(q_k) + \boldsymbol{\lambda}_{k+1}^\top \boldsymbol{\phi}(q_{k+1})) \right] = 0. \quad (4.73)$$

The translational and rotational components are treated separately, following the maximal-coordinate variational-integrator structure. In [BM21b], the variation of the translational component is given as

$$\delta_{\mathbf{x}_{i,k+1}} S_{d,c} = \left\langle \left(\frac{1}{h} \mathbf{M}_i (\mathbf{v}_{i,k} - \mathbf{v}_{i,k+1}) + g \mathbf{M}_i \mathbf{e}_2 + \mathbf{G}_{\mathbf{x}_{i,k+1}}^\top \boldsymbol{\lambda}_{k+1} \right) h, \delta \mathbf{x}_{i,k+1} \right\rangle, \quad (4.74)$$

using the relation (4.67). $\mathbf{G}_{\mathbf{x}_{i,k+1}}$ is defined as the Jacobian of the constraints $\boldsymbol{\phi}(q_{k+1})$ with respect to $\mathbf{x}_{i,k+1}$. The constraint contribution is calculated as

$$\mathbf{G}_{\mathbf{x}_{i,k+1}}^\top \boldsymbol{\lambda}_{k+1} = \begin{bmatrix} \frac{\partial \phi_{0,k+1}}{\partial \mathbf{x}_{1,k+1}} & \frac{\partial \phi_{0,k+1}}{\partial \mathbf{x}_{2,k+1}} \\ \frac{\partial \phi_{12,k+1}}{\partial \mathbf{x}_{1,k+1}} & \frac{\partial \phi_{12,k+1}}{\partial \mathbf{x}_{2,k+1}} \end{bmatrix}^\top \begin{bmatrix} \boldsymbol{\lambda}_{0,k+1} \\ \boldsymbol{\lambda}_{12,k+1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\lambda}_{0,k+1} + \boldsymbol{\lambda}_{12,k+1} \\ -\boldsymbol{\lambda}_{12,k+1} \end{bmatrix}. \quad (4.75)$$

with the discretized constraints

$$\boldsymbol{\phi}(q_{k+1}) = \begin{bmatrix} \phi_{0,k+1} \\ \phi_{12,k+1} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_{1,k+1} + \mathbf{R}_{1,k+1} \boldsymbol{\rho}_{1,0} - \mathbf{p}_0 \\ \mathbf{x}_{1,k+1} + \mathbf{R}_{1,k+1} \boldsymbol{\rho}_{1,12} - \mathbf{x}_{2,k+1} - \mathbf{R}_{2,k+1} \boldsymbol{\rho}_{2,12} \end{bmatrix}. \quad (4.76)$$

Therefore, substituting the gradient (4.75) into (4.74), the translational discrete Euler-Lagrange equations for link 1 and link 2 are

$$\frac{1}{h} \mathbf{M}_1 (\mathbf{v}_{1,k+1} - \mathbf{v}_{1,k}) - g \mathbf{M}_1 \mathbf{e}_2 - (\boldsymbol{\lambda}_{0,k+1} + \boldsymbol{\lambda}_{12,k+1}) = \mathbf{0}, \quad (4.77a)$$

$$\frac{1}{h} \mathbf{M}_2 (\mathbf{v}_{2,k+1} - \mathbf{v}_{2,k}) - g \mathbf{M}_2 \mathbf{e}_2 + \boldsymbol{\lambda}_{12,k+1} = \mathbf{0}. \quad (4.77b)$$

For the rotational part, the derivatives of (4.9) have to be taken for each link, analogously to the 3D-pendulum of Section 4.2.1. Firstly, the directional derivative $\langle D_{\mathbf{F}_{i,k}} L_d(\mathbf{F}_{i,k}), \delta \mathbf{F}_{i,k} \rangle$ is given by applying Proposition (5), which yields

$$\begin{aligned} \left\langle T_e^* \mathbb{L}_{\mathbf{F}_{i,k}} D_{\mathbf{F}_{i,k}} L_{d,R,i}, \widehat{\mathbf{X}}_k \right\rangle &= \frac{d}{d\epsilon} \bigg|_{\epsilon=0} L_d \left(\mathbf{F}_{i,k} \exp(\epsilon \widehat{\mathbf{X}}_k) \right) \\ &= -\frac{1}{h} \text{tr} \left[\mathbf{F}_{i,k} \widehat{\mathbf{X}}_k \mathbf{J}_{d,i} \right] \\ &= \left\langle \frac{1}{h} (\mathbf{J}_{d,i} \mathbf{F}_{i,k} - \mathbf{F}_{i,k}^\top \mathbf{J}_{d,i}), \widehat{\mathbf{X}}_k \right\rangle \\ &= \left\langle \widehat{\boldsymbol{\Pi}}_{i,k}, \widehat{\mathbf{X}}_k \right\rangle, \end{aligned} \quad (4.78)$$

corresponding to the derivative of the single body 3D-pendulum derived in Section 4.2.1. Here,

$$\widehat{\Pi}_{i,k} = \frac{1}{h} (\mathbf{J}_{d,i} \mathbf{F}_{i,k} - \mathbf{F}_{i,k}^\top \mathbf{J}_{d,i}), \quad i \in \{1, 2\}, \quad (4.79)$$

denotes the discrete angular momentum of link i associated with the relative rotation $\mathbf{F}_{i,k}$. It is the analogue of the discrete Legendre momentum (4.42a) used for the 3D-pendulum in Section 4.2.1, now defined separately for each rigid link [Lee08].

Since $SO(2)$ is abelian, it follows that $\text{Ad}_{\mathbf{F}_{i,k}} = \mathbf{I}$ [Lee08]. Therefore, the coadjoint term is equal to (4.78) at time step $k+1$, which gives

$$\text{Ad}_{\mathbf{F}_{i,k+1}}^* (T_e^* \mathbb{L}_{\mathbf{F}_{i,k+1}} D_{\mathbf{F}_{i,k+1}} L_d(\mathbf{F}_{i,k+1})) = \frac{1}{h} (\mathbf{J}_{d,i} \mathbf{F}_{i,k+1} - \mathbf{F}_{i,k+1}^\top \mathbf{J}_{d,i}). \quad (4.80)$$

It remains to derive the directional derivative of the rotations $\mathbf{R}_{i,k+1}$. From (4.73), it is observed that only the constraints $\phi(q_k)$ are dependent on the rotation matrices $\mathbf{R}_{1,k}$ and $\mathbf{R}_{2,k}$. Hence, the variation of the constraint contribution

$$\langle D_{\mathbf{R}_{i,k+1}} (h \boldsymbol{\lambda}_{k+1}^\top \phi_{k+1}), \delta \mathbf{R}_{i,k+1} \rangle = \langle T_e^* \mathbb{L}_{\mathbf{R}_{i,k+1}} D_{\mathbf{R}_{i,k+1}} (h \boldsymbol{\lambda}_{k+1}^\top \phi_{k+1}), \widehat{\boldsymbol{\eta}}_{i,k+1} \rangle,$$

has to be evaluated for each link i , using Proposition (4). For link $i = 1$, the variation of the constraint contribution is given by

$$\begin{aligned} & \langle T_e^* \mathbb{L}_{\mathbf{R}_{1,k+1}} D_{\mathbf{R}_{1,k+1}} (h \boldsymbol{\lambda}_{k+1}^\top \phi_{k+1}), \widehat{\boldsymbol{\eta}}_{1,k+1} \rangle \\ &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} h [\boldsymbol{\lambda}_{0,k+1}^\top \phi_{0,k+1}^\epsilon + \boldsymbol{\lambda}_{12,k+1}^\top \phi_{12,k+1}^\epsilon] \\ &= h [\boldsymbol{\lambda}_{0,k+1}^\top \mathbf{R}_{1,k+1} \widehat{\boldsymbol{\eta}}_{1,k+1} \boldsymbol{\rho}_{1,0} + \boldsymbol{\lambda}_{12,k+1}^\top \mathbf{R}_{1,k+1} \widehat{\boldsymbol{\eta}}_{1,k+1} \boldsymbol{\rho}_{1,12}] \\ &= h [(\boldsymbol{\rho}_{1,0} \times \mathbf{R}_{1,k+1}^\top \boldsymbol{\lambda}_{0,k+1}) + (\boldsymbol{\rho}_{1,12} \times \mathbf{R}_{1,k+1}^\top \boldsymbol{\lambda}_{12,k+1})] \cdot \boldsymbol{\eta}_{1,k+1} \\ &= \left\langle h [(\boldsymbol{\rho}_{1,0} \times \mathbf{R}_{1,k+1}^\top \boldsymbol{\lambda}_{0,k+1}) + (\boldsymbol{\rho}_{1,12} \times \mathbf{R}_{1,k+1}^\top \boldsymbol{\lambda}_{12,k+1})]^\wedge, \widehat{\boldsymbol{\eta}}_{1,k+1} \right\rangle, \end{aligned} \quad (4.81)$$

whose calculation corresponds to the derivative of the 3D-pendulum with respect to the rotation matrix $\mathbf{R}_k$ derived in Section 4.2.1. Here, the planar cross product of $\mathbf{a}, \mathbf{b} \in \mathbb{R}^2$ is defined as $\mathbf{a}^\top \widehat{\boldsymbol{\eta}} \mathbf{b} = (\mathbf{b} \times \mathbf{a}) \cdot \boldsymbol{\eta}$, using the $SO(2)$ hat map (2.7). In contrast, the variation of the constraint contribution for link $i = 2$ only depends on $\phi_{12,k+1}$ and is derived as

$$\begin{aligned} & \langle T_e^* \mathbb{L}_{\mathbf{R}_{2,k+1}} D_{\mathbf{R}_{2,k+1}} (h \boldsymbol{\lambda}_{k+1}^\top \phi_{k+1}), \widehat{\boldsymbol{\eta}}_{2,k+1} \rangle \\ &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} h [\boldsymbol{\lambda}_{12,k+1}^\top \phi_{12,k+1}^\epsilon] \\ &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} h [\boldsymbol{\lambda}_{12,k+1}^\top (-\mathbf{R}_{2,k+1} \exp(\epsilon \widehat{\boldsymbol{\eta}}_{2,k+1}) \boldsymbol{\rho}_{2,12})] \\ &= -h [\boldsymbol{\lambda}_{12,k+1}^\top \mathbf{R}_{2,k+1} \widehat{\boldsymbol{\eta}}_{2,k+1} \boldsymbol{\rho}_{2,12}] \\ &= -h (\boldsymbol{\rho}_{2,12} \times \mathbf{R}_{2,k+1}^\top \boldsymbol{\lambda}_{12,k+1}) \cdot \boldsymbol{\eta}_{2,k+1} \\ &= \left\langle -h (\boldsymbol{\rho}_{2,12} \times \mathbf{R}_{2,k+1}^\top \boldsymbol{\lambda}_{12,k+1})^\wedge, \widehat{\boldsymbol{\eta}}_{2,k+1} \right\rangle. \end{aligned} \quad (4.82)$$

The full rotational contribution is described by substituting (4.78), (4.80), (4.81) and (4.82) for each link $i = 1, 2$ into (4.9). Moreover, (4.77) gives the translational description. Finally, the full discrete Euler-Lagrange equations of the unforced Acrobot in maximal coordinates are expressed on $(\mathbb{R}^2 \times SO(2))^2$ as

Translation:

$$\frac{1}{h} \mathbf{M}_1 (\mathbf{v}_{1,k+1} - \mathbf{v}_{1,k}) - g \mathbf{M}_1 \mathbf{e}_2 - (\boldsymbol{\lambda}_{0,k+1} + \boldsymbol{\lambda}_{12,k+1}) = \mathbf{0}, \quad (4.83a)$$

$$\frac{1}{h} \mathbf{M}_2 (\mathbf{v}_{2,k+1} - \mathbf{v}_{2,k}) - g \mathbf{M}_2 \mathbf{e}_2 + \boldsymbol{\lambda}_{12,k+1} = \mathbf{0}, \quad (4.83b)$$

Rotation:

$$\hat{\boldsymbol{\Pi}}_{1,k} - \hat{\boldsymbol{\Pi}}_{1,k+1} + h \left[\left(\boldsymbol{\rho}_{1,0} \times \mathbf{R}_{1,k+1}^\top \boldsymbol{\lambda}_{0,k+1} \right) + \left(\boldsymbol{\rho}_{1,12} \times \mathbf{R}_{1,k+1}^\top \boldsymbol{\lambda}_{12,k+1} \right) \right]^\wedge = \mathbf{0}, \quad (4.83c)$$

$$\hat{\boldsymbol{\Pi}}_{2,k} - \hat{\boldsymbol{\Pi}}_{2,k+1} - h \left[\left(\boldsymbol{\rho}_{2,12} \times \mathbf{R}_{2,k+1}^\top \boldsymbol{\lambda}_{12,k+1} \right) \right]^\wedge = \mathbf{0}, \quad (4.83d)$$

Constraints:

$$\phi_{0,k+1} = \mathbf{x}_{1,k+1} + \mathbf{R}_{1,k+1} \boldsymbol{\rho}_{1,0} - \mathbf{p}_0 = \mathbf{0}, \quad (4.83e)$$

$$\phi_{12,k+1} = \mathbf{x}_{1,k+1} + \mathbf{R}_{1,k+1} \boldsymbol{\rho}_{1,12} - \mathbf{x}_{2,k+1} - \mathbf{R}_{2,k+1} \boldsymbol{\rho}_{2,12} = \mathbf{0}, \quad (4.83f)$$

Kinematics:

$$\mathbf{R}_{1,k+1} = \mathbf{R}_{1,k} \mathbf{F}_{1,k}, \quad (4.83g)$$

$$\mathbf{R}_{2,k+1} = \mathbf{R}_{2,k} \mathbf{F}_{2,k}. \quad (4.83h)$$

subject to the constraints (4.59).

Controlled Acrobot As in the controlled 3D-pendulum case of Section 4.2.1, the trapezoidal approximation distributes the virtual work equally to both endpoints of the interval (4.45). Therefore, the discrete Legendre force contributions of (4.23) are given by

$$\boldsymbol{\tau}_{d,k}^- = \frac{h}{2} \boldsymbol{\tau}_k, \quad \boldsymbol{\tau}_{d,k}^+ = \frac{h}{2} \boldsymbol{\tau}_{k+1}, \quad (4.84)$$

The Acrobot is actuated by a single motor located at the elbow joint [BM21a]. In contrast to the controlled 3D-pendulum, the input is therefore not an external moment acting on one rigid body, but an internal joint torque acting between the two links. The generalized torque vector is defined as

$$\boldsymbol{\tau}_k = \begin{bmatrix} -u_k \\ u_k \end{bmatrix}, \quad (4.85)$$

where $u_k \in \mathbb{R}$ is the scalar elbow torque that applies opposite moments between the two links.

Since the Acrobot is actuated by a motor at the elbow joint, the control input is a pure rotational torque acting between the two links. Hence, there is no external translational force acting directly on the centers of mass, and the translational force contribution is set to

$$\mathbf{u}_{1,k}^x = \mathbf{u}_{2,k}^x = \mathbf{0}. \quad (4.86)$$

Using the discrete Lagrange-d'Alembert principle (4.21), the generalized torque contribution (4.84) under the definition of torque vector $\boldsymbol{\tau}_k$ (4.85) is added only to the unforced rotational equations (4.83c) and (4.83d). In consequence, the controlled rotational discrete Euler-Lagrange equations are derived as

$$\hat{\mathbf{\Pi}}_{1,k} - \hat{\mathbf{\Pi}}_{1,k+1} + h [(\boldsymbol{\rho}_{1,0} \times \mathbf{R}_{1,k+1}^\top \boldsymbol{\lambda}_{0,k+1}) + (\boldsymbol{\rho}_{1,12} \times \mathbf{R}_{1,k+1}^\top \boldsymbol{\lambda}_{12,k+1}) - u_{k+1}]^\wedge = \mathbf{0}, \quad (4.87a)$$

$$\hat{\mathbf{\Pi}}_{2,k} - \hat{\mathbf{\Pi}}_{2,k+1} + h [u_{k+1} - (\boldsymbol{\rho}_{2,12} \times \mathbf{R}_{2,k+1}^\top \boldsymbol{\lambda}_{12,k+1})]^\wedge = \mathbf{0}. \quad (4.87b)$$

For $u_{k+1} = 0$, these equations reduce to the corresponding unforced rotational equations derived previously.

Chapter 5

Certifiable Trajectory Optimization via Semidefinite Relaxation

This chapter formulates the discrete rigid-body models derived in Chapter 4 as finite-horizon trajectory optimization problems. Section 5.1 introduces the geometric motion planning formulation of [TJVG25], which expresses the Lie group variational integrators and their kinematic constraints by polynomial equalities and inequalities. Section 5.2 gives an overall introduction to semidefinite programming and presents the Moment-SOS hierarchy and the exploitation of correlative and term sparsity. Thereby, a nonconvex polynomial optimization problem is relaxed to a convex semidefinite program. Finally, Sections 5.3 and 5.4 define the optimization formulation for the controlled 3D-pendulum and the Acrobot. The resulting construction follows the sparse semidefinite trajectory optimization framework of [KXS⁺26, KLY25]. For the Acrobot, the sparse SDP relaxation is embedded in a model predictive control loop. Because the prediction-horizon relaxation is not rank one, the extracted predicted trajectory is not certified as globally optimal or feasible. Nevertheless, applying the first extracted control and simulating the controlled LGVI produces a dynamically consistent numerical closed-loop trajectory.

5.1 Discrete Motion Planning of Rigid-Body Systems

A general robot motion planning problem is modeled as a mechanical system consisting of n_b rigid links, indexed by $i \in \mathcal{B} := \{1, \dots, n_b\}$, with N steps starting at $k = 0$. The robot configuration state $\mathbf{Y}_k$ consists of the center-of-mass positions $\mathbf{x}_{i,k}$, rotations $\mathbf{R}_{i,k}$. The relative rotations $\mathbf{F}_{i,k}$ are reconstructed from the kinematic update (4.1). The control input is denoted by $\mathbf{U}_{k+1}$, with $\boldsymbol{\lambda}_{k+1}$ being the Lagrange multipliers of the holonomic joint constraints. Adapted from the discrete kinody-

dynamic motion planning problem of [TJVG25], the general multi-body problem is written as

$$\min_{\{\mathbf{Y}_k\}_{k=0}^N, \{\mathbf{U}_k, \boldsymbol{\lambda}_k\}_{k=1}^N} \Psi(\mathbf{Y}_N) + \sum_{k=0}^{N-1} \psi(\mathbf{Y}_k, \mathbf{U}_{k+1}, \boldsymbol{\lambda}_{k+1}) \quad (5.1a)$$

$$\text{s.t. } \mathcal{D}_i(\mathbf{Y}_{k+1}, \mathbf{Y}_k, \mathbf{U}_{k+1}, \boldsymbol{\lambda}_{k+1}) = 0, \quad i \in \mathcal{B}, \quad (5.1b)$$

$$\mathbf{Y}_k \in \mathcal{Y}_i, \quad (5.1c)$$

$$\mathcal{H}(\mathbf{Y}_{k+1}, \mathbf{U}_{k+1}, \boldsymbol{\lambda}_{k+1}) \geq 0, \quad (5.1d)$$

$$\mathbf{U}_{\min} \leq \mathbf{U}_{k+1} \leq \mathbf{U}_{\max}, \quad (5.1e)$$

$$\mathbf{Y}_0 = \mathbf{Y}_{\text{init}}, \quad (5.1f)$$

$$k = 0, \dots, N-1.$$

Here, $\Psi(\cdot)$ denotes the terminal costs and $\psi(\cdot)$ the running costs. The equality constraint set $\mathcal{D}_i = 0$ represents the forced discrete Euler–Lagrange equations of link i derived in Chapter 4, including the set of control inputs $\mathbf{U}_{k+1}$ and Lagrange multipliers $\boldsymbol{\lambda}_{k+1}$. Additionally, for multi-body systems, the set $\mathcal{D}_i$ contains the holonomic joint constraints (4.50). The inequalities $\mathcal{H} \geq 0$ describe rotation-step limits, geometric constraints such as collision avoidance, and bounds introduced to obtain a compact feasible set [KXS⁺26]. Control limits are set by $\mathbf{U}_{\min} \leq \mathbf{U}_k \leq \mathbf{U}_{\max}$ in (5.1).

The set $\mathcal{Y}_i$ enforces the matrix Lie-group structure of the set of states $\mathbf{Y}_k$. Thereby, the multiplicative kinematic update

$$\mathbf{R}_{i,k+1} = \mathbf{R}_{i,k} \mathbf{F}_{i,k} \quad (5.2)$$

preserves the corresponding Lie group structure for each link i .

For rigid-body motion on $SO(3)$, the rotation of link i is given by

$$\mathbf{R}_{i,k} = [\mathbf{r}_{i,k}^1, \mathbf{r}_{i,k}^2, \mathbf{r}_{i,k}^3] \in \mathbb{R}^{3 \times 3}.$$

Following [TJVG25], $\mathbf{R}_{i,k} \in SO(3)$ is represented by the 15 quadratic polynomial equalities

$$\begin{aligned} \|\mathbf{r}_{i,k}^1\|_2^2 - 1 &= \|\mathbf{r}_{i,k}^2\|_2^2 - 1 = \|\mathbf{r}_{i,k}^3\|_2^2 - 1 = 0, \\ (\mathbf{r}_{i,k}^1)^\top \mathbf{r}_{i,k}^2 &= (\mathbf{r}_{i,k}^2)^\top \mathbf{r}_{i,k}^3 = (\mathbf{r}_{i,k}^1)^\top \mathbf{r}_{i,k}^3 = 0, \\ \mathbf{r}_{i,k}^1 \times \mathbf{r}_{i,k}^2 - \mathbf{r}_{i,k}^3 &= \mathbf{r}_{i,k}^2 \times \mathbf{r}_{i,k}^3 - \mathbf{r}_{i,k}^1 = \mathbf{r}_{i,k}^3 \times \mathbf{r}_{i,k}^1 - \mathbf{r}_{i,k}^2 = \mathbf{0}_{3 \times 1}. \end{aligned} \quad (5.3)$$

The same constraints are imposed on $\mathbf{F}_{i,k}$ by replacing $\mathbf{r}_{i,k}^1, \mathbf{r}_{i,k}^2, \mathbf{r}_{i,k}^3$ with its columns $\mathbf{f}_{i,k}^1, \mathbf{f}_{i,k}^2, \mathbf{f}_{i,k}^3$.

For rigid-body motion on $SO(2)$, the standard polynomial parametrization is

$$\mathbf{R}_{i,k} = \begin{bmatrix} c_{i,k} & -s_{i,k} \\ s_{i,k} & c_{i,k} \end{bmatrix}, \quad \mathbf{F}_{i,k} = \begin{bmatrix} a_{i,k} & -b_{i,k} \\ b_{i,k} & a_{i,k} \end{bmatrix}, \quad (5.4)$$

where $c_{i,k} = \cos \theta_{i,k}$, $s_{i,k} = \sin \theta_{i,k}$, $a_{i,k} = \cos \Delta \theta_{i,k}$, and $b_{i,k} = \sin \Delta \theta_{i,k}$ [Hal00]. Applying (5.6) to the kinematic update for each link i in (5.2), its planar version is given by

$$c_{i,k+1} - c_{i,k}a_{i,k} + s_{i,k}b_{i,k} = 0, \quad s_{i,k+1} - s_{i,k}a_{i,k} - c_{i,k}b_{i,k} = 0, \quad (5.5)$$

for planar motion. In contrast to (5.3), the planar rotation matrices $\mathbf{R}_{i,k}, \mathbf{F}_{i,k} \in \text{SO}(2)$ are imposed by the two quadratic polynomial equalities

$$c_{i,k}^2 + s_{i,k}^2 = 1, \quad a_{i,k}^2 + b_{i,k}^2 = 1, \quad (5.6)$$

[TJVG25].

The matrix in (5.6) describes a rotation in the xy -plane and therefore about the z -axis. In general, rotations about a fixed axis are

$$\mathbf{R}_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix}, \quad \mathbf{R}_y = \begin{bmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{bmatrix}, \quad \mathbf{R}_z = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (5.7)$$

with $c = \cos \theta$ and $s = \sin \theta$ [TJVG25]. Consequently, planar motion in the yz -, xz -, or xy -plane corresponds to a rotation about the x -, y -, or z -axis.

Finally, from a given initial configuration $\mathbf{Y}_k$, the target configuration $\mathbf{Y}_N$ is reached by minimizing the terminal cost $\Psi(\cdot)$.

5.2 Semidefinite Optimization and Relaxation

The discrete motion planning problem (5.1) is considered as a polynomial optimization problem (POP) with polynomial objective, equality, and inequality constraints [TJVG25]. Thereby, $\mathbf{z} \in \mathbb{R}^{n_z}$ is the complete set of states $\mathbf{Y}_k$, control inputs $\mathbf{U}_k$, and Lagrange multipliers $\boldsymbol{\lambda}_{k+1}$ over the planning horizon. Equivalently, the POP can be written in the general form

$$p^* = \min_{\mathbf{z} \in \mathbb{R}^{n_z}} p(\mathbf{z}) \quad (5.8a)$$

$$\text{s.t.} \quad h_l(\mathbf{z}) = 0, \quad l = 1, \dots, m_h, \quad (5.8b)$$

$$g_j(\mathbf{z}) \geq 0, \quad j = 1, \dots, m_g, \quad (5.8c)$$

with the basic closed semialgebraic feasible set

$$\mathcal{K} := \{\mathbf{z} \in \mathbb{R}^{n_z} \mid h_l(\mathbf{z}) = 0, \quad l = 1, \dots, m_h, \quad g_j(\mathbf{z}) \geq 0, \quad j = 1, \dots, m_g\}. \quad (5.9)$$

Here, p , h_l , and g_j are real multivariate polynomials. In general, a multivariate polynomial is defined as

$$p(\mathbf{z}) = \sum_{\boldsymbol{\alpha}} p_{\boldsymbol{\alpha}} \mathbf{z}^{\boldsymbol{\alpha}}, \quad \mathbf{z}^{\boldsymbol{\alpha}} := \prod_{i=1}^{n_z} z_i^{\alpha_i}, \quad (5.10)$$

where α_i are nonnegative integers and p_α are nonzero [Yan24]. Although (5.8) is generally nonconvex, the Moment-SOS hierarchy constructs a sequence of convex semidefinite relaxations whose optimal values are certified lower bounds on the optimal solution p^* [Yan24].

5.2.1 Semidefinite Relaxation and the Moment-SOS Hierarchy

Semidefinite programming (SDP) Let $\mathbb{S}^n$ denote the vector space of symmetric $n \times n$ matrices. A matrix $\mathbf{X} \in \mathbb{S}^n$ is positive semidefinite, written as $\mathbf{X} \succeq 0$, if $\mathbf{v}^\top \mathbf{X} \mathbf{v} \geq 0$ for all $\mathbf{v} \in \mathbb{R}^n$. A semidefinite program (SDP) has the standard primal form

$$\begin{aligned} p_{\text{SDP}}^* &= \min_{\mathbf{X} \in \mathbb{S}^n} \langle \mathbf{C}, \mathbf{X} \rangle \\ \text{s.t. } &\mathcal{A}(\mathbf{X}) = \mathbf{b}, \\ &\mathbf{X} \succeq 0. \end{aligned} \quad (5.11)$$

where $\mathbf{C} \in \mathbb{S}^n$ and $\mathbf{b} \in \mathbb{R}^m$ [Yan24]. The linear map $\mathcal{A} : \mathbb{S}^n \rightarrow \mathbb{R}^m$ is defined as

$$\mathcal{A}(\mathbf{X}) := \begin{bmatrix} \langle \mathbf{A}_1, \mathbf{X} \rangle \\ \vdots \\ \langle \mathbf{A}_i, \mathbf{X} \rangle \\ \vdots \\ \langle \mathbf{A}_m, \mathbf{X} \rangle \end{bmatrix}. \quad (5.12)$$

Here, $\langle \mathbf{C}, \mathbf{X} \rangle = \text{tr}(\mathbf{C}\mathbf{X})$. In contrast to (5.8), the objective and the constraints in (5.11) are convex. In the following, the Moment-SOS relaxation approximates the nonconvex polynomial problem by a convex SDP [Yan24].

Sum-of-squares (SOS) relaxation [Yan24] defines a monomial of the optimization variables $\mathbf{z} = (z_1, \dots, z_{n_z})$ as

$$\mathbf{z}^\alpha := \prod_{i=1}^{n_z} z_i^{\alpha_i}, \quad (5.13)$$

with $\alpha = (\alpha_1, \dots, \alpha_{n_z}) \in \mathbb{N}_0^{n_z}$. Thereby, $|\alpha| := \sum_{i=1}^{n_z} \alpha_i$ is the total degree of the monomial $\mathbf{z}^\alpha$. Moreover, the standard basis of monomials with degree up to $|\alpha| \leq d$ is given by

$$[\mathbf{z}]_d := \begin{bmatrix} 1 & z_1 & \cdots & z_{n_z} & z_1^2 & z_1 z_2 & \cdots & z_{n_z}^d \end{bmatrix}^\top = [\mathbf{z}^\alpha]_{\alpha \in \mathbb{N}_0^{n_z}},$$

The vector has dimension

$$s(n_z, d) = \binom{n_z + d}{d}. \quad (5.14)$$

[Yan24] defines a polynomial p of degree at most $2d$ as a sum of squares (SOS) if it can be written as $p(\mathbf{z}) = \sum_i q_i(\mathbf{z})^2$. Here, each q_i has degree at most d and the

polynomial p is globally nonnegative. Equivalently, p is SOS if the Gram-matrix $\mathbf{Q} \in \mathbb{S}^{s(n_z, d)}$ satisfies

$$p(\mathbf{z}) = [\mathbf{z}]_d^\top \mathbf{Q} [\mathbf{z}]_d, \quad \mathbf{Q} \succeq 0. \quad (5.15)$$

Finding an SOS representation can be formulated as an SDP feasibility problem. For the POP (5.8), any value γ satisfying $p(\mathbf{z}) - \gamma \geq 0$ on the feasible set $\mathcal{K}$ is a lower bound on p^* [Yan24]. Since verifying this nonnegativity directly is difficult, the SOS relaxation searches for the certificate

$$p(\mathbf{z}) - \gamma = \sigma_0(\mathbf{z}) + \sum_{j=1}^{m_g} \sigma_j(\mathbf{z}) g_j(\mathbf{z}) + \sum_{l=1}^{m_h} \chi_l(\mathbf{z}) h_l(\mathbf{z}), \quad (5.16)$$

where σ_0 and σ_j are SOS polynomials and χ_l are arbitrary polynomials [PV99]. On $\mathcal{K}$, the inequalities satisfy $g_j \geq 0$ and the equalities satisfy $h_l = 0$. Therefore, the right-hand side of (5.16) is nonnegative and proves $p(\mathbf{z}) \geq \gamma$.

By restricting the degree of $\sigma_0, \sigma_j g_j, \chi_l h_l$ in (5.16) to at most 2κ gives the finite-dimensional order- κ SOS relaxation

$$\gamma_\kappa^* = \max_{\gamma, \{\sigma_j\}, \{\chi_l\}} \gamma \quad (5.17a)$$

$$\text{s.t. } p - \gamma = \sigma_0 + \sum_{j=1}^{m_g} \sigma_j g_j + \sum_{l=1}^{m_h} \chi_l h_l, \quad (5.17b)$$

$$\deg(\sigma_0), \deg(\sigma_j g_j), \deg(\chi_l h_l) \leq 2\kappa, \quad (5.17c)$$

$$\sigma_j \in \Sigma[\mathbf{z}], \quad \chi_l \in \mathbb{R}[\mathbf{z}]. \quad (5.17d)$$

Here, $\Sigma[\mathbf{z}]$ denotes the set of SOS polynomials in $\mathbf{z}$, and $\mathbb{R}[\mathbf{z}]$ denotes the set of real polynomials in $\mathbf{z}$. κ is called the relaxation order. Consequently, every feasible certificate provides the lower bound $\gamma_\kappa^* \leq p^*$ [Yan24].

Moment relaxation The moment relaxation is the SDP dual of the SOS relaxation. [Yan24] introduces a scalar variable y_α , called a moment, for each monomial $\mathbf{z}^\alpha$. Additionally, the linear functional $\mathcal{L}_y : \mathbb{R}[\mathbf{z}] \rightarrow \mathbb{R}$ is defined as

$$q(\mathbf{z}) = \sum_{\alpha} q_{\alpha} \mathbf{z}^{\alpha} \rightarrow \mathcal{L}_y(q) = \sum_{\alpha} q_{\alpha} y_{\alpha}, \quad (5.18)$$

which replaces the monomials $\mathbf{z}^\alpha$ in $q(\mathbf{z})$ to obtain a numerical value [TJVG25]. The moment matrix $\mathbf{M}_\kappa(\mathbf{y})$ is defined as

$$\mathbf{M}_\kappa(\mathbf{y}) := \mathcal{L}_y([\mathbf{z}]_\kappa [\mathbf{z}]_\kappa^\top). \quad (5.19)$$

The polynomial constraints are included similarly. For an inequality $g_j(\mathbf{z}) \geq 0$, let $d_j := \lceil \deg(g_j)/2 \rceil$. [TJVG25] defines the corresponding localizing matrix as

$$\mathbf{M}_{\kappa-d_j}(g_j \mathbf{y}) := \mathcal{L}_y(g_j(\mathbf{z}) [\mathbf{z}]_{\kappa-d_j} [\mathbf{z}]_{\kappa-d_j}^\top). \quad (5.20)$$

Requiring this matrix to be positive semidefinite represents the inequality $g_j \geq 0$. Similarly, for an equality $h_l(\mathbf{z}) = 0$, with $e_l := \lceil \deg(h_l)/2 \rceil$, the corresponding localization matrix is

$$\mathbf{M}_{\kappa-e_l}(h_l \mathbf{y}) := \mathcal{L}_{\mathbf{y}}(h_l(\mathbf{z}))[\mathbf{z}]_{\kappa-e_l}[\mathbf{z}]_{\kappa-e_l}^\top. \quad (5.21)$$

For relaxation order κ , a solution is approximated by sequences y_α with finite order $|\alpha| \leq 2\kappa$. This gives the order- κ moment relaxation

$$\beta_\kappa^* = \min_{\mathbf{y}} \mathcal{L}_{\mathbf{y}}(p) \quad (5.22a)$$

$$\text{s.t. } y_0 = 1, \quad (5.22b)$$

$$\mathbf{M}_\kappa(\mathbf{y}) \succeq 0, \quad (5.22c)$$

$$\mathbf{M}_{\kappa-d_j}(g_j \mathbf{y}) \succeq 0, \quad j = 1, \dots, m_g, \quad (5.22d)$$

$$\mathbf{M}_{\kappa-e_l}(h_l \mathbf{y}) = \mathbf{0} \quad l = 1, \dots, m_h. \quad (5.22e)$$

with $y_0 = 1$ being the monomial $\mathbf{z}^0 = 1$ [Yan24].

The moment and SOS relaxations form a primal-dual SDP pair. Therefore, the Moment-SOS hierarchy provides the sequence of lower bounds

$$\gamma_\kappa^* \leq \beta_\kappa^* \leq p^*. \quad (5.23)$$

If the SDP pair has no duality gap, then $\gamma_\kappa^* = \beta_\kappa^*$ [Las01].

Convergence Solving the SOS and moment relaxations for increasing orders κ defines Lasserre's hierarchy. Higher orders give tighter lower bounds but larger SDPs. For a nonempty feasible set, both bound sequences increase monotonically and converge to the global optimum p^* , if the quadratic module is Archimedean, for example through $R^2 - \|\mathbf{z}\|_2^2 \geq 0$. This convergence is asymptotic

$$\gamma_\kappa^* \leq \gamma_{\kappa+1}^* \leq p^*, \quad \beta_\kappa^* \leq \beta_{\kappa+1}^* \leq p^*, \quad \lim_{\kappa \rightarrow \infty} \gamma_\kappa^* = \lim_{\kappa \rightarrow \infty} \beta_\kappa^* = p^*, \quad (5.24)$$

described in [Las01].

In addition, under additional suitable optimality conditions, convergence is finite given by

$$\gamma_{\kappa^*}^* = \beta_{\kappa^*}^* = p^* \quad (5.25)$$

for some finite relaxation order κ^* . This property is referred to as finite convergence [Nie14, Las01].

Solution extraction Let $\mathbf{y}^*$ denote an optimal moment sequence of the order- κ moment relaxation. A simple finite-order certificate is obtained from the optimal moment matrix $\mathbf{M}_\kappa(\mathbf{y}^*)$ [Yan24]. If it has rank one $\text{rank}(\mathbf{M}_\kappa(\mathbf{y}^*)) = 1$, then

$$\mathbf{M}_\kappa(\mathbf{y}^*) = [\mathbf{z}^*]_\kappa [\mathbf{z}^*]_\kappa^\top. \quad (5.26)$$

Therefore, the relaxation is tight, and a globally optimal decision vector is recovered from the first-order moments as

$$z_r^* = y_{\mathbf{e}_r}^*, \quad r = 1, \dots, n_z, \quad (5.27)$$

where $\mathbf{e}_r$ is the r -th unit multi-index [Yan24]. If the moment matrix has higher rank, the rank-one condition does not certify that the relaxation is tight [TJVG25]. Moreover, any feasible trajectory $\hat{\mathbf{z}}$ provides an upper bound, such that

$$\gamma_\kappa^* \leq p^* \leq p(\hat{\mathbf{z}}). \quad (5.28)$$

5.2.2 Sparse Relaxations

The dense moment matrix in (5.19) has $\binom{n_z + \kappa}{\kappa}$ rows and columns. Since n_z increases with the number of links and the planning horizon N , a dense relaxation quickly becomes computationally intractable. However, each cost or constraint in a trajectory optimization problem usually depends only on a small subset of the complete decision vector. Sparse Moment–SOS relaxations exploit this structure at the variable level by correlative sparsity (CS) and at the monomial level by term sparsity (TS) [KLY25, KXS⁺26, TJVG25].

Correlative sparsity and chordal extension The correlative sparsity pattern (CSP) is represented by an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Its vertices represent the scalar variables $z_1, \dots, z_{n_z}$, and two vertices are connected if their variables occur together in one monomial of the objective or in one constraint polynomial. A clique $\mathcal{C}$ is a subset of vertices $\mathcal{C} \subset \mathcal{V}$ in which every pair of vertices is connected by an edge. Thus, the variables of each constraint are contained in one clique. Thereby, a clique is considered maximal if it is not properly contained within any other clique inside the graph $\mathcal{G}$ [KLY25].

In addition, a graph is chordal if every cycle with at least four vertices has an edge between two non-adjacent vertices in the cycle. However, if the graph is not chordal, edges are added to obtain a chordal extension. Its maximal cliques then define the variable groups of the sparse relaxation. Finding an extension with the minimum number of edges to $\mathcal{G}$ is computationally difficult. Therefore, heuristic extensions such as minimum degree (MD) and minimum fill (MF) are commonly used [KLY25]. Larger cliques generally produce a tighter but more expensive relaxation, such that the extension determines a trade-off between relaxation strength and SDP block size.

Figure 5.1 illustrates a chordal extension. The original graph in Figure 5.1(a) is non-chordal. The MD extension adds the fill edges (B, D) and (B, F) in (b), and the resulting maximal cliques are illustrated in (c) [KLY25].

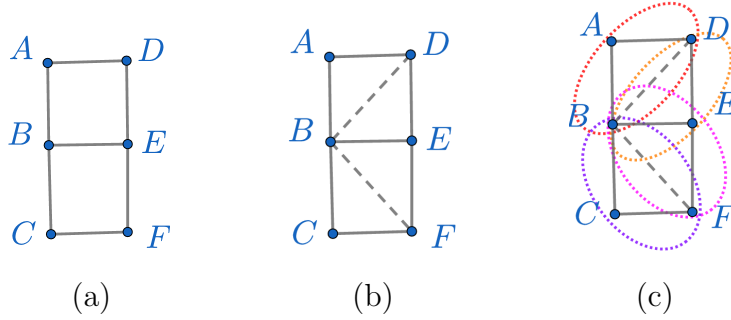


Figure 5.1: An example of a chordal extension and its maximal cliques. (a) non-chordal. (b) MD chordal extension. (c) maximal cliques. Adapted from [KLY25].

Chain-like sparsity of trajectory optimization For the motion planning formulation of Section 5.1, the vectorized decision variables can be ordered as

$$\mathbf{z} = [\mathbf{Y}_0 \quad \mathbf{U}_1 \quad \boldsymbol{\lambda}_1 \quad \mathbf{Y}_1 \quad \cdots \quad \mathbf{U}_N \quad \boldsymbol{\lambda}_N \quad \mathbf{Y}_N]^\top. \quad (5.29)$$

[KXS⁺26] defines the index set $\mathcal{I}_k$, which selects the scalar variables of the k -th trajectory clique,

$$\mathbf{z}(\mathcal{I}_k) = [\mathbf{Y}_{k-1} \quad \mathbf{U}_k \quad \boldsymbol{\lambda}_k \quad \mathbf{Y}_k]^\top, \quad k = 1, \dots, N. \quad (5.30)$$

For multibody rigid-body systems, each link constraint and stage cost for transition k is contained inside $\mathcal{I}_k$. Additionally, the temporal cliques satisfy

$$\bigcup_{k=1}^N \mathcal{I}_k = \{1, \dots, n_z\}, \quad \left(\bigcup_{i=1}^{k-1} \mathcal{I}_i \right) \cap \mathcal{I}_k = \mathcal{I}_{k-1} \cap \mathcal{I}_k, \quad k = 2, \dots, N. \quad (5.31)$$

Their adjacent overlap contains the shared boundary state $\mathbf{Y}_{k-1}$. Therefore, the property (5.31) results from the Markov structure of the discretized dynamics [TJVG25].

Figure 5.2 illustrates the chain-like sparsity pattern for the clique (5.30) and the resulting multi-block SDP, forming a chain-like graph. Thereby, $\psi(\mathbf{Y}_{k-1}, \mathbf{U}_k, \boldsymbol{\lambda}_k)$ denotes the stage cost, $\mathcal{D}(\mathbf{Y}_k, \mathbf{Y}_{k-1}, \mathbf{U}_k, \boldsymbol{\lambda}_k) = 0$ the polynomial equality constraints, and $H(\mathbf{Y}_k, \mathbf{U}_k, \boldsymbol{\lambda}_k) \geq 0$ the polynomial inequality constraints.

Multibody rigid-body-specific sparsity Many robotic systems have local dependencies. [KLY25] gives the example of a one-dimensional two-link chain in discrete time with the three states $x_{1,k}, x_{2,k}, x_{3,k}$ and the corresponding control inputs $u_{1,k}, u_{2,k}, u_{3,k}$,

$$x_{1,k+1} = x_{1,k} + u_{1,k}, \quad (5.32)$$

$$x_{2,k+1} = x_{2,k} + u_{2,k}, \quad (5.33)$$

$$x_{3,k+1} = x_{3,k} + u_{3,k}, \quad (5.34)$$

$$(x_{1,k} - x_{2,k})^2 = r^2, \quad (5.35)$$

$$(x_{2,k} - x_{3,k})^2 = r^2. \quad (5.36)$$

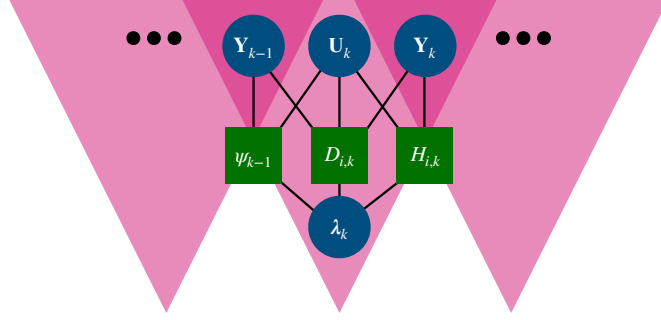


Figure 5.2: Illustration of a chain-like sparsity pattern. Adapted from [KXS⁺26].

Using the chain-like sparsity condition (5.31) therefore gives the nine-variable clique vector

$$\mathbf{z}(\mathcal{I}_k^{\text{full}}) = [x_{1,k} \ x_{2,k} \ x_{3,k} \ x_{1,k+1} \ x_{2,k+1} \ x_{3,k+1} \ u_{1,k} \ u_{2,k} \ u_{3,k}]^\top. \quad (5.37)$$

[KLY25] elaborates that each input affects only its corresponding state and the geometric constraints couple only neighboring states. Therefore, this clique vector can be decomposed further into

$$\mathbf{z}(\mathcal{I}_{\text{dyn},i,k}) = [x_{i,k} \ x_{i,k+1} \ u_{i,k}]^\top, \quad i = 1, 2, 3, \quad (5.38)$$

$$\mathbf{z}(\mathcal{I}_{\text{geom},i,m}) = [x_{i,m} \ x_{i+1,m}]^\top, \quad i = 1, 2, \quad m \in \{k, k+1\}. \quad (5.39)$$

In consequence, the original clique of size nine is replaced by three cliques of size three and four cliques of size two.

Sparse Moment–SOS relaxation After defining the N_{cl} variable cliques $\mathcal{I}_1, \dots, \mathcal{I}_{N_{\text{cl}}}$, $p(\mathbf{z}) = \sum_{i=1}^{N_{\text{cl}}} p_i(\mathbf{z}(\mathcal{I}_i))$ and every constraint polynomial is assigned to a clique containing all of its variables, the κ -order moment relaxation problem (5.22) can be solved. For the sparse Moment-SOS relaxation, the moment and localizing conditions are imposed separately on the local monomial bases [KLY25]. Consequently, instead of one matrix of dimension $\binom{n_z+\kappa}{\kappa}$, the sparse relaxation contains blocks of dimension $\binom{|\mathcal{I}_i|+\kappa}{\kappa}$. Using the chain-like cliques (5.30), its size remains independent of N if the per-stage model size is fixed. Only the number of blocks grows linearly with N [KXS⁺26].

Figure 5.3 illustrates an example from [KLY25]. A reduction for a POP with two variables $\mathbf{x} = (x_1, x_2)$ and relaxation order $\kappa = 2$. The dense moment matrix has dimension $\binom{2+2}{2} = 6$. If the variables are separated into the cliques $\{x_1\}$ and $\{x_2\}$, the sparse relaxation instead contains two moment matrices of dimension $\binom{1+2}{2} = 3$. Consequently, one 6×6 positive-semidefinite constraint is replaced by two 3×3 positive-semidefinite constraints [KLY25].

At a fixed order κ , the sparse lower-bound may be further away from the optimal solution p^* than the dense bound. Still, if all optimal clique moment matrices have

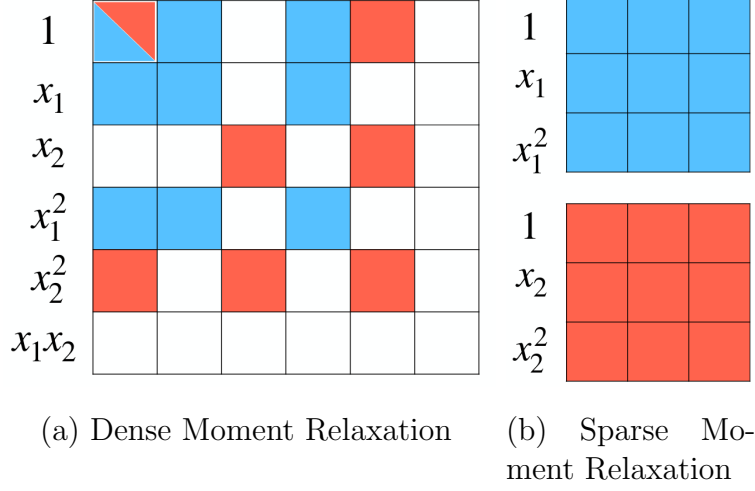


Figure 5.3: Comparison of dense and sparse moment matrices for a two-variable POP at relaxation order $\kappa = 2$. Reproduced from [KLY25].

rank one, a globally optimal trajectory $\mathbf{z}^*$ can be extracted. However, rank one of only the first clique does not certify the complete horizon. Following [TJVG25], proximity to rank one is evaluated by

$$\delta_k := \frac{\nu_{2,k}}{\nu_{1,k}}, \quad \delta_{\max} := \max_{k=1,\dots,N_{\text{cl}}} \delta_k. \quad (5.40)$$

Here, $\nu_{1,k} \geq \nu_{2,k} \geq \dots \geq 0$ are the eigenvalues of $\mathbf{M}_{\kappa,k}$. $\delta_{\max}$ close to zero indicates an approximately rank-one solution. Nevertheless, it remains a numerical tightness diagnostic and is not an exact certificate.

Sparse Polynomial Optimization Toolbox (SPOT) The sparse Moment–SOS relaxations in this thesis are constructed using the Sparse Polynomial Optimization Toolbox (SPOT) developed by [KLY25]. Given a POP, SPOT automatically constructs its correlative sparsity graph, applies a chordal extension, identifies the maximal variable cliques, groups the polynomial constraints, and generates the corresponding sparse moment or SOS relaxation. For both CS and TS, SPOT provides maximal (MAX), minimum-degree (MD), and minimum-fill (MF) chordal-extension options, as well as user-defined (SELF) cliques. TS further decomposes the monomial bases within the variable cliques, but is not considered in this thesis, because of limited computational capabilities. In this thesis, the controlled 3D-pendulum of Section 4.2.1 uses automatic CS with the MD extension, whereas the Acrobot of Section 4.3.1 uses problem-specific SELF cliques. Their exact clique constructions and relaxation orders are specified in the following sections.

5.3 Discrete Motion Planning for the 3D-Pendulum

In order to optimize the trajectory of the 3D pendulum, the discrete motion planning problem (5.1) has to be formulated as a polynomial optimization problem (POP), using the derived discrete Lie group dynamics of the 3D pendulum in Section 4.2.1. The optimization variables are the entries of $\mathbf{R}_0, \dots, \mathbf{R}_N$, $\mathbf{F}_0, \dots, \mathbf{F}_{N-1}$, and the control parameters $\mathbf{u}_{p,1}, \dots, \mathbf{u}_{p,N-1}$, given [Lee08] definition of the control moment

$$\mathbf{u}_k = \mathbf{R}_k^\top \mathbf{e}_3 \times \mathbf{u}_{p,k}. \quad (5.41)$$

For improved numerical conditioning, the variables have to be scaled to the range $[-1, 1]$. Therefore, the control input is scaled by the scalar control limit as

$$\mathbf{u}_{p,k} = u_{\max} \bar{\mathbf{u}}_{p,k}, \quad u_{\max} > 0, \quad (5.42)$$

so that the normalized Euclidean control bound becomes

$$1 - \bar{\mathbf{u}}_{p,k}^\top \bar{\mathbf{u}}_{p,k} \geq 0. \quad (5.43)$$

In contrast, the entries of $\mathbf{R}_k$ and $\mathbf{F}_k$ are already bounded by the Lie-group constraints (5.3).

[KXS⁺26] defines the cost function as a weighted sum of the terminal and stage costs

$$J_{\text{LQR}} = P_f (\mathbf{x}_N - \mathbf{x}_f)^\top \mathbf{Q}_x (\mathbf{x}_N - \mathbf{x}_f) + \sum_{k=0}^{N-1} \left\{ (\mathbf{x}_k - \mathbf{x}_f)^\top \mathbf{Q}_x (\mathbf{x}_k - \mathbf{x}_f) + \mathbf{u}_k^\top \mathbf{Q}_u \mathbf{u}_k \right\}, \quad (5.44)$$

where $\mathbf{Q}_x$ and $\mathbf{Q}_u$ are the identity matrices after rescaling of the variables. In addition to (5.44), this thesis uses individual weights w_R , w_F for the terminal costs and α_R , α_F , w_u for the stage costs.

In the next step, the constraints given by (5.1) have to be identified for the 3D-pendulum. The equality constraints $\mathcal{D}$ are given by the discrete forced dynamics (4.46a) and the kinematics update (4.46b). Moreover, the Lie-group constraints $\mathcal{Y}$ defined in Section 5.1 by (5.3) have to be included for $\mathbf{R}_k$ and $\mathbf{F}_k$. Finally, by using the inequality relation $\mathcal{H}$ given in [GQ20],

$$\text{tr}(\mathbf{F}_k) = 1 + 2 \cos(\Delta\theta_k) \quad (5.45)$$

the relative rotation during one step $\mathbf{F}_k$ is bounded by the angle $0 \leq \Delta\theta_{\max} \leq \pi$, and the control input by (5.43). In summary, the full controlled 3D-pendulum POP

is defined as

$$p_P^* = \min_{\substack{\{\mathbf{R}_k\}_{k=0}^N, \{\mathbf{F}_k\}_{k=0}^{N-1} \\ \{\bar{\mathbf{u}}_{p,k}\}_{k=1}^{N-1}}} w_R \|\mathbf{R}_N - \mathbf{R}_N^{\text{des}}\|_F^2 + w_F \|\mathbf{F}_{N-1} - \mathbf{F}^{\text{des}}\|_F^2 \\ + \alpha_R \sum_{k=0}^{N-1} \|\mathbf{R}_k - \mathbf{R}_N^{\text{des}}\|_F^2 + \alpha_F \sum_{k=0}^{N-2} \|\mathbf{F}_k - \mathbf{F}^{\text{des}}\|_F^2 \\ + \frac{1}{w_u} \sum_{k=1}^{N-1} \|\bar{\mathbf{u}}_{p,k}\|_2^2 \quad (5.46a)$$

$$\text{s.t. } \frac{1}{h} (\mathbf{F}_{k+1} \mathbf{J}_d - \mathbf{J}_d \mathbf{F}_{k+1}^\top - \mathbf{J}_d \mathbf{F}_k + \mathbf{F}_k^\top \mathbf{J}_d) = h (\boldsymbol{\tau}_{g,k+1} + \mathbf{u}_{k+1})^\wedge, \quad (5.46b)$$

$$\mathbf{R}_{k+1} = \mathbf{R}_k \mathbf{F}_k, \quad (5.46c)$$

$$\begin{aligned} \|\mathbf{r}_{1,k}\|_2^2 - 1 &= \|\mathbf{r}_{2,k}\|_2^2 - 1 = \|\mathbf{r}_{3,k}\|_2^2 - 1 = 0, \\ \mathbf{r}_{1,k}^\top \mathbf{r}_{2,k} &= \mathbf{r}_{2,k}^\top \mathbf{r}_{3,k} = \mathbf{r}_{1,k}^\top \mathbf{r}_{3,k} = 0, \\ \mathbf{r}_{1,k} \times \mathbf{r}_{2,k} - \mathbf{r}_{3,k} &= \mathbf{r}_{2,k} \times \mathbf{r}_{3,k} - \mathbf{r}_{1,k} = \mathbf{r}_{3,k} \times \mathbf{r}_{1,k} - \mathbf{r}_{2,k} = \mathbf{0}_{3 \times 1}, \end{aligned} \quad (5.46d)$$

$$\begin{aligned} \|\mathbf{f}_{1,k}\|_2^2 - 1 &= \|\mathbf{f}_{2,k}\|_2^2 - 1 = \|\mathbf{f}_{3,k}\|_2^2 - 1 = 0, \\ \mathbf{f}_{1,k}^\top \mathbf{f}_{2,k} &= \mathbf{f}_{2,k}^\top \mathbf{f}_{3,k} = \mathbf{f}_{1,k}^\top \mathbf{f}_{3,k} = 0, \\ \mathbf{f}_{1,k} \times \mathbf{f}_{2,k} - \mathbf{f}_{3,k} &= \mathbf{f}_{2,k} \times \mathbf{f}_{3,k} - \mathbf{f}_{1,k} = \mathbf{f}_{3,k} \times \mathbf{f}_{1,k} - \mathbf{f}_{2,k} = \mathbf{0}_{3 \times 1}, \end{aligned} \quad (5.46e)$$

$$\frac{\text{tr}(\mathbf{F}_k) - 1}{2} \geq \cos(\Delta\theta_{\max}) \quad (5.46f)$$

$$1 - \bar{\mathbf{u}}_{p,k}^\top \bar{\mathbf{u}}_{p,k} \geq 0 \quad (5.46g)$$

$$\mathbf{R}_0 = \mathbf{R}^{\text{init}}, \quad \mathbf{F}_0 = \mathbf{F}^{\text{init}} \quad (5.46h)$$

with the initial conditions $\mathbf{R}^{\text{init}}$ and $\mathbf{F}^{\text{init}}$. Here, (5.46b) holds for $k = 0, \dots, N-2$, (5.46c) and (5.46f) hold for $k = 0, \dots, N-1$, (5.46d) holds for $k = 0, \dots, N$, (5.46e) holds for $k = 0, \dots, N-1$, and (5.46g) holds for $k = 1, \dots, N-1$. Moreover, h is the time step, $\mathbf{J}_d$ is the nonstandard inertia matrix (3.22), and $\boldsymbol{\tau}_{g,k}$ denotes the gravity moment in the body-fixed frame (4.38). The definition of $\mathbf{r}_{i,k}$ and $\mathbf{f}_{i,k}$ is given in (5.3). Finally, (5.46) gives the full POP of the 3d-pendulum with polynomials until degree 2.

In order to construct the sparse moment relaxation, the chain-like sparsity structure of the 3D-pendulum POP has to be identified. The local decision vector associated with the pendulum clique index set $\mathcal{I}_k^P$ is ordered as

$$\mathbf{z}_P(\mathcal{I}_k^P) = [\mathbf{R}_{k+1} \quad \mathbf{F}_{k+1} \quad \mathbf{R}_k \quad \mathbf{F}_k \quad \bar{\mathbf{u}}_{p,k+1}]^\top, \quad k = 0, \dots, N-2, \quad (5.47)$$

which is consistent with the chain-like sparsity condition (5.31). Thereby, the dynamics and kinematics only couple neighboring time steps. Therefore, the clique vector $\mathbf{z}_P(\mathcal{I}_k^P)$ contains $n_{\text{cl}} = 39$ scalar variables. [TJVG25] identifies for a clique size of n_{cl} , the order- κ moment matrix dimension as

$$s(n_{\text{cl}}, \kappa) = \binom{n_{\text{cl}} + \kappa}{\kappa}. \quad (5.48)$$

Consequently, at $\kappa = 2$, a 39-variable interior clique $\mathbf{z}_P(\mathcal{I}_k^P)$ yields $s(39, 2) = 820$. The corresponding moment matrix therefore has size 820×820 . In contrast, the full decision vector contains

$$n_{z,P} = 9(N + 1) + 9N + 3(N - 1) = 21N + 6$$

variables. A dense second-order relaxation would consequently require a moment matrix of size

$$s(n_{z,P}, 2) = \binom{21N + 8}{2}.$$

For the implemented horizon $N = 20$, this corresponds to a $91\,378 \times 91\,378$ moment matrix, which is computationally not feasible. Nevertheless, the sparse relaxation (5.47) consists of larger moment matrices, compared to the moment matrices constructed using MD, which will be used for the Evaluation. In the following Section 6.2, SPOT is used to construct the second-order sparse moment relaxation from the 3D-pendulum's POP (5.46) [KLY25]. The resulting SDP is solved with the interior-point solver MOSEK [MOS26]. The scalar equations used in the implementation are listed in Appendix A.1. Finally, the implementation is documented in the GitHub repository [Els26].

5.4 Discrete Motion Planning for an Acrobot

The following Section applies the discrete motion planning formulation (5.1) to the controlled Acrobot derived in Section 4.3.1. First, the trajectory optimization problem is solved in Section 5.4.1 using a sparse semidefinite relaxation. Subsequently, the same relaxation is embedded into a model predictive control framework to obtain a closed-loop controller in Section 5.4.2.

5.4.1 Trajectory Optimization via SDP Relaxation

To optimize the Acrobot trajectory, the discrete motion planning problem (5.1) has to be formulated as a POP using the controlled discrete Euler-Lagrange equations of Section 4.3.1. For the two links $i = 1, 2$, the planar rotations are represented around the z-axis by (5.4). The Acrobot model in maximal coordinates contains additional velocity variables $\mathbf{v}_{i,k} \in \mathbb{R}^2$, which results in a higher number of optimization variables. Instead, the maximal coordinate model can be reduced by eliminating the translational positions and velocities while the constraints still remain polynomial. Therefore, the holonomic constraints (4.57) and (4.58) eliminate the center-of-mass positions exactly according to

$$\mathbf{x}_{1,k} = \mathbf{p}_0 - \mathbf{R}_{1,k}\boldsymbol{\rho}_{1,0}, \quad \mathbf{x}_{2,k} = \mathbf{p}_0 + \mathbf{R}_{1,k}(\boldsymbol{\rho}_{1,12} - \boldsymbol{\rho}_{1,0}) - \mathbf{R}_{2,k}\boldsymbol{\rho}_{2,12}. \quad (5.49)$$

Therefore, the velocity $\mathbf{v}_{i,k}$ can be approximated from the rotations by

$$\mathbf{v}_{1,k} \approx \frac{(\mathbf{R}_{1,k} - \mathbf{R}_{1,k+1})\boldsymbol{\rho}_{1,0}}{h}, \quad (5.50a)$$

$$\mathbf{v}_{2,k} \approx \frac{(\mathbf{R}_{1,k+1} - \mathbf{R}_{1,k})(\boldsymbol{\rho}_{1,12} - \boldsymbol{\rho}_{1,0}) - (\mathbf{R}_{2,k+1} - \mathbf{R}_{2,k})\boldsymbol{\rho}_{2,12}}{h}. \quad (5.50b)$$

substituting (5.49) into (4.67). In addition, the kinematic update (4.83g)-(4.83h) can be rewritten as

$$\mathbf{R}_{i,k-1} = \mathbf{R}_{i,k} \mathbf{F}_{i,k-1}^\top. \quad (5.51)$$

Consequently, using (4.83g)-(4.83h), (5.51) and (5.50), the translational dynamics (4.83a)-(4.83b) are reduced to a polynomial form

Translation:

$$-\mathbf{M}_1 \mathbf{R}_{1,k} \mathbf{A}_{1,k} \boldsymbol{\rho}_{1,0} - h^2 g \mathbf{M}_1 \mathbf{e}_2 - h^2 (\boldsymbol{\lambda}_{0,k} + \boldsymbol{\lambda}_{12,k}) = \mathbf{0}, \quad (5.52a)$$

$$\mathbf{M}_2 \left[\mathbf{R}_{1,k} \mathbf{A}_{1,k} (\boldsymbol{\rho}_{1,12} - \boldsymbol{\rho}_{1,0}) - \mathbf{R}_{2,k} \mathbf{A}_{2,k} \boldsymbol{\rho}_{2,12} \right] - h^2 g \mathbf{M}_2 \mathbf{e}_2 + h^2 \boldsymbol{\lambda}_{12,k} = \mathbf{0}, \quad (5.52b)$$

with $\mathbf{A}_{i,k} = \mathbf{F}_{i,k} - 2\mathbf{I}_2 + \mathbf{F}_{i,k-1}^\top$. Therefore, the reduced translational dynamics only depend on the relative rotations $\mathbf{F}_{i,k}$ and $\mathbf{F}_{i,k-1}$, and the attitude $\mathbf{R}_{i,k}$. The physical properties of the reduced model are validated and compared to the full model in the Appendix B.

Similar to the 3D pendulum in Section 5.3, for numerical conditioning, the control and constraint multipliers are scaled to the interval $[-1, 1]$ componentwise,

$$u_k^{\text{phys}} = u_{\max} \bar{u}_k, \quad \boldsymbol{\lambda}_{0,k} = \lambda_{\max} \bar{\boldsymbol{\lambda}}_{0,k}, \quad \boldsymbol{\lambda}_{12,k} = \lambda_{\max} \bar{\boldsymbol{\lambda}}_{12,k}. \quad (5.53)$$

The normalized control and componentwise multiplier bounds are

$$1 - \bar{u}_k^2 \geq 0, \quad 1 - (\bar{\lambda}_{0,k}^{(j)})^2 \geq 0, \quad 1 - (\bar{\lambda}_{12,k}^{(j)})^2 \geq 0, \quad j = 1, 2. \quad (5.54)$$

Moreover, since $\frac{1}{2} \text{tr}(\mathbf{F}_{i,k}) = a_{i,k} = \cos(\Delta\theta_{i,k})$, the relative-rotation bound is

$$a_{i,k} \geq \cos(\Delta\theta_{i,\max}). \quad (5.55)$$

The entries of $\mathbf{R}_{i,k}$ and $\mathbf{F}_{i,k}$ are already bounded by the $SO(2)$ constraints.

The optimization variables are the entries of $\mathbf{R}_{i,0}, \dots, \mathbf{R}_{i,N}$, $\mathbf{F}_{i,0}, \dots, \mathbf{F}_{i,N-1}$, and the normalized variables $\bar{\boldsymbol{\lambda}}_{0,k}, \bar{\boldsymbol{\lambda}}_{12,k} \in \mathbb{R}^2$ and $\bar{u}_k \in \mathbb{R}$, $k = 1, \dots, N-1$. Therefore, the reduced full-horizon decision vector contains $n_{z,A}^{\text{reduced}} = 13N - 1$ scalar variables, compared to the original $n_{z,A}^{\text{full}} = 17N + 3$ variables of the full model using position variables $\mathbf{x}_{i,k}$.

Similar to the 3D-pendulum objective introduced in (5.44) (see Section 5.3), the Acrobot objective uses individual terminal weights w_R, w_F , and stage weights α_R, α_F , and w_u . The multipliers $\bar{\boldsymbol{\lambda}}_{0,k}$ and $\bar{\boldsymbol{\lambda}}_{12,k}$ are not penalized. In the notation

of the general problem (5.1), the equality constraints $\mathcal{D}_i$ consist of the reduced translational (5.52a)-(5.52b) and rotational dynamics (4.87a)-(4.87b), together with the kinematic update (4.83g)-(4.83h). The $\text{SO}(2)$ Lie-group constraints (5.6) form $\mathcal{Y}_i$, whereas $\mathcal{H}$ contains the increment (5.55), control, and multiplier bounds (5.54). In summary, the full controlled Acrobot POP is defined as

$$\begin{aligned} p_A^* = \min_{\mathbf{z}_A} \quad & w_R \sum_{i=1}^2 \|\mathbf{R}_{i,N} - \mathbf{R}_i^{\text{des}}\|_F^2 + w_F \sum_{i=1}^2 \|\mathbf{F}_{i,N-1} - \mathbf{F}_i^{\text{des}}\|_F^2 \\ & + \alpha_R \sum_{k=0}^{N-1} \sum_{i=1}^2 \|\mathbf{R}_{i,k} - \mathbf{R}_i^{\text{des}}\|_F^2 + \alpha_F \sum_{k=0}^{N-2} \sum_{i=1}^2 \|\mathbf{F}_{i,k} - \mathbf{F}_i^{\text{des}}\|_F^2 \\ & + \frac{1}{w_u} \sum_{k=1}^{N-1} \bar{u}_k^2 \end{aligned} \quad (5.56a)$$

$$\text{s.t. } -\mathbf{M}_1 \mathbf{R}_{1,k} \mathbf{A}_{1,k} \boldsymbol{\rho}_{1,0} - h^2 g \mathbf{M}_1 \mathbf{e}_2 - h^2 \bar{\boldsymbol{\lambda}}_{\max} (\bar{\boldsymbol{\lambda}}_{0,k} + \bar{\boldsymbol{\lambda}}_{12,k}) = \mathbf{0} \quad (5.56b)$$

$$\begin{aligned} & \mathbf{M}_2 \left[\mathbf{R}_{1,k} \mathbf{A}_{1,k} (\boldsymbol{\rho}_{1,12} - \boldsymbol{\rho}_{1,0}) - \mathbf{R}_{2,k} \mathbf{A}_{2,k} \boldsymbol{\rho}_{2,12} \right] \\ & - h^2 g \mathbf{M}_2 \mathbf{e}_2 + h^2 \bar{\boldsymbol{\lambda}}_{\max} \bar{\boldsymbol{\lambda}}_{12,k} = \mathbf{0} \end{aligned} \quad (5.56c)$$

$$\begin{aligned} & \hat{\boldsymbol{\Pi}}_{1,k} - \hat{\boldsymbol{\Pi}}_{1,k+1} \\ & + h \left[(\boldsymbol{\rho}_{1,0} \times \mathbf{R}_{1,k+1}^\top \bar{\boldsymbol{\lambda}}_{\max} \bar{\boldsymbol{\lambda}}_{0,k+1}) \right. \\ & \left. + (\boldsymbol{\rho}_{1,12} \times \mathbf{R}_{1,k+1}^\top \bar{\boldsymbol{\lambda}}_{\max} \bar{\boldsymbol{\lambda}}_{12,k+1}) - u_{\max} \bar{u}_{k+1} \right]^\wedge = \mathbf{0} \end{aligned} \quad (5.56d)$$

$$\hat{\boldsymbol{\Pi}}_{2,k} - \hat{\boldsymbol{\Pi}}_{2,k+1} + h \left[u_{\max} \bar{u}_{k+1} - (\boldsymbol{\rho}_{2,12} \times \mathbf{R}_{2,k+1}^\top \bar{\boldsymbol{\lambda}}_{\max} \bar{\boldsymbol{\lambda}}_{12,k+1}) \right]^\wedge = \mathbf{0} \quad (5.56e)$$

$$\mathbf{R}_{i,k+1} = \mathbf{R}_{i,k} \mathbf{F}_{i,k}, \quad (5.56f)$$

$$c_{i,k}^2 + s_{i,k}^2 - 1 = 0, \quad (5.56g)$$

$$a_{i,k}^2 + b_{i,k}^2 - 1 = 0, \quad (5.56h)$$

$$a_{i,k} - \cos(\Delta\theta_{i,\max}) \geq 0, \quad (5.56i)$$

$$1 - \bar{u}_k^2 \geq 0, \quad (5.56j)$$

$$1 - (\bar{\lambda}_{0,k}^{(j)})^2 \geq 0, \quad 1 - (\bar{\lambda}_{12,k}^{(j)})^2 \geq 0, \quad (5.56k)$$

$$\mathbf{F}_{i,0} = \mathbf{F}_i^{\text{init}}, \quad \mathbf{R}_{i,1} = \mathbf{R}_i^{\text{init}}. \quad (5.56l)$$

Table 5.1 describes the index ranges for the constraints in the controlled Acrobot POP. Since the center-of-mass positions are reconstructed from the rotations (5.49), the holonomic constraints (4.83e)-(4.83f) are satisfied identically and therefore do not appear in the reduced POP. Moreover, $\mathbf{F}_{i,0} = \mathbf{F}_i^{\text{init}}$ and $\mathbf{R}_{i,1} = \mathbf{R}_i^{\text{init}}$ specify the initial rotation increment and attitude of each link, respectively. Instead of using fixed boundary conditions, the desired terminal values are penalized. Finally, the full POP of the Acrobot on $\text{SO}(2)$ in (5.56) is a polynomial of degree less than or equal to two.

Chain-like sparsity The reduced dynamics and kinematics couple neighboring time steps. Following the chain-like sparsity pattern introduced in Section 5.2, the

Table 5.1: Index ranges for the constraints in the controlled Acrobot POP.

Constraint	Equation(s)	Indexing
Translation	(5.56b), (5.56c)	$k = 1, \dots, N - 1$
Rotation	(5.56d), (5.56e)	$k = 0, \dots, N - 2$
Kinematics	(5.56f)	$i = 1, 2, k = 0, \dots, N - 1$
$SO(2)$ for $\mathbf{R}$	(5.56g)	$i = 1, 2, k = 0, \dots, N$
$SO(2)$ for $\mathbf{F}$	(5.56h)	$i = 1, 2, k = 0, \dots, N - 1$
Step bound	(5.56i)	$i = 1, 2, k = 1, \dots, N - 1$
Control bound	(5.56j)	$k = 1, \dots, N - 1$
Multiplier bound	(5.56k)	$j = 1, 2, k = 1, \dots, N - 1$
Initial conditions	(5.56l)	$i = 1, 2$

first complete stage clique is

$$\mathbf{z}_A(\mathcal{I}_1^A) = [\mathbf{R}_{1,0}, \mathbf{R}_{2,0}, \mathbf{R}_{1,1}, \mathbf{R}_{2,1}, \mathbf{R}_{1,2}, \mathbf{R}_{2,2}, \mathbf{F}_{1,0}, \mathbf{F}_{2,0}, \mathbf{F}_{1,1}, \mathbf{F}_{2,1}, \bar{\boldsymbol{\lambda}}_{0,1}, \bar{\boldsymbol{\lambda}}_{12,1}, \bar{u}_1]^\top, \quad (5.57)$$

whereas the remaining complete stage cliques are

$$\mathbf{z}_A(\mathcal{I}_k^A) = [\mathbf{R}_{1,k}, \mathbf{R}_{2,k}, \mathbf{R}_{1,k+1}, \mathbf{R}_{2,k+1}, \mathbf{F}_{1,k-1}, \mathbf{F}_{2,k-1}, \mathbf{F}_{1,k}, \mathbf{F}_{2,k}, \bar{\boldsymbol{\lambda}}_{0,k}, \bar{\boldsymbol{\lambda}}_{12,k}, \bar{u}_k]^\top, \quad (5.58)$$

for $k = 2, \dots, N - 1$. For the Acrobot, each rotation and relative rotation is represented by two scalar variables $c_{i,k}$ and $s_{i,k}$ or $a_{i,k}$ and $b_{i,k}$, following the $SO(2)$ definition of (5.6). The larger first clique also covers the initial values (5.56l) and the kinematic update (5.56f) at $k = 0$ and $k = 1$. Moreover, the terminal variables are contained inside $\mathbf{z}_A(\mathcal{I}_{N-1}^A)$. Therefore, no additional terminal clique is required. The size of the first clique is $|\mathcal{I}_1^A| = 25$, whereas the interior cliques have size $|\mathcal{I}_k^A| = 21$. Here, the moment matrix dimensions are

$$s(25, 2) = \binom{27}{2} = 351, \quad s(21, 2) = \binom{23}{2} = 253, \quad (5.59)$$

using (5.48) for the second-order relaxation $\kappa = 2$. Consequently, the chain-like sparsity formulation contains one 351×351 moment matrix and $N - 2$ interior matrices of size 253×253 .

Special multi-body sparsity The Acrobot dynamics couple both rigid links through $\bar{\boldsymbol{\lambda}}_{12,k}$, but the kinematic equations (5.56f) do not contain the multipliers or the input $\bar{u}_k$. The special multi-body sparsity construction exploits this additional structure. Therefore, the interior cliques can be decomposed into the smaller cliques

$$\mathbf{z}_A(\mathcal{I}_{\text{dyn},k}^A) = [\mathbf{R}_{1,k}, \mathbf{F}_{1,k-1}, \mathbf{F}_{1,k}, \mathbf{R}_{2,k}, \mathbf{F}_{2,k-1}, \mathbf{F}_{2,k}, \bar{\boldsymbol{\lambda}}_{0,k}, \bar{\boldsymbol{\lambda}}_{12,k}, \bar{u}_k]^\top \in \mathbb{R}^{17}, \quad (5.60a)$$

$$\mathbf{z}_A(\mathcal{I}_{\text{kin},k}^A) = [\mathbf{R}_{1,k}, \mathbf{R}_{2,k}, \mathbf{R}_{1,k+1}, \mathbf{R}_{2,k+1}, \mathbf{F}_{1,k}, \mathbf{F}_{2,k}]^\top \in \mathbb{R}^{12}, \quad (5.60b)$$

for $k = 3, \dots, N-1$. Here, $\mathbf{z}_A(\mathcal{I}_{\text{dyn},k}^A)$ covers the coupled translational and rotational dynamics, whereas $\mathbf{z}_A(\mathcal{I}_{\text{kin},k}^A)$ covers the kinematics of both links. Keeping both links together avoids discarding their physical coupling. However, for this thesis, $\mathbf{z}_A(\mathcal{I}_1^A)$ (5.57) and $\mathbf{z}_A(\mathcal{I}_2^A)$ (5.58) are taken as the first two cliques, to strengthen the relaxation around the first control decision, which will be used in the MPC implementation of Section 5.4.2.

At $\kappa = 2$, the moment matrices using the special multi-body (5.60a) and (5.60b) cliques have dimensions

$$s(17, 2) = \binom{19}{2} = 171, \quad s(12, 2) = \binom{14}{2} = 91. \quad (5.61)$$

The first two complete clique vectors $\mathbf{z}_A(\mathcal{I}_1^A)$ and $\mathbf{z}_A(\mathcal{I}_2^A)$ are retained around the first control decision, resulting in moment matrices of size 351×351 and 253×253 , respectively. Nevertheless, for each time step $k = 3, \dots, N-1$ there are two moment matrices with the size 171×171 and 91×91 .

For comparison, a dense relaxation has a single clique containing all $n_{z,A} = 13N - 1$ variables. Its order-two moment matrix has the dimension

$$s(13N - 1, 2) = \binom{13N + 1}{2}. \quad (5.62)$$

For the $N = 60$ open-loop experiments reported in Section 6.3.1, this gives $n_{z,A} = 779$ and a dense moment matrix of size $304\,590 \times 304\,590$. Consequently, the chain-like and special multi-body sparsity formulations significantly reduce the size of the moment matrices, by replacing one horizon-dependent positive-semidefinite block by blocks whose dimensions remain independent of N .

In the following Section 6.3.1, SPOT is used to construct the second-order sparse moment relaxation from the Acrobot POP (5.56) [KLY25]. Thereby, the chain-like and special multi-body sparsity structures are exploited by using the defined cliques (5.57)–(5.58) and (5.60). The resulting SDPs are solved with the interior-point solver MOSEK [MOS26]. Finally, the complete scalar POP used in the implementation is listed in Appendix A.2, and the corresponding code is documented in the GitHub repository [Els26].

5.4.2 Trajectory Optimization via Semidefinite Relaxation using Model Predictive Control

For the more difficult Acrobot swing-up problems, the SDP relaxation does not satisfy the rank-one condition over the complete prediction horizon. Consequently, the extracted moments do not provide a certified feasible trajectory [TJVG25]. However, the numerical results in Section 6.3.1 show that the first clique moment matrix is considerably closer to rank one than the later matrices. This empirical observation motivates applying only the first extracted control and solving the SDP again from the resulting state inside a closed loop.

Consequently, the sparse relaxation is embedded into a model predictive control (MPC) framework. Only the first extracted control is applied, the Acrobot is simulated over one control interval, and the problem is solved again from the new simulated state. This converts open-loop trajectory optimization into closed-loop control with implicit feedback [KXS⁺26]. A similar SDP-based MPC formulation obtains a candidate input sequence from an SDP relaxation and uses it inside a receding horizon [HKD⁺26]. Repeated sparse SDP replanning has shown experimental robustness to disturbances and model mismatch [KLY25]. However, the MPC loop does not make the SDP relaxation tight and the predicted SDP trajectory remains uncertified. Nevertheless, the applied controls generate a dynamically consistent closed-loop trajectory using the LGVI of Section 4.3.1. Figure 5.4 summarizes the implemented closed-loop semidefinite-relaxation loop, which will be elaborated in the following paragraphs.

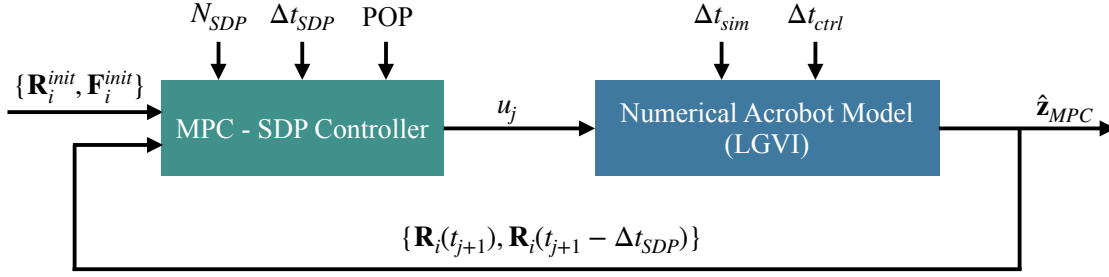


Figure 5.4: SDP-based model predictive control loop for the reduced Acrobot.

Receding-horizon In the following, $j \in \mathbb{N}_0$ denotes the MPC iteration. Therefore, the corresponding simulation time is

$$t_j = j \Delta t_{\text{ctrl}} \quad (5.63)$$

In addition, $k \mid j$ is the SDP prediction node k of the problem solved at MPC iteration j . At every iteration, the second-order sparse Moment-SOS relaxation and clique construction introduced in Section 5.4.1 are used. Only the initial rotation data $\mathbf{R}_{i,1|j}$, $\mathbf{F}_{i,0|j}$, and $\mathbf{R}_{i,0|j}$ are updated from the simulated state, while the dynamics, cost function, bounds, prediction horizon, and desired state remain unchanged.

For any two consecutive prediction nodes, the relative rotation satisfies

$$\mathbf{F}_{i,k|j} = \mathbf{R}_{i,k|j}^\top \mathbf{R}_{i,k+1|j}, \quad \mathbf{R}_{i,k+1|j} = \mathbf{R}_{i,k|j} \mathbf{F}_{i,k|j}. \quad (5.64)$$

At the boundary of each newly solved SDP problem, node $k = 1$ represents the current simulated attitude at t_j , whereas node $k = 0$ represents the required history state one SDP interval earlier. Therefore, with $h = \Delta t_{\text{SDP}}$, the updated boundary

data are

$$\begin{aligned} \mathbf{R}_{i,1|j} &= \mathbf{R}_i(t_j), \\ \mathbf{F}_{i,0|j} &= \mathbf{R}_i(t_j - \Delta t_{\text{SDP}})^\top \mathbf{R}_i(t_j), \\ \mathbf{R}_{i,0|j} &= \mathbf{R}_{i,1|j} \mathbf{F}_{i,0|j}^\top, \end{aligned} \quad (5.65)$$

for links $i = 1, 2$ and with the requirement $\frac{\Delta t_{\text{SDP}}}{\Delta t_{\text{sim}}} \in \mathbb{N}$. The reconstruction $\mathbf{F}_{i,0|j} = \mathbf{R}_i(t_j - \Delta t_{\text{SDP}})^\top \mathbf{R}_i(t_j)$ ensures that the incoming relative rotation always spans exactly one SDP interval Δt_{SDP} . This is important when the control interval Δt_{ctrl} or the simulation interval Δt_{sim} differs from Δt_{SDP} . However, during the first MPC iterations, the simulated state does not yet contain a full history of past rotations for $\Delta t_{\text{SDP}} > t_j$, because the time $t_j - \Delta t_{\text{SDP}}$ lies before the simulation start. Therefore, a prehistory is assumed by

$$\mathbf{F}_{i,0|j} = \begin{cases} \mathbf{R}_i(0)^\top \mathbf{R}_i(t_j), & 0 \leq t_j < \Delta t_{\text{SDP}}, \\ \mathbf{R}_i(t_j - \Delta t_{\text{SDP}})^\top \mathbf{R}_i(t_j), & t_j \geq \Delta t_{\text{SDP}}. \end{cases} \quad (5.66)$$

Control application After updating the boundary data at iteration j , the reduced Acrobot POP of Section 5.4.1 is converted into a second-order sparse SDP using SPOT and solved with MOSEK. Ordered extraction of the degree-one moments provides the normalized first control $\bar{u}_{1|j}^*$ from the solution vector

$$\{\bar{u}_{1|j}^*, \bar{u}_{2|j}^*, \dots, \bar{u}_{N_{\text{SDP}}-1|j}^*\}.$$

The corresponding physical torque is

$$u_j = u_{\max} \bar{u}_{1|j}^*. \quad (5.67)$$

Then, this first control u_j is applied to the numerical Acrobot simulation of Section 6.1.2, while the remaining predicted control sequence is discarded. The control input u_j is kept constant over the control interval $t_{j+1} = t_j + \Delta t_{\text{ctrl}}$. During this interval, the Acrobot is simulated using the reduced LGVI of Section 4.3.1. The simulated trajectory at $j+1$ is then stored inside $\hat{\mathbf{z}}_{\text{MPC},j+1}$. The resulting rotations and rotation-steps are then used to construct the initial data for the next SDP problem as described above.

After the simulation interval, the rotation error $e_{R,j}$ and the relative-step rotation error $e_{F,j}$ are evaluated. The step-angle error calculated from the last fine simulation step and is measured in degrees and degrees per second. The Acrobot is declared stabilized if

$$e_{R,j} \leq \varepsilon_R, \quad e_{F,j} \leq \varepsilon_F \quad (5.68)$$

holds for n_{stable} consecutive MPC updates. In this case, the MPC loop terminates and the full simulated trajectory is saved inside $\hat{\mathbf{z}}_{\text{MPC},j+1}$. Otherwise, j is increased by one by the MPC-SDP controller, the initial data is updated from the new simulated state, and the SDP is solved again. If the SDP is infeasible or no finite first control can be extracted, the loop terminates before another simulation interval.

Prediction horizon In summary, the MPC implementation uses

$$\Delta t_{\text{SDP}}, \quad \Delta t_{\text{sim}}, \quad n_{\text{sim}} = \frac{\Delta t_{\text{ctrl}}}{\Delta t_{\text{sim}}} \in \mathbb{N}. \quad (5.69)$$

Here, Δt_{SDP} is the discretization interval of the SDP prediction model, Δt_{sim} is the fine LGVI simulation step, and Δt_{ctrl} is the control interval between two MPC updates. Consequently, the numerical plant performs n_{sim} LGVI steps before the SDP is solved again. The intervals Δt_{SDP} and Δt_{ctrl} can differ from each other. If the two intervals are equal, $\Delta t_{\text{SDP}} = \Delta t_{\text{ctrl}}$, one MPC update corresponds to one predicted SDP interval.

Increasing N_{SDP} provides a longer look-ahead for the swing-up but also increases the size and solution time of the SDP. The complete decision logic of the SDP-based MPC loop is summarized in Figure 5.5.

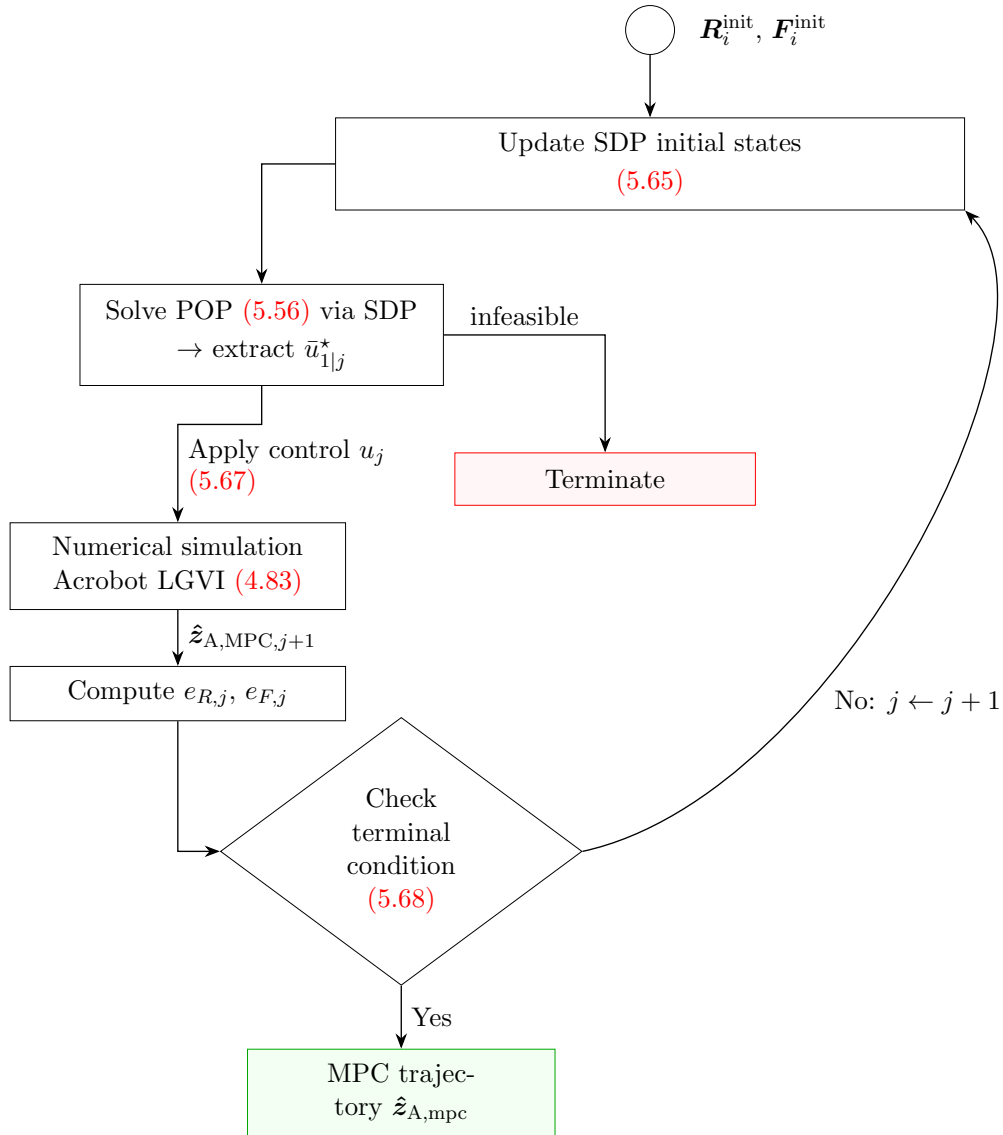


Figure 5.5: Decision logic of the SDP-based model predictive control loop for the reduced Acrobot.

Chapter 6

Evaluation

This chapter evaluates the proposed modeling and optimization framework of Chapter 5. Section 6.1 first validates the derived LGVIs for the 3D pendulum and the Acrobot through numerical simulations. Sections 6.2 and 6.3 then assess the corresponding sparse Moment-SOS trajectory optimization problems. Finally, Section 6.3.2 examines the closed-loop performance of the SDP-MPC formulation for the Acrobot.

6.1 Numerical Simulation Results

In this section, numerical simulations are conducted to evaluate the correctness of the derived discrete integrators for mechanical models on Lie groups. In particular, the structure-preserving properties of the variational discretizations derived in Section 4.1.3 are examined. The analysis focuses especially on the preservation of the momentum map associated with the symmetry of the system, the evolution on the Lie-group configuration manifold, and the long-time behaviour of the total energy. This validation is important since discrete mechanical models obtained by variational integration provide a reliable basis for certifiable trajectory optimization. If the discrete flow violates the manifold structure or introduces artificial drift in conserved quantities such as momentum, the resulting optimization model becomes less faithful to the underlying mechanics [Lee08]. Additionally, for the 3D-pendulum, the numerical simulations are compared with classical fourth-order Runge-Kutta methods and projected Runge-Kutta methods on $SO(3)$ as reference integrators [CO03, Hai00]. (see Table 6.1).

Table 6.1: Comparison of the considered integration methods properties.

Method	Symplectic	Manifold Preserving
Variational Integrator	✓	✓
RK4	✗	✗
Projected RK4	✗	✓

Runge–Kutta 4th order As a baseline integrator, the classical explicit fourth-order Runge-Kutta (RK4) method is applied to the continuous rigid-body equations written as a first-order system $(\dot{\mathbf{R}}, \dot{\boldsymbol{\omega}}) = \mathbf{f}(\mathbf{R}, \boldsymbol{\omega})$. For a system of the form

$$\dot{\mathbf{y}} = \mathbf{f}(\mathbf{y}), \quad (6.1)$$

the classical RK4 scheme computes the four stage values

$$(\mathbf{k}_1^R, \mathbf{k}_1^\omega) = \mathbf{f}(\mathbf{R}_n, \boldsymbol{\omega}_n), \quad (6.2a)$$

$$(\mathbf{k}_2^R, \mathbf{k}_2^\omega) = \mathbf{f}\left(\mathbf{R}_n + \frac{h}{2}\mathbf{k}_1^R, \boldsymbol{\omega}_n + \frac{h}{2}\mathbf{k}_1^\omega\right), \quad (6.2b)$$

$$(\mathbf{k}_3^R, \mathbf{k}_3^\omega) = \mathbf{f}\left(\mathbf{R}_n + \frac{h}{2}\mathbf{k}_2^R, \boldsymbol{\omega}_n + \frac{h}{2}\mathbf{k}_2^\omega\right), \quad (6.2c)$$

$$(\mathbf{k}_4^R, \mathbf{k}_4^\omega) = \mathbf{f}(\mathbf{R}_n + h\mathbf{k}_3^R, \boldsymbol{\omega}_n + h\mathbf{k}_3^\omega), \quad (6.2d)$$

and updates the numerical solution according to

$$\mathbf{R}_{n+1} = \mathbf{R}_n + \frac{h}{6}(\mathbf{k}_1^R + 2\mathbf{k}_2^R + 2\mathbf{k}_3^R + \mathbf{k}_4^R), \quad (6.3a)$$

$$\boldsymbol{\omega}_{n+1} = \boldsymbol{\omega}_n + \frac{h}{6}(\mathbf{k}_1^\omega + 2\mathbf{k}_2^\omega + 2\mathbf{k}_3^\omega + \mathbf{k}_4^\omega), \quad (6.3b)$$

for each discrete-time step n [But16].

The explicit RK4 method does not preserve the orthogonality constraint $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$ and leaves the configuration manifold $SO(3)$, since the rotation matrix $\mathbf{R}_n$ is updated additively in the vector space of $\mathbb{R}^{3 \times 3}$ [CO03].

RK4 projected to $SO(3)$ The explicit RK4 method is combined with a projection of the updated discrete-time rotation matrix $\mathbf{R}_n$ onto $SO(3)$ to satisfy the orthogonality constraint

$$SO(3) = \{\mathbf{R} \in \mathbb{R}^{3 \times 3} \mid \mathbf{R}^\top \mathbf{R} = \mathbf{I}_3, \det(\mathbf{R}) = 1\}.$$

After performing the RK4 step in $\mathbb{R}^{3 \times 3}$, the singular value decomposition of the obtained rotation matrix $\tilde{\mathbf{R}}_{n+1}$ is computed as

$$\tilde{\mathbf{R}}_{n+1} = \mathbf{U}\Sigma\mathbf{V}^\top, \quad (6.4)$$

followed by a reconstruction of the updated rotation matrix

$$\mathbf{R}_{n+1} = \mathbf{U}\mathbf{V}^\top, \quad (6.5)$$

which is projected onto $SO(3)$ [YTT11]. Consequently, $\mathbf{R}_{n+1}$ satisfies the orthogonality constraint, while the angular velocity $\boldsymbol{\omega}_{n+1}$ remains unchanged in the vector space $\mathbb{R}^3$. However, this method does not inherit structure-preserving properties of the Lie group variational integrator, for example, symplecticity or momentum conservation [CO03].

6.1.1 3D Pendulum

For evaluating the preservation of the geometric properties of the previously defined 3D-pendulum (see Section 4.2.1), numerical simulations are conducted for the unforced and controlled cases. The rigid body is modeled as a uniform solid cylinder of length $l = 1$ m and radius $r = 5$ cm, with its center of mass at $\frac{l}{2}$. Thereby, the physical parameters characterizing the configuration manifold are chosen as

$$m = 1 \text{ kg}, \quad g = 9.81 \text{ m s}^{-2}, \quad (6.6)$$

$$\boldsymbol{\rho}_c = [0 \ 0 \ 0.5]^\top \text{ m}, \quad \mathbf{e}_3 = [0 \ 0 \ 1]^\top. \quad (6.7)$$

The inertia matrix about the center of mass can be taken from the inertia table given in [Ric]. The rotations about the x - and y -axes are about the central diameters, with $J_{11,com} = J_{22,com} = \frac{1}{12}m(3r^2 + l^2)$, while the rotation about the gravity-direction z -axis has $J_{33,com} = \frac{1}{2}mr^2$. The off-diagonal terms are equal to zero, because the body-fixed frame is aligned with the principal axes of the cylindrically symmetric body [Lee08]. Additionally, the inertia matrix about the pivot in body-frame is obtained by the parallel axis theorem of [BR18] given by

$$\mathbf{J} = \mathbf{J}_{com} + m (\|\boldsymbol{\rho}_c\|^2 \mathbf{I} - \boldsymbol{\rho}_c \boldsymbol{\rho}_c^\top). \quad (6.8)$$

After substituting the chosen values, the 3D-pendulum's inertia matrix is given by

$$\mathbf{J} = \begin{bmatrix} 0.33396 & 0 & 0 \\ 0 & 0.33396 & 0 \\ 0 & 0 & 0.00125 \end{bmatrix} \text{ kg m}^2. \quad (6.9)$$

Given (3.22), the non-standard inertia matrix is calculated as

$$\mathbf{J}_d = \begin{bmatrix} 0.000625 & 0 & 0 \\ 0 & 0.000625 & 0 \\ 0 & 0 & 0.333333 \end{bmatrix} \text{ kg m}^2. \quad (6.10)$$

Using the introduced computational approach of Section 4.1.5, the derived discrete-time implicit Hamilton equations of the unforced (4.43) and controlled (4.47) 3D-pendulum are converted to an equivalent vector equation in $\mathbb{R}^3$ using the Cayley transformation. From [CJ12], the corresponding nonlinear vector equation $\mathbf{0} = \mathbf{A}(\mathbf{f})$ of the 3D-pendulum is taken as

$$\mathbf{0} = \mathbf{a} + \mathbf{a} \times \mathbf{f} + \mathbf{f}(\mathbf{a}^\top \mathbf{f}) - 2\mathbf{J}\mathbf{f} = \mathbf{A}(\mathbf{f}), \quad (6.11)$$

with $\mathbf{a} = h(\boldsymbol{\Pi}_k + \frac{h}{2}\boldsymbol{\tau}_{g,k})$. Additionally, [CJ12] defines its Jacobian $\nabla \mathbf{A}(\mathbf{f})$ as

$$\nabla \mathbf{A}(\mathbf{f}) = \hat{\mathbf{a}} + (\mathbf{a}^\top \mathbf{f})\mathbf{I} + \mathbf{f}\mathbf{a}^\top - 2\mathbf{J}. \quad (6.12)$$

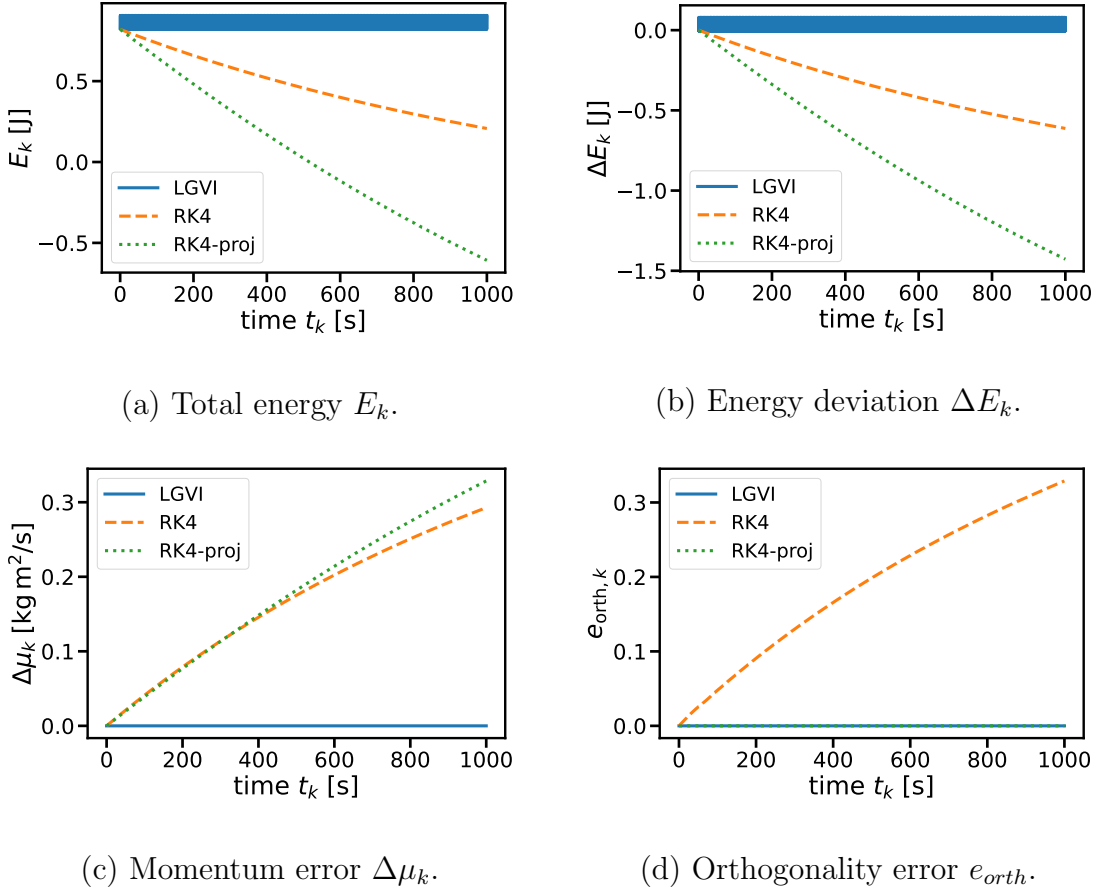


Figure 6.1: Comparison of the numerical preservation properties for the unforced 3D pendulum.

The tolerance for the Newton iteration is defined by $\|\mathbf{f}^{(i+1)} - \mathbf{f}^{(i)}\| < 10^{-12}$. The initial conditions are taken from [Lee08]

$$\mathbf{f}^0 = (2\mathbf{J} - \hat{\mathbf{a}})^{-1}\mathbf{a}, \quad \mathbf{R}_0 = \mathbf{I}, \quad \boldsymbol{\omega}_0 = [4.14 \ 4.14 \ 4.14]^\top \text{ rad s}^{-1}, \quad (6.13)$$

where $\mathbf{f}^0$ is the linearized solution of (6.11). In the following, the numerical simulation is plotted against time $t_k = kh$. The corresponding implementation is documented in the GitHub repository [Els26].

Unforced 3D-Pendulum

The results of the numerical simulation for the unforced case, with $\mathbf{u}_k = \mathbf{0}$, are shown in Fig. 6.1. The simulation is performed over 20000 discrete-time steps with a step size of $h = 0.05$ s. Thereby, the accuracy of the different integrator methods given in Table 6.1 is compared by plotting the corresponding total energy

$$E_k = \frac{1}{2}\boldsymbol{\omega}_k^\top \mathbf{J} \boldsymbol{\omega}_k - mg \mathbf{e}_3^\top \mathbf{R}_k \boldsymbol{\rho}_c,$$

and its deviation $\Delta E_k = E_k - E_0$, the angular momentum error along the gravity direction

$$\Delta \mu_k = \mathbf{e}_3^\top \mathbf{R}_k \boldsymbol{\Pi}_k - \mathbf{e}_3^\top \mathbf{R}_0 \boldsymbol{\Pi}_0,$$

defined in [Lee08], and the orthogonality error

$$e_{\text{orth},k} = \|\mathbf{I} - \mathbf{R}_k^\top \mathbf{R}_k\|_F.$$

The total energy (see Fig. 6.1 (a)) and the corresponding energy deviation (see Fig. 6.1 (b)) illustrate the energy-preserving behavior of the Lie Group Variational Integrator (LGVI) over a long-time discrete flow. The energy remains close to the reference level with only small bounded oscillations because the symplectic method exactly solves a perturbed Hamiltonian [Hai94]. However, RK4 and projected RK4 display a clear drift of the energy away from its true value, which remains constant for this unforced model.

Moreover, the momentum error (see Fig. 6.1 (c)) confirms that the LGVI method preserves the momentum, as validated for continuous time in 3.2.2. The computational results for RK4 and its projection onto $SO(3)$ show that these methods drift from the initial momentum value because they do not preserve the total angular momentum [Lee08]. In particular, projected RK4 has a larger discrepancy because it enforces $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$ while leaving the angular velocity $\boldsymbol{\omega}$ unchanged, which influences the momentum.

Lastly, the orthogonality error highlights that the LGVI preserves the structure of the Lie group configuration manifold, as depicted in Fig. 6.1 (d). A similar result is obtained for projected RK4, while classical explicit RK4 violates the orthogonality constraint. For RK4, e_{orth} increases linearly with the number of simulation time steps.

Consequently, only the LGVI preserves the manifold structure together with the momentum and energy behaviour over a long-time horizon. Although RK4 can be projected onto the Lie group manifold $SO(3)$, it cannot preserve the momentum and exhibits long-term energy drift.

Controlled 3D-Pendulum

In the controlled case, the previously derived free dynamics of the 3D-pendulum are extended by an external body-fixed control moment. Following the control structure defined in (4.44), the time-dependent control parameter is chosen as

$$\mathbf{u}_p(t) = \begin{bmatrix} 0.15 \sin(0.5t) \\ 0.10 \cos(0.25t) \\ 0 \end{bmatrix}. \quad (6.14)$$

The corresponding discrete control input is obtained by evaluating the control law at the discrete time nodes $\mathbf{u}_{p,k} = \mathbf{u}_p(t_k)$. Thus, the control input acts as a body-fixed external moment that is orthogonal to the gravity direction expressed in the body

frame, given by $\mathbf{R}_k^\top \mathbf{e}_3$. Consequently, the control moment has no component along this direction.

For the RK4 simulation, the continuous-time Hamilton equations (3.31) are used, with the control input $\mathbf{u}(t)$ added according to the Lagrange–d’Alembert principle of Section 4.1.4. With the body angular momentum defined by $\mathbf{\Pi} = \mathbf{J}\boldsymbol{\omega}$, the forced continuous-time equations of motion are written as

$$\dot{\mathbf{\Pi}} = mg \boldsymbol{\rho}_c \times \mathbf{R}^\top \mathbf{e}_3 - \mathbf{J}^{-1} \mathbf{\Pi} \times \mathbf{\Pi} + \mathbf{u}(t), \quad (6.15a)$$

$$\dot{\mathbf{R}} = \mathbf{R} (\mathbf{J}^{-1} \mathbf{\Pi})^\wedge. \quad (6.15b)$$

In contrast to the unforced case, the external input acts on the system. Therefore, the total energy and the momentum map are no longer expected to be conserved.

For the Lie group variational integrator, the forced discrete Hamilton equations derived in (4.47) are applied. The implicit equation for the relative update $\mathbf{F}_k$ is again solved by using the Cayley transformation. Compared to the unforced case, only the vector $\mathbf{a}$ in (6.11) changes due to the additional control input. For the controlled dynamics, it is given by

$$\mathbf{a} = h \left(\mathbf{\Pi}_k + \frac{h}{2} (\boldsymbol{\tau}_{g,k} + \mathbf{u}_k) \right). \quad (6.16)$$

Therefore, the nonlinear vector equation (6.11) and its Jacobian (6.12) are solved equivalently to the unforced simulation, with the modified vector $\mathbf{a}$ including the control contribution. After computing $\mathbf{f}_k$, the relative attitude update is obtained from the Cayley transformation and the discrete-time states are propagated according to the forced LGVI. The corresponding implementation is documented in the GitHub repository [Els26].

The numerical simulation of the controlled 3D-pendulum compares the classical RK4 method with the forced LGVI. Since the purpose of this experiment is to evaluate the behaviour of the forced variational discretization under an external input, the projected RK4 method is not considered in this comparison. The same physical parameters and initial conditions as in the unforced case are used.

Fig. 6.2 shows the numerical results for the controlled 3D-pendulum. The simulation is performed over $N = 100000$ discrete-time steps with step size $h = 0.001$ s. This shorter time interval is chosen because the differences between the RK4 method and the forced LGVI already become visible after a small number of time steps. During approximately the first eight seconds, both methods show almost identical behaviour, which indicates that the derived forced equations and their implementation are consistent. As the simulation progresses, however, the trajectories begin to separate.

The position error in Fig. 6.2 (a) is computed from the center-of-mass position $\mathbf{x}_k = \mathbf{R}_k \boldsymbol{\rho}_c$, as

$$\|\Delta \mathbf{x}_k\|_2 = \|\mathbf{x}_{k,\text{RK4}} - \mathbf{x}_{k,\text{LGVI}}\|_2. \quad (6.17)$$

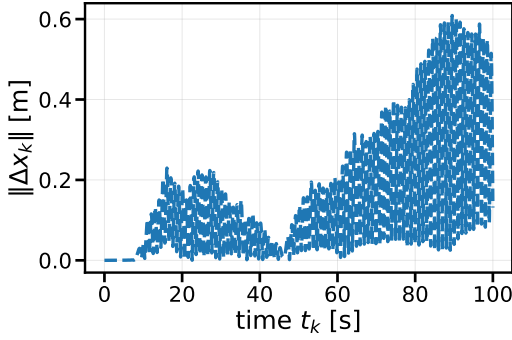
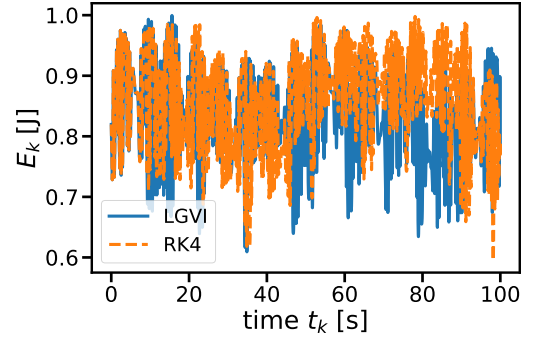
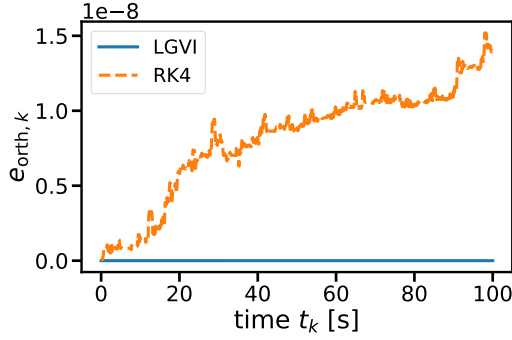
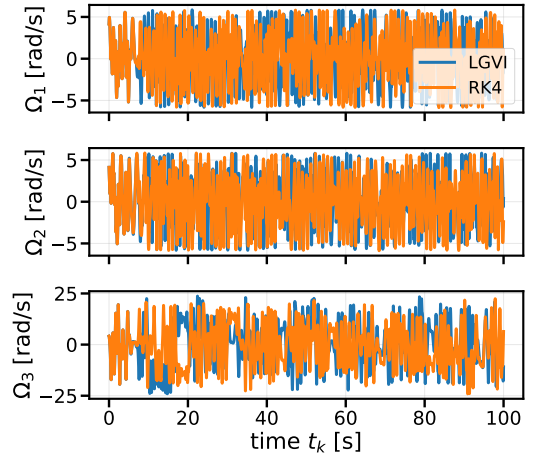
(a) Position error norm $\|\Delta \mathbf{x}_k\|$.(b) Total energy E_k .(c) Orthogonality error $e_{\text{orth},k}$.(d) Angular velocity components $\boldsymbol{\omega}_k$.

Figure 6.2: Comparison of the forced LGVI and RK4 simulation for the controlled 3D pendulum.

The plot shows that the position error $\Delta \mathbf{x}_k$ is initially close to zero, but increases over the simulation. Therefore, both methods reproduce the same short-time motion, while their trajectories separate after a certain number of time steps. This behaviour is expected for forced rigid-body dynamics, where small numerical differences in the attitude $\mathbf{R}_k$ and momentum update $\boldsymbol{\Pi}_k$ can accumulate over time.

The total energy is shown in Fig. 6.2 (b). In contrast to the unforced case, the total mechanical energy is not expected to be conserved, since the external input $\mathbf{u}_{p,k}$ acts on the system. It is observed that the energetic behaviour of LGVI and RK4 under the same applied control input follows a similar trend at the beginning, but small deviations appear as the trajectories begin to separate, which is consistent with the position error.

The orthogonality error $e_{\text{orth},k}$ is illustrated in Fig. 6.2 (c). Again, the plot

confirms that the LGVI preserves the Lie group structure by staying on the manifold and updating the attitude through $\mathbf{R}_{k+1} = \mathbf{R}_k \mathbf{F}_k$ with $\mathbf{F}_k \in SO(3)$. For RK4, the attitude is updated additively in the surrounding vector space using the discrete update map (6.3). Over the short interval and due to the small step size h , the orthogonality error remains small, but it is not preserved and increases over time.

Finally, Fig. 6.2 (d) compares the angular velocity components $\boldsymbol{\omega}_k$. In the discrete Hamiltonian formulation, the angular velocity is recovered from the body angular momentum by

$$\hat{\boldsymbol{\omega}}_k = (\mathbf{J}^{-1} \boldsymbol{\Pi}_k)^\wedge, \quad (6.18)$$

discretizing and solving (3.30) for $\hat{\boldsymbol{\omega}}_k$. The angular velocity components of LGVI and RK4 are equal at the beginning of the simulation, but begin to differ after a small number of steps. This confirms the same behaviour observed in the position error $\Delta \mathbf{x}_k$ and the plotted energy E_k . The 3D-Pendulum numerical simulation using external control inputs produces consistent short-time dynamics of LGVI and RK4, while the numerical trajectories produced by the two methods separate as the simulation continues.

6.1.2 Acrobot

To evaluate the preservation of the geometric properties of the previously defined Acrobot (see Section 4.3.1), numerical simulations are conducted for the unforced and controlled cases similar to those conducted for the 3D-pendulum in Section 6.1.1. In contrast to the 3D-pendulum, the Acrobot is a constrained multi-body system in maximal coordinates with the configuration

$$q_k = (\mathbf{x}_{1,k}, \mathbf{R}_{1,k}, \mathbf{x}_{2,k}, \mathbf{R}_{2,k}) \in (\mathbb{R}^2 \times SO(2))^2,$$

Therefore, the validation focuses not only on the preservation of the Lie-group structure of the attitude variables, but also on the preservation of the holonomic joint constraints $\phi_0(q_k)$ and $\phi_{12}(q_k)$. The physical parameters are chosen as two identical uniform rods with

$$m_1 = m_2 = 1 \text{ kg}, \quad l_1 = l_2 = 1.0 \text{ m}, \quad g = 9.81 \text{ m/s}^2,$$

and the center of mass of each link is located at $\frac{l_i}{2}$. Since the maximal-coordinate formulation already represents the translational motion of the link centers of mass, the rotational inertia is taken about the corresponding center of mass. [BR18] defines the rotational inertia of a uniform rod about its center of mass as

$$J_i^{\text{com}} = \frac{1}{12} m_i l_i^2.$$

By substituting the defined values of m_i and l_i into (4.62), the non-standard inertia matrix of each link is given as

$$\mathbf{J}_{d,1} = \mathbf{J}_{d,2} = \begin{bmatrix} 0.04167 & 0 \\ 0 & 0.04167 \end{bmatrix} \text{ kg m}^2,$$

The base point is fixed at

$$\mathbf{p}_0 = [0 \ 0]^\top.$$

Using the body-fixed link convention from Section 4.3.1, the joint vectors are

$$\boldsymbol{\rho}_{1,0} = [0 \ 0.5]^\top \text{ m}, \quad \boldsymbol{\rho}_{1,12} = [0 \ -0.5]^\top \text{ m}, \quad \boldsymbol{\rho}_{2,12} = [0 \ 0.5]^\top \text{ m},$$

illustrated in Fig. 4.3.

The implementation of the numerical simulation follows Section 4.1.5. In contrast to (6.11), a unknown variable vector

$$\mathbf{z}_{A,k} = [a_{1,k} \ b_{1,k} \ a_{2,k} \ b_{2,k} \ \boldsymbol{\lambda}_{0,k}^\top \ \boldsymbol{\lambda}_{12,k}^\top]^\top,$$

is defined. The nonlinear residual is written compactly as

$$\mathbf{A}_{\text{acro}}(\mathbf{z}_{A,k}) = \begin{bmatrix} \mathbf{A}_{T,k} \\ \mathbf{A}_{R,k} \\ a_{1,k}^2 + b_{1,k}^2 - 1 \\ a_{2,k}^2 + b_{2,k}^2 - 1 \end{bmatrix} = \mathbf{0},$$

where $\mathbf{A}_{T,k}$ contains the translational dynamics (4.83a)-(4.83b) and $\mathbf{A}_{R,k}$ contains the rotational dynamics (4.83c)-(4.83d). After the nonlinear system has converged, the relative rotations $\mathbf{F}_{i,k}$ are reconstructed and the link attitudes are updated by $\mathbf{R}_{i,k+1} = \mathbf{R}_{i,k} \mathbf{F}_{i,k}$.

As in the controlled 3D-pendulum simulation, a classical fourth-order Runge-Kutta method is used as a reference integrator. The continuous-time Acrobot dynamics are implemented from the standard joint-coordinate equations given by [Ted23]. The resulting absolute link angles $\theta_{1,k}$ and $\theta_{2,k}$ are mapped directly to the link rotations $\mathbf{R}_{1,k}$ and $\mathbf{R}_{2,k}$ using (4.54). The complete implementation is documented in the GitHub repository [Els26].

Additionally, the following metrics are used for the Acrobot simulation. The total mechanical energy is evaluated as

$$E_k = \sum_{i=1}^2 \left(\frac{1}{2} \mathbf{v}_{i,k}^\top \mathbf{M}_i \mathbf{v}_{i,k} + \frac{1}{2} J_i^{\text{com}} \omega_{i,k}^2 - g \mathbf{e}_2^\top \mathbf{M}_i \mathbf{x}_{i,k} \right).$$

where $\mathbf{M}_i = m_i \mathbf{I}$ denotes the mass matrix of link i . The position difference between the LGVI trajectory and the reconstructed RK4 reference trajectory is computed by

$$\|\Delta \mathbf{x}_k\|_2 = \|\mathbf{X}_{k,\text{RK4}} - \mathbf{X}_{k,\text{LGVI}}\|_2,$$

with $\mathbf{X}_k = [\mathbf{x}_{1,k}^\top \ \mathbf{x}_{2,k}^\top]^\top$. The holonomic constraint residuals are evaluated separately as $\|\boldsymbol{\phi}_{0,k}\|_2$, and $\|\boldsymbol{\phi}_{12,k}\|_2$, where $\boldsymbol{\phi}_{0,k}$ and $\boldsymbol{\phi}_{12,k}$ are defined in (4.57) and (4.58). Finally, the Lie-group preservation of both link attitudes is measured by

$$e_{\text{orth},i,k} = \|\mathbf{I}_2 - \mathbf{R}_{i,k}^\top \mathbf{R}_{i,k}\|_F, \quad i \in 1, 2.$$

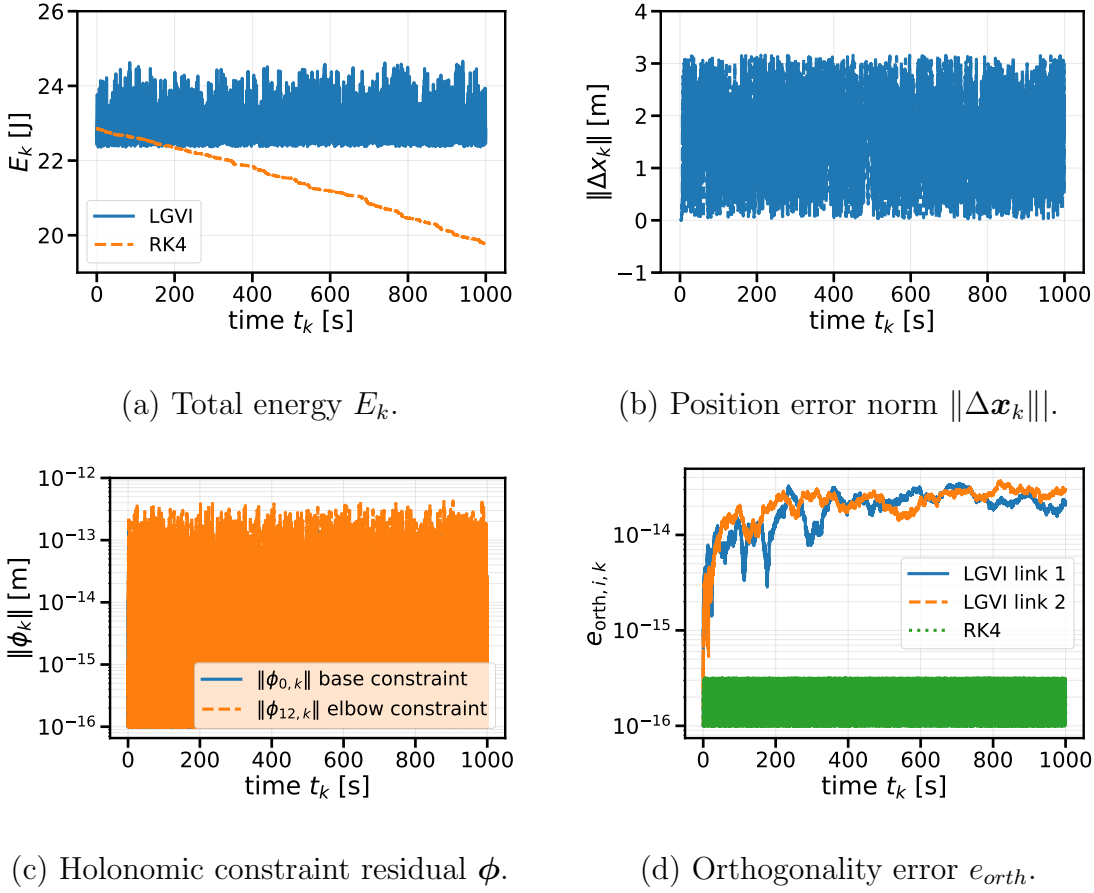


Figure 6.3: Comparison of the numerical preservation properties for the unforced Acrobot.

In contrast to the 3D-pendulum, no momentum-map error is evaluated for the Acrobot, as the system does not possess the same rotational symmetry about the gravity direction as the 3D-pendulum. For the unforced and controlled Acrobot simulations, both links start from the identity attitude with the same absolute angular velocity,

$$\mathbf{R}_{i,0} = \mathbf{I}, \quad \omega_{i,0} = 4.14 \text{ rad s}^{-1}, \quad i \in \{1, 2\}.$$

Unforced Acrobot

The first Acrobot simulation considers the unforced case with $u_k = 0$. The simulation is performed with the step size $h = 0.01$ s over the final time $t_f = 1000$ s, which corresponds to $N = 100000$ discrete-time steps.

The total energy E_k is shown in Fig.6.3(a). Since the system is unforced, the exact mechanical energy is expected to remain constant. The LGVI energy remains close to its initial value and shows bounded oscillations over the long-time horizon.

This behaviour is consistent with the symplectic structure of variational integrators introduced in Section 4.1.3. In contrast, the RK4 reference shows a visible drift of the energy over time. This agrees with the observation from the unforced 3D-pendulum simulation in Fig.6.1.

Fig. 6.3(b) shows the position error norm $\|\Delta \mathbf{x}_k\|$ between the LGVI trajectory and the reconstructed RK4 reference. It subsequently grows and oscillates over the long-time horizon, reaching a maximum of approximately 3.16 m. The deviation is reasoned, by the numerical differences between the two integrators LGVI and RK4.

The holonomic constraint residuals $\|\phi_{0,k}\|_2$ and $\|\phi_{12,k}\|_2$ are shown in Fig.6.3(c). Both the base constraint residual $\|\phi_{0,k}\|_2$ and the elbow constraint residual $\|\phi_{12,k}\|_2$ remain close to machine precision over the complete simulation ($< 10^{-12}$). This confirms that the constrained LGVI in maximal-coordinates (4.83) enforces the fixed-base joint and the elbow joint at the position level.

Finally, Fig.6.3(d) shows the orthogonality error $e_{orth,i,k}$ of both link rotations. The LGVI preserves the $SO(2)$ structure of both attitudes by the multiplicative update $\mathbf{R}_{i,k+1} = \mathbf{R}_{i,k} \mathbf{F}_{i,k}$ with $\mathbf{F}_{i,k} \in SO(2)$. The RK4 reference also remains very small, because it is integrated in relative coordinates and then reconstructed into rotation matrices. However, the LGVI shows a slightly larger error, which is caused by the finite tolerance of the Newton iteration used to compute the Cayley parameters. Nevertheless, the orthogonality error remains below 10^{-13} over the complete simulation.

Overall, the unforced Acrobot simulation confirms that the derived constrained LGVI preserves the geometric structure of the maximal-coordinate model. The link attitudes remain on $SO(2)$, the joint constraints remain satisfied, and the energy shows bounded long-time behaviour.

Controlled Acrobot

The Acrobot is actuated by a scalar torque at the elbow joint. As derived in (4.85), this torque is an internal joint torque acting between the two links. The control input is chosen as the sinusoidal elbow torque

$$u(t) = 0.15 \sin(0.7t) \text{ N m.}$$

The discrete input is obtained by evaluating the continuous input at the time nodes, $u_k = u(t_k)$. The forced simulation uses the same physical parameters and initial condition as the unforced case. The simulation is performed with

$$h = 0.001 \text{ s}, \quad t_f = 100 \text{ s},$$

corresponding to $N = 100000$ discrete-time steps. The same metrics as in the unforced case are evaluated.

The total energy E_k of the controlled Acrobot is shown in Fig. 6.4(a). As for the controlled 3D pendulum, the LGVI and RK4 energies agree closely during approximately the first 5 s and subsequently deviate as the numerical trajectories separate.

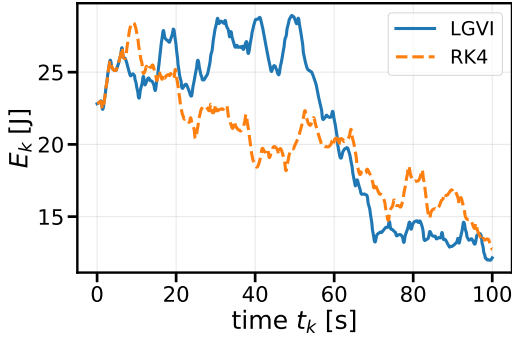
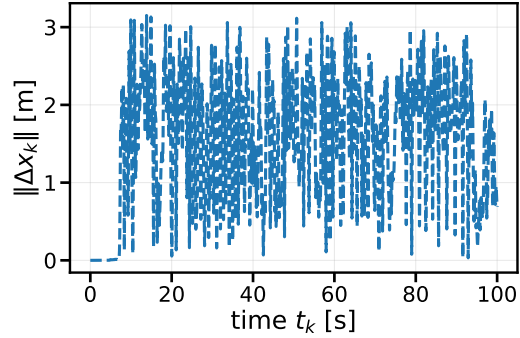
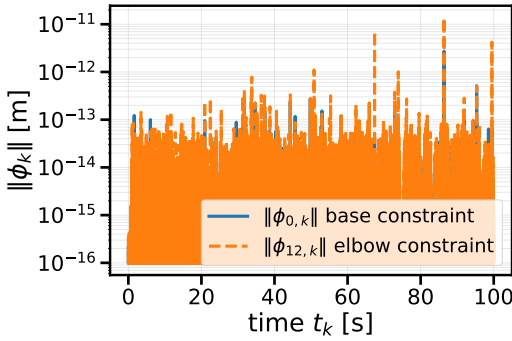
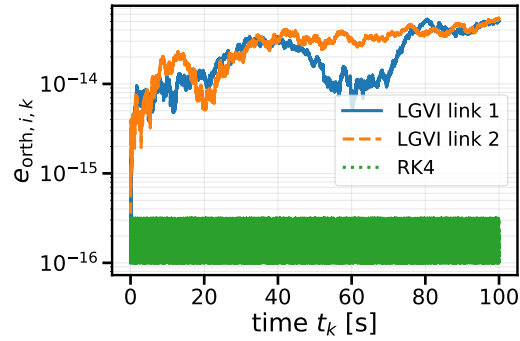
(a) Total energy E_k .(b) Position error norm $\|\Delta \mathbf{x}_k\|$.(c) Holonomic constraint residual ϕ .(d) Orthogonality error e_{orth} .

Figure 6.4: Numerical simulation results for the controlled Acrobot.

In contrast to the unforced case, the applied torque performs work on the system, and the mechanical energy is therefore neither conserved nor expected to remain bounded around its initial value.

Fig. 6.4(b) shows the position error norm $\|\Delta \mathbf{x}_k\|$ between the LGVI trajectory and the RK4 reference. The error is initially zero and remains close to zero over the first 5s, confirming that both methods reproduce the same short-time response to the sinusoidal elbow input. It then increases and oscillates as the trajectories diverge, following the same behaviour observed for the controlled 3D pendulum, because of the numerical differences between the two methods.

The holonomic constraint residuals $\|\phi_{0,k}\|_2$ and $\|\phi_{12,k}\|_2$ are plotted in Fig. 6.4(c). The base and elbow constraint residuals are negligible ($< 10^{-11}$) over the complete simulation. In the forced case, the maximal-coordinate LGVI therefore also prevents constraint drift and keeps the Acrobot links physically connected, which is essential for the later SDP optimization conducted in Section 6.3.

Finally, Fig. 6.4(d) shows the orthogonality error $e_{orth,i,k}$ of both link rotations. As in the unforced simulation, the LGVI preserves the $SO(2)$ structure through the

attitude update (4.83g) and (4.83h). As in the unforced case, the RK4 reference remains close to zero, since its rotations are represented as rotation matrices. Therefore, the controlled Acrobot simulation confirms that both the forcing term and the constrained Lie-group update are numerically consistent.

Overall, Fig. 6.3 and Fig. 6.4 show that the Acrobot LGVI also behaves consistently for a constrained multi-body system, introduced in Section 4.3. The method preserves the $SO(2)$ structure, keeps the holonomic constraints small, and aligns closely with the RK4 reference for the tested timestep.

6.2 3D-Pendulum Optimization Results

This section evaluates the controlled 3D-pendulum polynomial optimization problem (POP) (5.46) introduced in Section 5.3. The POP is converted into a sparse second-order Moment-SOS relaxation using SPOT and is solved with MOSEK [KLY25]. The implementation is documented in [Els26]. All experiments were conducted on the Harvard FASRC Cannon cluster using Intel Xeon Sapphire Rapids CPUs and 512 GB of RAM. Each optimization was executed on a single compute node using 8 CPU cores.

Optimization Configuration The physical parameters are identical to those used for the 3D pendulum in Section 6.1.1. For all optimization runs, the relaxation order, SDP time step, number of time steps, and prediction horizon are

$$\kappa = 2, \quad \Delta t_{\text{SDP}} = 0.1 \text{ s}, \quad N = 20, \quad T = N\Delta t_{\text{SDP}} = 2.0 \text{ s}. \quad (6.19)$$

The initial attitude is $\mathbf{R}^{\text{init}} = \mathbf{R}_x(5^\circ)$ and the initial relative rotation is $\mathbf{F}^{\text{init}} = \mathbf{I}$. Starting from the same initial state, the desired rotation angle is varied as $\theta_{x,\text{des}} \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ\}$ around the x-axis, with $\mathbf{R}_{\text{des}} = \mathbf{R}_x(\theta_{x,\text{des}})$. The terminal desired relative rotation is chosen to be $\mathbf{F}_{\text{des}} = \mathbf{I}$, which represents zero velocity at step N of the 3D-pendulum. The scalar control and relative-rotation limits of the POP are set as

$$u_{\max} = 5, \\ \cos(\Delta\theta_{\max}) = 0.5.$$

The objective parameters of (5.46a) remain constant for all target rotations and are identical to the Acrobot optimization configuration in Table 6.3.

The correlative sparsity pattern is completed with the minimum-degree (MD) chordal-extension heuristic introduced in Section 5.2.2 [KLY25]. For the present problem, the largest clique generated by SPOT contains 24 scalar variables. Consequently, its order-two moment matrix has $\binom{24+2}{2} = 325$ rows and columns. In comparison, the complete 39-variable temporal chain-like clique $\mathbf{z}_P(\mathcal{I}_k^P)$ would produce an 820×820 matrix, while the dense relaxation for $N = 20$ would require a $91,378 \times 91,378$ matrix (see Section 5.3). Therefore, the MD chordal extension substantially reduces the dense or chain-like moment-matrix in this experiment.

Evaluation metrics The angular rotation error at time step k is evaluated from the ordered SDP extraction in degrees as

$$e_{R,k}^P = \frac{180}{\pi} \theta_k^{\text{rad}}, \quad \theta_k^{\text{rad}} = \cos^{-1} \left(\frac{\text{tr}(\mathbf{R}_{\text{des}}^\top \mathbf{R}_k) - 1}{2} \right), \quad (6.20)$$

where θ_k^{rad} is the relative rotation angle in radians and $\mathbf{R}_{\text{des}}^\top \mathbf{R}_k$ is the relative rotation to the target attitude [GQ20]. Thus, $e_{R,k}^P$ is reported in degrees. Thereby, the terminal angular error at N is denoted as $e_{R,N}^P$. In addition, the maximum orthogonality violation over the optimization horizon is defined as

$$e_{\text{orth}}^{P,\max} = \max_{0 \leq k \leq N} \|\mathbf{R}_k^\top \mathbf{R}_k - \mathbf{I}\|_F. \quad (6.21)$$

In order to evaluate the optimality of the lower-bound solution γ_κ^* , a feasible upper bound $p(\hat{\mathbf{z}})$ can be generated through a nonlinear programming (NLP) solver [TJVG25]. Thereby, the corresponding objectives J_{SDP} and J_{NLP} can be calculated, whose suboptimality gap is given by

$$\epsilon = \frac{|J_{\text{NLP}} - J_{\text{SDP}}|}{|J_{\text{NLP}}| + 10^{-6}}. \quad (6.22)$$

Here, the SDP trajectory extraction is used as a warm start for the NLP solver, using the GitHub code of [TJVG25]. The resulting suboptimality gap is denoted by ϵ_{warm} . The numerical rank condition $\delta_{\max}$ is evaluated according to (5.40) over the sub-moment matrices used for the ordered extraction. The runtime given in Table 6.2 is the MOSEK optimizer time.

Table 6.2: Results of the controlled 3D pendulum for desired x -axis rotations.

$\theta_{x,\text{des}}$	0°	45°	90°	135°	180°
Terminal angular error $e_{R,N}^P$	0.0389°	13.735°	19.496°	10.614°	0.1418°
Manifold violation $e_{\text{orth}}^{P,\max}$	1.2×10^{-9}	4.6×10^{-8}	1.1×10^{-11}	2.6×10^{-9}	4.5×10^{-8}
Suboptimality gap ϵ_{warm}	1.8×10^{-6}	8.8×10^{-8}	5.6×10^{-10}	2.4×10^{-9}	2.8×10^{-9}
Rank condition $\delta_{\max}$	3.0×10^{-9}	7.3×10^{-8}	1.1×10^{-11}	3.2×10^{-9}	2.3×10^{-8}
Runtime (s)	3349.46	3706.36	4032.72	4128.29	3873.03

Results Table 6.2 shows that $e_{\text{orth}}^{P,\max} \leq 4.6 \times 10^{-8}$ for all target rotations. Therefore, the extracted rotation matrices satisfy the orthogonality constraints. Moreover, $\epsilon_{\text{warm}} \leq 1.8 \times 10^{-6}$ and $\delta_{\max} \leq 7.3 \times 10^{-8}$. The feasible NLP upper bounds and the SDP lower bounds therefore nearly coincide, while the maximal rank condition $\delta_{\max}$ of the evaluated sub-moment matrices is numerically close to rank one. Consequently, the suboptimality gap ϵ_{warm} and rank metric validate the tightness of the

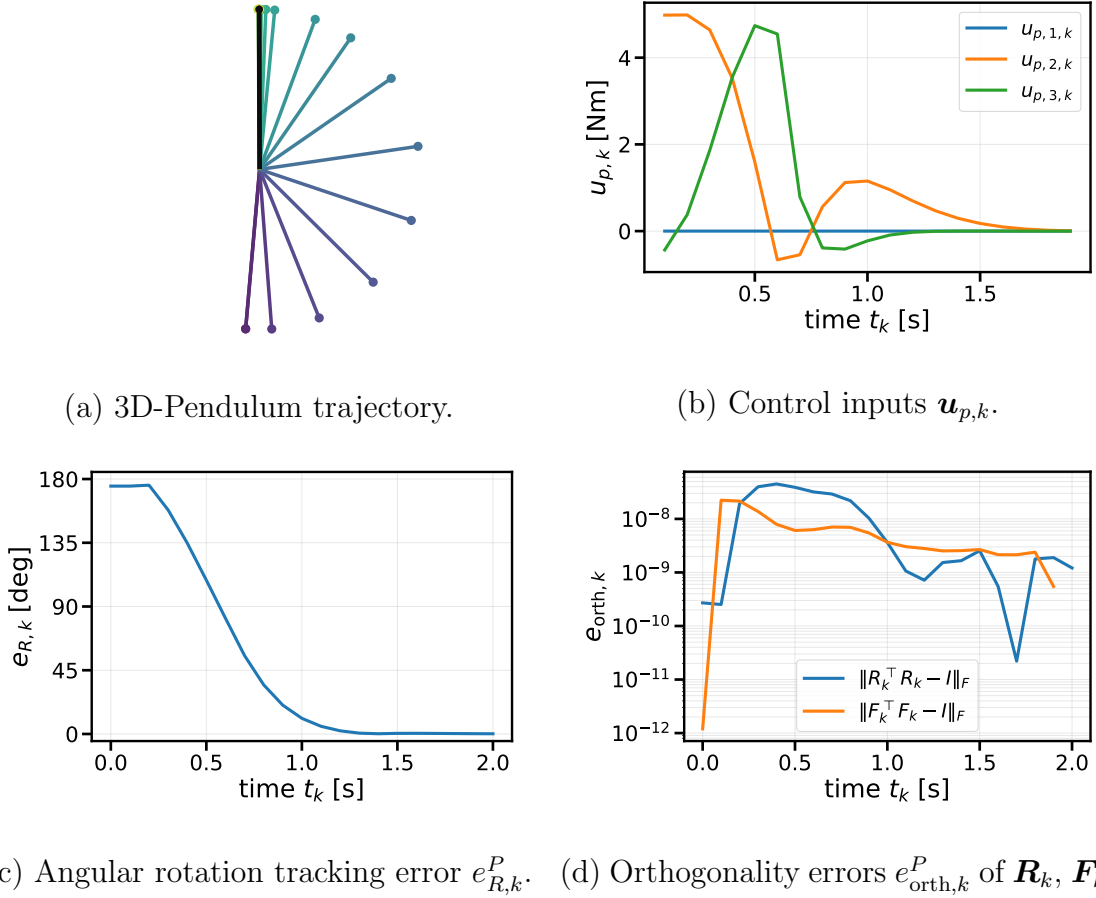


Figure 6.5: SPOT optimization results for the controlled 3D pendulum performing a 180° rotation about the x -axis using the MD sparsity setting: (a) optimization trajectory, (b) control inputs $\mathbf{u}_{p,k}$, (c) angular rotation tracking error $e_{R,k}^P$, and (d) orthogonality errors $e_{\text{orth},k}^P$

second-order MD relaxation for the considered 3D-pendulum test cases, certifying the feasibility of the extracted solutions.

The terminal angular error does not increase monotonically with the target rotation $\theta_{x,\text{des}}$. The smallest errors are obtained for the 0° and 180° targets, with $e_{R,N}^P = 0.0389^\circ$ and $e_{R,N}^P = 0.1418^\circ$. In contrast, the intermediate targets result in errors of 13.735° , 19.496° , and 10.614° for 45° , 90° , and 135° . Consequently, the 90° target produces the largest terminal tracking error. The reason for this behaviour is that the corresponding POP (5.46) does not require the optimal trajectory to reach the desired attitude exactly, as $\mathbf{R}_{\text{des}}$ is not considered as a hard terminal constraint but weighted in the objective. Furthermore, the 0° and 180° attitudes are equilibrium configurations of the considered pendulum for which the gravitational moment vanishes. In contrast, the intermediate attitudes require a control moment to stay at $\mathbf{R}_{\text{des}}$, which contributes to their larger terminal errors.

Fig.6.5 illustrates the ordered extraction for a desired rotation of $\theta_{x,\text{des}} = 180^\circ$. The optimized trajectory follows a smooth path from the initial to the inverted configuration. As the rotation is around the x-axis, $u_{p,1,k}$ remains zero. When reaching the final horizon time at $T = 2$ s, all three control inputs approach zero. Correspondingly, the angular tracking error $e_{R,k}^P$ decreases from approximately 175° and reaches 0.1418° at the terminal time. Finally, the orthogonality errors of both $\mathbf{R}_k$ and $\mathbf{F}_k$ remain less than 10^{-8} , which validates the feasibility of the extracted solution. The four panels therefore show that the extracted trajectory performs the desired rotational motion while remaining on the Lie group.

6.3 Acrobot Optimization Results

This section evaluates the controlled Acrobot polynomial optimization problem (POP) introduced in Section 5.4. As for the 3D pendulum, SPOT constructs a sparse second-order Moment-SOS relaxation, which is solved using MOSEK [KLY25]. First, the semidefinite optimization of Section 5.4.1 is investigated. For the larger desired rotations, its ordered extraction does not provide a certified feasible trajectory over the complete horizon. This motivates the closed-loop SDP-MPC formulation of Section 5.4.2, which applies only the first extracted control before solving the problem again. The complete implementation is documented in [Els26]. All experiments were conducted on the Harvard FASRC Cannon cluster using Intel Xeon Sapphire Rapids CPUs and 512 GB of RAM. Each optimization was executed on a single compute node using 8 CPU cores.

6.3.1 Semidefinite Optimization Results

Optimization configuration The physical parameters are identical to those of the Acrobot simulation in Section 6.1.2. For all optimization runs, the relaxation order, SDP time step, and number of time steps are

$$\kappa = 2, \quad \Delta t_{\text{SDP}} = 0.1 \text{ s}, \quad N = 60, \quad N \Delta t_{\text{SDP}} = 6.0 \text{ s}. \quad (6.23)$$

The complete discretization window from the history node $k = 0$ to the terminal node $k = N$ therefore spans 6.0 s.

Both links start from the same initial attitude and relative rotation,

$$\mathbf{R}_1^{\text{init}} = \mathbf{R}_2^{\text{init}} = \mathbf{R}(5^\circ), \quad \mathbf{F}_1^{\text{init}} = \mathbf{F}_2^{\text{init}} = \mathbf{I}, \quad (6.24)$$

where $\mathbf{R}(\theta) \in SO(2)$ denotes the planar rotation matrix defined in (4.54). Starting from this state, the same desired absolute rotation is assigned to both links

$$\theta_1^{\text{des}} = \theta_2^{\text{des}} = \theta_A^{\text{des}} \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ\}, \quad (6.25)$$

with desired relative rotation $\mathbf{F}_1^{\text{des}} = \mathbf{F}_2^{\text{des}} = \mathbf{I}_2$ after $N = 60$. In particular, $\theta_A^{\text{des}} = 180^\circ$ represents the inverted equilibrium with both links at rest, which is

considered as a final stabilization target for the Acrobot. The physical control and multiplier scales of (5.53), together with the link-specific rotation increment bounds in (5.56i), are chosen as

$$u_{\max} = 15 \text{ N m}, \quad \lambda_{\max} = 40 \text{ N}, \quad \Delta\theta_{1,\max} = 70^\circ, \quad \Delta\theta_{2,\max} = 60^\circ. \quad (6.26)$$

Thus, the polynomial step constraints are $a_{1,k} \geq \cos(70^\circ) \approx 0.3420$ and $a_{2,k} \geq \cos(60^\circ) = 0.5$ for $k = 1, \dots, N-1$. The objective parameters of the cost function (5.56a) are summarized in Table 6.3.

Table 6.3: Objective parameters used for the 3D-pendulum and Acrobot optimization experiments.

w_R	w_F	α_R	α_F	w_u
20	8	15	2	0.09

Two user-defined clique constructions are compared. The first follows the complete chain-like sparsity of (5.57)–(5.58). At $\kappa = 2$, its first clique $\mathbf{z}_A(\mathcal{I}_1^A)$ produces a 351×351 moment matrix, while every subsequent complete clique $\mathbf{z}_A(\mathcal{I}_k^A)$ produces a 253×253 matrix, as given in (5.59). The second construction uses the special multi-body sparsity in (5.60). It keeps the complete cliques $\mathbf{z}_A(\mathcal{I}_1^A)$ and $\mathbf{z}_A(\mathcal{I}_2^A)$, but separates the later coupled dynamics $\mathbf{z}_A(\mathcal{I}_{\text{dyn},k}^A)$ and kinematics $\mathbf{z}_A(\mathcal{I}_{\text{kin},k}^A)$ into moment matrices of size 171×171 and 91×91 (5.61). Consequently, the larger cliques around the first control decision are preserved, whereas the semidefinite blocks over the remaining horizon are reduced.

Evaluation metrics Similar to the 3D-pendulum evaluation (6.20), the terminal attitude error is reported in degrees. The combined angular error of both links is defined as

$$e_{R,k}^A := \frac{180^\circ}{\pi} \left(\sum_{i=1}^2 (\theta_{i,k}^{\text{rad}})^2 \right)^{1/2}, \quad (6.27)$$

$$\theta_{i,k}^{\text{rad}} := \cos^{-1} \left(\frac{\text{tr}((\mathbf{R}_i^{\text{des}})^\top \mathbf{R}_{i,k})}{2} \right),$$

for $i \in \{1, 2\}$. The angular error $e_{R,k}^A$ is the Euclidean combination of the two link-wise angular errors. Thereby, $e_{R,N}^A$ is interpreted as a terminal matrix error at $k = N$. The maximum orthogonality violation is evaluated over the configuration of both links by

$$e_{\text{orth}}^{\text{A,max}} = \max_{\substack{i \in \{1,2\} \\ 0 \leq k \leq N-1}} \|\mathbf{R}_{i,k}^\top \mathbf{R}_{i,k} - \mathbf{I}_2\|_F. \quad (6.28)$$

The relative-rotation errors of $\mathbf{F}_{i,k}$ are smaller than $\mathbf{R}_{i,k}$ for all k . Therefore, the orthogonality of $\mathbf{F}_{i,k}$ is not evaluated.

The maximum numerical rank condition $\delta_{\max}$ is evaluated according to (5.40) [TJVG25]. Additionally, the rank condition of the first moment matrix, which belongs to the clique vector $\mathbf{z}_A(\mathcal{I}_1^A)$, is denoted by

$$\delta_1 := \frac{\nu_{2,1}}{\nu_{1,1}}, \quad \nu_{1,1} \geq \nu_{2,1} \geq 0. \quad (6.29)$$

Here, $\nu_{1,1}$ and $\nu_{2,1}$ are the two largest eigenvalues of the moment matrix associated with $\mathbf{z}_A(\mathcal{I}_1^A)$. The quantity δ_1 is reported in addition to $\delta_{\max}$ because $\mathbf{z}_A(\mathcal{I}_1^A)$ contains the first normalized control $\bar{u}_1$ used later by the subsequent MPC controller.

In contrast to the 3D pendulum, the extracted SDP trajectories for the Acrobot POP with $N = 60$ were not certifiable for all targets. Therefore, cold-start NLP upper bounds were generated instead of using the SDP extraction as a warm start. The reported suboptimality gap ϵ_{cold} is computed according to (6.22) with the cold-start NLP objective. The runtimes in Tables 6.4 and 6.5 are the solver operation times reported by the implementation.

Table 6.4: Results of the Acrobot trajectory optimization using the complete chain-like cliques $\mathbf{z}_A(\mathcal{I}_k^A)$. All trajectory metrics are evaluated from the ordered extraction.

$\theta_1^{\text{des}} = \theta_2^{\text{des}}$	0°	45°	90°	135°	180°
Terminal angular error $e_{R,N}^A$	3.2815°	54.975°	64.629°	36.377°	3.1×10^{-4} °
Manifold violation $e_{\text{orth}}^{A,\max}$	2.2×10^{-8}	1.9×10^{-8}	1.0368	1.3688	1.4053
Suboptimality gap ϵ_{cold}	2.0×10^{-6}	4.8×10^{-10}	0.0717	0.4126	0.5233
Rank condition $\delta_{\max}$	1.4×10^{-8}	1.1×10^{-8}	0.3542	0.4696	0.5596
Rank condition δ_1	3.1×10^{-9}	2.8×10^{-10}	0.003747	0.1020	0.2066
Runtime [s]	2675.54	3581.13	6806.96	5913.05	5421.42

Table 6.5: Results of the Acrobot trajectory optimization using the special multi-body clique construction $\mathbf{z}_A(\mathcal{I}_1^A)$, $\mathbf{z}_A(\mathcal{I}_2^A)$, $\mathbf{z}_A(\mathcal{I}_{\text{dyn},k}^A)$, and $\mathbf{z}_A(\mathcal{I}_{\text{kin},k}^A)$. All metrics are defined as in Table 6.4.

$\theta_1^{\text{des}} = \theta_2^{\text{des}}$	0°	45°	90°	135°	180°
Terminal angular error $e_{R,N}^A$	3.2815°	54.975°	65.192°	36.378°	3.6×10^{-4} °
Manifold violation $e_{\text{orth}}^{A,\max}$	3.0×10^{-12}	3.4×10^{-6}	1.0225	1.4103	1.4561
Suboptimality gap ϵ_{cold}	1.1×10^{-10}	9.2×10^{-9}	0.0778	0.4500	0.5668
Rank condition $\delta_{\max}$	2.1×10^{-12}	2.4×10^{-6}	0.4216	0.4046	0.6603
Rank condition δ_1	2.6×10^{-13}	2.4×10^{-8}	0.0010	0.0973	0.2172
Runtime [s]	904.83	1058.08	1356.14	1524.25	1869.36

Results Tables 6.4 and 6.5 show the same qualitative behavior for both clique constructions. For the 0° and 45° targets, $\delta_{\max}$ and $e_{\text{orth}}^{\text{A,max}}$ remain of order 10^{-6} . Moreover, all corresponding cold-start suboptimality gaps satisfy $\epsilon_{\text{cold}} \leq 2.0 \times 10^{-6}$. Thus, the feasible NLP upper bounds and the SDP lower bounds nearly coincide, while the evaluated moment matrices are numerically close to rank one. The extracted matrices satisfy the $SO(2)$ constraints up to a small tolerance. Together, these metrics provide strong numerical evidence that the second-order relaxations are tight and provide a certificate for these two cases.

The complete chain-like and special multi-body constructions give combined terminal angular errors of 54.975° for the 45° target. This comparatively large terminal error does not contradict the observed tightness because the desired attitudes are included as objective terms and are not imposed as terminal constraints. Similar to the 3D pendulum results in Section 6.2, this target is not an equilibrium configuration of the Acrobot.

For desired rotations of 90° , 135° , and 180° , the definition of chain-like cliques gives terminal angular errors of 64.629° , 36.377° , and $3.1449 \times 10^{-4}^\circ$. The corresponding values for the special multi-body construction are 65.192° , 36.378° , and $3.6388 \times 10^{-4}^\circ$. However, these terminal orientation errors alone do not establish feasibility.

For these three targets, the values of $\delta_{\max}$ lie between 0.3542 and 0.6603, while $e_{\text{orth}}^{\text{A,max}}$ lies between 1.0225 and 1.4561. Consequently, the raw ordered extraction does not represent a feasible $SO(2)$ trajectory over the complete horizon, and the rank-one condition does not provide tightness. In particular, the 180° results illustrate why both metrics must be considered together. Although the projected terminal angular error is below $4 \times 10^{-4}^\circ$, it has the highest suboptimality gap and the largest manifold violation among all target rotations.

The cold-start NLP solutions provide valid upper bounds for these cases. Nevertheless, the suboptimality gaps ϵ_{cold} are 0.0717 and 0.0778 for the 90° target, 0.4126 and 0.4500 for the 135° target, and 0.5233 and 0.5668 for the 180° target for the complete chain-like and special multi-body constructions. These large suboptimality gaps alone do not prove that the relaxation is non-tight as it may also be a result of a suboptimal NLP upper bound [TJVG25]. However, the higher-rank moment matrices and infeasible ordered extractions show that global optimality is not certified for these three cases.

Nevertheless, δ_1 is smaller than $\delta_{\max}$ in every test case. For example, the 90° target gives $\delta_1 = 0.0037$ compared with $\delta_{\max} = 0.3542$ for the chain-like cliques. Moreover, the special multi-body clique construction gives $\delta_1 = 0.0010$ compared with $\delta_{\max} = 0.4216$. Consequently, the first moment block is considerably closer to rank one than $\delta_{\max}$, even if the moment blocks with values between 0.10 and 0.22 for the 135° and 180° targets are generally not rank one.

The special multi-body construction produces terminal, manifold, suboptimality-gap, and rank metrics that are comparable to the complete chain-like construction, while its runtime is approximately 3 to 5 times smaller. Therefore, it is used for the

following closed-loop evaluation in Section 6.3.2.

6.3.2 Model Predictive SDP Control Results

The open-loop results of Section 6.3.1 show that the first-clique rank metric δ_1 is substantially smaller than the worst rank metric $\delta_{\max}$ for all test cases (see Table 6.5). This observation motivates the SDP-MPC formulation of Section 5.4.2. At every MPC iteration, the second-order sparse Moment-SOS relaxation of the Acrobot POP (5.56) is solved, only the first extracted control u_j in (5.67) is applied, and the numerical LGVI dynamics (4.83) are propagated (see Section 6.1.2). The remaining predicted trajectory is ignored and the boundary data are updated from the simulated state before the next SDP is solved. The purpose of the following experiments is to determine whether this repeated first-step extraction can stabilize the Acrobot at the inverted equilibrium of

$$\theta_A^{\text{des}} = 180^\circ, \quad \mathbf{F}_1^{\text{des}} = \mathbf{F}_2^{\text{des}} = \mathbf{I}, \quad (6.30)$$

even when the complete predicted SDP trajectory is not certifiable.

MPC configuration for $u_{\max} = 5 \text{ N m}$ The physical Acrobot parameters are identical to those used for the numerical simulation in Section 6.1.2, and the objective parameters are those in Table 6.3. The second-order relaxation uses only the special multi-body clique construction in (5.60). As shown in the preceding open-loop comparison, this construction preserves the complete first two cliques ($\mathbf{z}_A(\mathcal{I}_1^A)$, $\mathbf{z}_A(\mathcal{I}_2^A)$), including the first applied control, while reducing the sizes of the moment blocks at $k = 3, \dots, N_{\text{SDP}} - 1$ using $\mathbf{z}_A(\mathcal{I}_{\text{dyn},k}^A)$ and $\mathbf{z}_A(\mathcal{I}_{\text{kin},k}^A)$.

Both links start at rest from an absolute rotation of 5°

$$\mathbf{R}_1^{\text{init}} = \mathbf{R}_2^{\text{init}} = \mathbf{R}(5^\circ), \quad \mathbf{F}_1^{\text{init}} = \mathbf{F}_2^{\text{init}} = \mathbf{I}_2. \quad (6.31)$$

The relaxation order, time scales, and horizon are

$$\kappa = 2, \quad \Delta t_{\text{SDP}} = \Delta t_{\text{sim}} = \Delta t_{\text{ctrl}} = 0.1 \text{ s}, \quad N_{\text{SDP}} = 30. \quad (6.32)$$

Consequently, the complete discretization during each SDP iteration spans 3.0 s. The bounds are

$$u_{\max} = 5 \text{ N m}, \quad \lambda_{\max} = 40 \text{ N}, \quad \Delta\theta_{1,\max} = 70^\circ, \quad \Delta\theta_{2,\max} = 60^\circ. \quad (6.33)$$

Therefore, only the torque bound $u_{\max}$ differs from the bounds used for the open-loop experiments in (6.26).

Stabilization is achieved if the simulated state (5.68) is reached. The attitude and relative-step rotation tolerances are $\varepsilon_R = 2^\circ$ and $\varepsilon_F = 2^\circ$. Both conditions must hold for three consecutive MPC updates $n_{\text{stable}} = 3$. No NLP correction or fallback controller is used. The closed-loop experiment is limited to a maximum of $N_{\text{MPC}} = 600$ MPC iterations, which corresponds to a maximum simulated duration of 60 s.

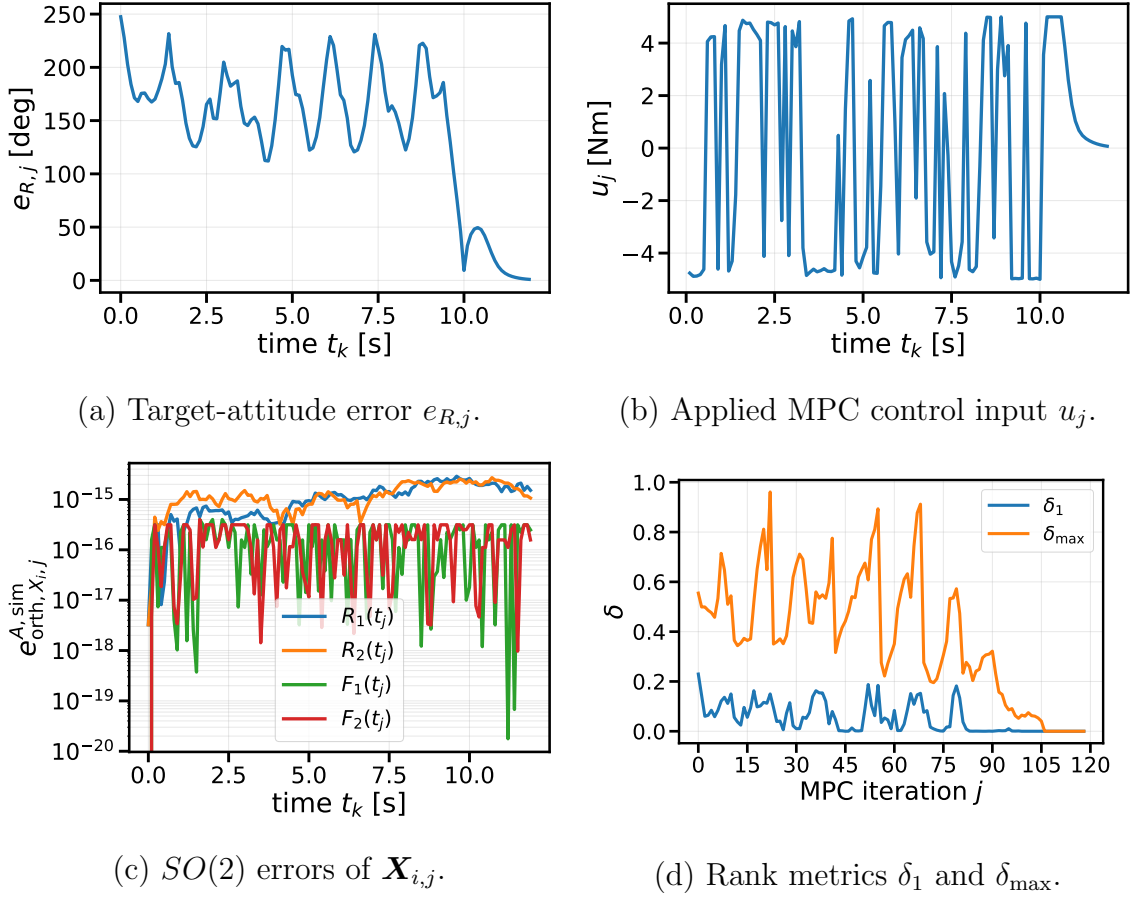


Figure 6.6: Closed-loop SDP-MPC results for the Acrobot with $u_{\max} = 5$ N m.

Results for $u_{\max} = 5$ N m Figure 6.6 shows one successful closed-loop swing-up and achieved stabilization. The combined starting attitude error $e_{R,1}^A$ at $j = 1$ is approximately 247.5° . The error remains large and oscillates during the swing up sequence. It begins to decrease rapidly after approximately 9.5 s and remains close to zero after approximately 11.9 s. Therefore, the controller generates a swing-up motion before stabilizing the Acrobot near the inverted equilibrium (see Fig 6.6(a)).

In Fig. 6.6(b), the control input repeatedly reaches both limits of ± 5 N m during most of the swing-up and decreases toward zero near the target. The torque constraint is therefore active and restricts the available corrective action.

Fig. 6.6(c) illustrates that the orthogonality errors of the simulated $\mathbf{R}_i$ and $\mathbf{F}_i$ remain below 10^{-15} without a growing drift. Therefore, the generated numerical LGVI trajectory remains on $SO(2)$. However, the raw predicted trajectories extracted from the individual SDP problems are not feasible.

The rank metrics in Fig. 6.6(d) gives the corresponding relaxation diagnostic. During a large part of the experiment, δ_1 assumes values up to approximately 0.2, while $\delta_{\max}$ reaches values close to 1. The first moment block is therefore generally

closer to rank one than the worst block over the horizon. Still it is not rank one throughout the full swing-up process. The two metrics are simultaneously close to zero only near stabilization.

Finally, the implemented closed-loop SDP optimization establishes a successful numerical closed-loop stabilization for the considered initial state. Nevertheless, the SDP-predicted trajectories do not provide a global optimality certificate for the applied control sequence.

The 5 N m bound was not sufficient across all additional tested initial configurations. For configurations that require stronger counteracting torques, the controller has limited capability to provide sufficient corrective action. The following evaluation therefore increases the bound to $u_{\max} = 15$ N m and investigates a grid of initial link orientations.

Empirical basin of attraction and time-discretization mismatch For this evaluation, the physical parameters, the relaxation order $\kappa = 2$, the target in (6.30), the objective parameters, the multiplier and step-angle bounds, and the special multi-body cliques remain unchanged. Of the optimization bounds, only the control bound is increased to $u_{\max} = 15$ N m. Three combinations of prediction, simulation, and control time scales are considered, as summarized in Table 6.6.

Table 6.6: Time-scale configurations used for the sampled SDP–MPC stabilization regions with $u_{\max} = 15$ N m.

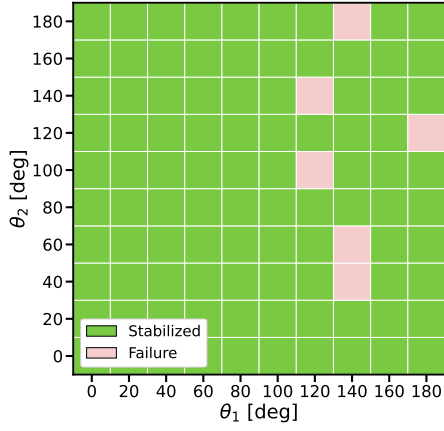
Case	Δt_{SDP}	Δt_{sim}	Δt_{ctrl}	N_{SDP}	Time scales
I	0.1 s	0.1 s	0.1 s	20	matched
II	0.05 s	0.05 s	0.05 s	40	matched
III	0.1 s	0.0025 s	0.025 s	20	mismatched

In both Cases I and II, the prediction, plant, and control intervals are matched. In both cases, the total prediction horizon is $N_{\text{SDP}}\Delta t_{\text{SDP}} = 2.0$ s. Case II therefore evaluates whether a consistent refinement of all time scales changes the sampled stabilization result while retaining approximately the same prediction horizon time. In Case III, the prediction model uses the same 0.1 s step as Case I, but the simulation model uses fine LGVI steps of $\Delta t_{\text{sim}} = 0.0025$ s and simulates for $\Delta t_{\text{ctrl}} = 0.025$ s, which corresponds to 10 LGVI steps per control interval. Therefore, Case III introduces a temporal-discretization mismatch between the SDP prediction and its fine closed-loop execution.

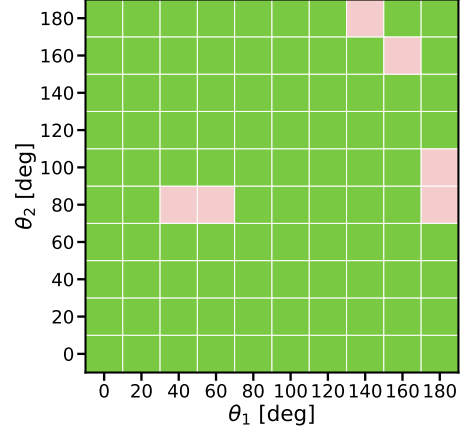
For each case, both links start at rest and their initial absolute orientations are selected from

$$\theta_1^{\text{init}}, \theta_2^{\text{init}} \in \{0^\circ, 20^\circ, \dots, 180^\circ\}. \quad (6.34)$$

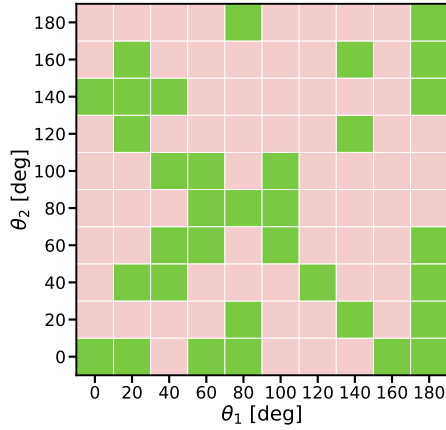
All $10 \times 10 = 100$ combinations are evaluated directly. The same stabilization tolerances $\varepsilon_R = 2^\circ$ and $\varepsilon_F = 2^\circ$ are used in all cases.



(a) Case I.



(b) Case II.

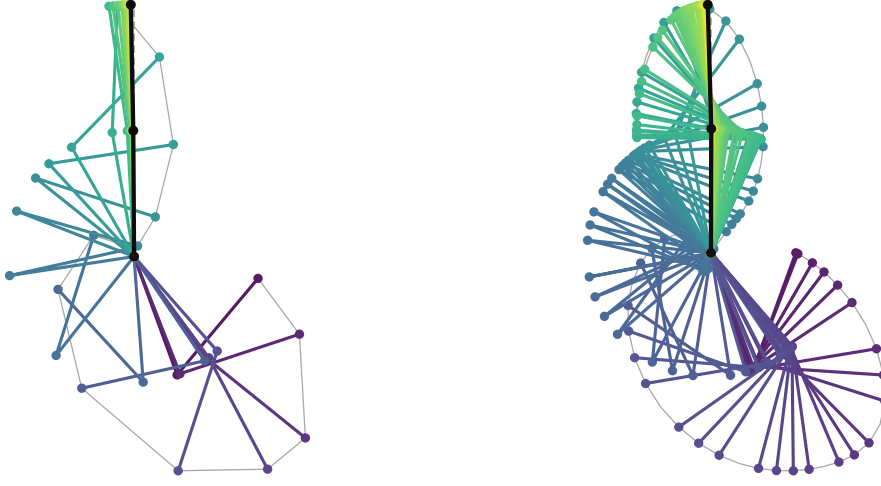


(c) Case III.

Figure 6.7: Empirical sampled stabilization regions of the SDP-MPC controller with $u_{\max} = 15 \text{ N m}$. The tuples $(\Delta t_{\text{SDP}}, \Delta t_{\text{sim}}, \Delta t_{\text{ctrl}})$ are $(0.1, 0.1, 0.1) \text{ s}$ in Case I, $(0.05, 0.05, 0.05) \text{ s}$ in Case II, and $(0.1, 0.0025, 0.025) \text{ s}$ in Case III.

Figures 6.7 (a)-(b) show that the Acrobot model stabilizes 94 of the 100 initial configurations in Case I. The matched time-scale configuration of Case II also contains 94 stabilized samples, although the locations of the six non-stabilized entries changed. Therefore, reducing all three time steps from 0.1 s to 0.05 s does not change the overall stabilization success. Here, the failures occur after the SDP optimization returns an infeasible solution.

In contrast, only 34 of the 100 samples are marked as stabilized in Case III, illustrated in Figure 6.7 (c). The successful samples form a scattered set rather than a contiguous region and occur primarily for easier configurations for $\theta_1 <$



(a) Case I: (0.1, 0.1, 0.1) s.

(b) Case III: (0.1, 0.0025, 0.025) s.

Figure 6.8: Closed-loop Acrobot tip trajectories and sampled link configurations for $(\theta_1^{\text{init}}, \theta_2^{\text{init}}) = (20^\circ, 140^\circ)$ in Cases I and III. Each tuple gives $(\Delta t_{\text{SDP}}, \Delta t_{\text{sim}}, \Delta t_{\text{ctrl}})$.

100° or configurations in which link $i = 1$ is already relatively close to the 180° target. However, both groups are interrupted by failures. Moreover, only five of the thirty samples in the region $120^\circ \leq \theta_1 \leq 160^\circ$ stabilize. The increase in failure states cannot be explained only by the smaller Δt_{sim} , as Case II also refines the prediction, simulation, and control intervals consistently, but without reducing the total number of stabilized samples. Instead, the comparison between Case I-II and Case III indicates that the mismatched time scales in Case III lead to increased failure rates. However, a possible case with $\Delta t_{\text{SDP}} = \Delta t_{\text{sim}} = \Delta t_{\text{ctrl}} = 0.0025$ s could not be evaluated due to the high computational cost of solving the SDP problems with $N_{\text{SDP}} = 800$.

For the successfully stabilized initial configuration $\theta_{1,\text{init}} = \theta_{2,\text{init}} = 80^\circ$, the complete SDP-MPC computation required approximately 3.1 h in Case I, 27.3 h in Case II, and 17.8 h in Case III. The runtime is determined first by Δt_{SDP} and N_{SDP} , which define the size and cost of each SDP problem, and second by the time-discretization mismatch, which can increase the number of MPC iterations required before stabilization.

The trajectory comparison in Fig. 6.8 illustrates the described behaviour for the common initial configuration $(\theta_1^{\text{init}}, \theta_2^{\text{init}}) = (20^\circ, 140^\circ)$. With matched 0.1 s time scales, the Acrobot follows a comparatively direct swing-up and satisfies the stabilization criterion after approximately 2.2 s. The mismatched Case III also reaches the inverted equilibrium, but requires approximately 4.1 s. Its link-tip paths contain considerably larger loops, and the repeated passages around the upright configuration indicate a more oscillatory transient. Thus, receding-horizon feedback can

compensate for the temporal mismatch in this particular case, but only at the cost of a slower and less direct closed-loop maneuver.

Summary For the evaluated initial states, the SDP-MPC controller can convert first-step extractions into on-manifold simulated swing-ups, although the complete-horizon moment relaxations are not generally numerically rank one. With $u_{\max} = 5 \text{ N m}$, stabilization is obtained for the symmetric 5° initial state, but the limited control and unsuccessful additional trials reveal limited control authority. Increasing the bound to 15 N m yields broad sampled stabilization regions for matched prediction and execution time scales, whereas the mismatched configuration stabilizes only about one third of the tested initial states. Although the SDP relaxations do not provide a certificate of global optimality, the results show that embedding the SDP optimization in a closed loop with numerical simulation can generate feasible stabilizing trajectories.

Chapter 7

Discussion

Chapter 6 examined whether the proposed LGVI-SDP framework preserves the geometric structure of the modeled rigid-body systems, produces certifiable open-loop trajectories, and can generate feasible trajectories for non-tight relaxations in a closed-loop implementation. The results provide a qualified answer to the research question in Section 1.1. The framework is suitable for Lagrangian variational integrators of single- and multibody systems and can certify particular discrete trajectory-optimization instances. Still tightness, feasible extraction, and computational cost depend strongly on the considered dynamics and remain challenging. In the following, the main findings are discussed and related to the state of the art. Finally, potential applications are elaborated.

7.1 Discussion of Main Findings

The simulation results in Section 6.1 establish that the derived discrete models of the 3D pendulum on $SO(3)$ and the underactuated Acrobot $(SO(2) \times \mathbb{R}^2)^2$ represent the intended dynamics while exhibiting energy-conservation properties over a long time horizon. For the unforced 3D pendulum, Figure 6.1 shows that the LGVI remains on $SO(3)$, preserves the momentum and exhibits bounded long-term energy error. In comparison, a standard RK4 integrator has errors in all three quantities, whereas projected RK4 restores the rotation constraint but accelerates energy drift. The constrained Acrobot results in Figures 6.3 and 6.4 extend this observation to a multibody system, where both link rotations remain on $SO(2)$, the holonomic link constraints are satisfied, and the unforced energy error remains bounded. These results demonstrate that the derived discrete Lagrangian variational integrators of the 3D-pendulum (4.46) and the Acrobot (4.83) preserve the relevant configuration and constraint geometry over the simulated horizon.

Single-body optimization The 3D pendulum demonstrates the complete method for a single rigid body. Its LGVI can be written as the quadratic POP in (5.46), but

a dense second-order relaxation would already require a $91,378 \times 91,378$ moment matrix for $N = 20$. The minimum-degree chordal extension generated by SPOT in Section 5.2.2 reduces the largest matrix to 325×325 , compared with 820×820 for the complete temporal clique. Therefore, the Moment-SOS relaxation can convert a nonconvex algebraic LGVI formulation into a sparse convex optimization problem. Still, the MD sparsity construction is required for computational application.

For all five targets $\theta_{x,des} = \{0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ\}$, Table 6.2 reports a rank metric below 7.3×10^{-8} , a manifold violation below 4.6×10^{-8} , and a suboptimality gap below 1.8×10^{-6} . Additionally, the solution extractions remain feasible. Therefore, these values indicate that the second-order relaxations are numerically tight and certify global optimality for the considered discrete POP. Figure 6.5 illustrates the $\theta_{x,des} = 180^\circ$ case, where the extracted trajectory reaches the desired attitude while remaining on $SO(3)$. Consequently, the single-body problem of a 3D pendulum on $SO(3)$ is globally certifiable using discrete LGVI dynamics and sparse SDP relaxations.

Still, it has to be noted that the certified solutions do not necessarily reach the desired states after exactly N steps, while still being certifiable. For example, the $\theta_{x,des} = 90^\circ$ case reaches a terminal state of $\theta_{x,final} \approx 70.5^\circ$, since the implemented target is not a hard constraint but a cost inside the objective function. Therefore, the certified solution is optimal for the discrete objective, but not for reaching the exact terminal state.

Multibody formulation and sparsity The Acrobot extends the framework to a constrained and underactuated multibody system. Its maximal-coordinate LGVI represents both links separately and imposes the joints through holonomic constraints and multipliers. This enables a quadratic polynomial structure. Moreover, the reduction introduced in Section 5.4 eliminates the position variables and expresses the velocities through rotation variables. Thereby, the size of the decision vector, which has to be optimized, decreases from $17N + 3$ to $13N - 1$ variables while preserving quadratic polynomial dynamics. It therefore reduces the SDP without introducing densely coupled dynamical representations when using minimal coordinates [BM21a].

Nevertheless, even after the reduction, the dense second-order moment matrix has the dimension $304,590 \times 304,590$ for $N = 60$. The complete chain-like sparsity construction (5.57)- (5.58) replaces the dense matrix with separate blocks of sizes 351×351 and 253×253 . The multibody-specific construction in (5.60) keeps the complete cliques at $k = 1, 2$, but separates later dynamics and kinematics into 171×171 and 91×91 matrices. The results of the SDP relaxation, shown in Tables 6.4 and 6.5, identify similar relaxation metrics for both structures. Still, the specialized cliques reduce the MOSEK runtime by a factor of approximately three to five. Consequently, the identification of problem-specific sparsity is necessary for solving multi-body systems using semidefinite programming to obtain fast and numerically reliable results [KLY25].

However, it is important to note that multibody-specific sparsity does not guarantee tighter relaxations than chain-like sparsity. Therefore, before applying the proposed framework to new multibody systems, it is necessary to evaluate the tightness of the relaxation for the specific problem.

Open-loop Acrobot optimization The open-loop SDP evaluation in Section 6.3.1 separates easy from difficult Acrobot problems. For the targets $\theta_A^{\text{des}} = 0^\circ$ and $\theta_A^{\text{des}} = 45^\circ$, Tables 6.4 and 6.5 report rank-one moment matrices, feasible extractions, and negligible suboptimality gaps. Therefore, in these cases, the solutions are certified. However, similar to the 3D-Pendulum, the certified 45° solution retains a terminal error of approximately 55° , because the target is only weighted in the objective.

For the targets $\theta_A^{\text{des}} = 90^\circ$, $\theta_A^{\text{des}} = 135^\circ$, and $\theta_A^{\text{des}} = 180^\circ$, the maximum rank metric increases to approximately 0.35-0.66, while the extracted rotations have manifold violations around 1.0-1.4. Therefore, the first moments do not define feasible $\text{SO}(2)$ trajectories and no certificate is obtained. The results indicate that the additional complexity of the multibody underactuated system increases the difficulty of the convex optimization problem.

For the runtime, the optimization using specialized multibody cliques (5.60) requires approximately 15-31 minutes, compared with 45-113 minutes for the chain-like cliques (5.57)- (5.58). Therefore, this demonstrates that the size of the moment matrices is the driving factor for the runtime of the SDP solver. Still, both optimization variants remain too slow for online control. A lower relaxation order would reduce matrix sizes but weaken the tightness of the relaxation. Moreover, instead of using the second order interior-point solver MOSEK [MOS26], a first-order SDP solver, for example CuADMM [XKS⁺26], could accelerate the same relaxation at the possible cost of less accurate certified solutions. Additionally, in this thesis, only second order $\kappa = 2$ relaxations are considered, as [TJVG25] states that second order relaxations are sufficient for most of the considered problems. In general, higher order relaxations can be used to tighten the relaxation (5.24), but at the cost of increased computational complexity.

Closed-loop Acrobot stabilization Tables 6.4 and 6.5 show that δ_1 is smaller than the worst-horizon metric δ_{max} inside the open-loop optimization. This motivates the receding-horizon implementation in Section 5.4.2, where only the first control from (5.67) is applied. The remaining non-tight prediction is discarded, and the SDP is solved again from the simulated state. However, δ_1 is not a rank-one condition and therefore not tight. Still, the implemented closed-loop control has a feedback structure, which can stabilize the system even if the open-loop solution is not tight (see Section 6.3.2).

Figure 6.6 illustrates that this procedure stabilizes the system starting from an initial rotation of 5° , after approximately 11.9s with $u_{\text{max}} = 5 \text{ N m}$. The simulated

trajectory remains on $\text{SO}(2)$ although the predicted blocks are not rank one throughout the whole maneuver. This demonstrates numerical closed-loop feasibility for the tested configuration. However, the solution cannot be considered globally optimal, since the optimization problem is not tight. Additionally, failures are also consistent with limited control torque availability for $u_{\max} = 5 \text{ N m}$.

With $u_{\max} = 15 \text{ N m}$, Figure 6.7 shows that the matched discretization ($\Delta t_{\text{SDP}} = \Delta t_{\text{sim}} = \Delta t_{\text{ctrl}}$) Cases I and II stabilize 94 of 100 initial configurations. Therefore, consistently refining all Δt_{SDP} and Δt_{sim} intervals from 0.1 s to 0.05 s does not change the success rate. Nevertheless, the different failed trials for the same initial conditions between Cases I and II indicate that the success is not guaranteed for all $\Delta t_{\text{SDP}} = \Delta t_{\text{sim}} = \Delta t_{\text{ctrl}}$ configurations.

In Case III, where there is a discretization mismatch between the SDP and simulation time steps ($\Delta t_{\text{SDP}} = 0.1 \text{ s}$, $\Delta t_{\text{sim}} = 0.0025 \text{ s}$, $\Delta t_{\text{ctrl}} = 0.025 \text{ s}$), the controller stabilizes only 34 of 100 configurations. Consequently, Figure 6.8(b) shows a more oscillatory behaviour near the target configuration before stabilization, compared to Case I in Figure 6.8(a) for the same initial condition of $(\theta_1^{\text{init}}, \theta_2^{\text{init}}) = (20^\circ, 140^\circ)$. Here, Case III requires 4.1 s instead of 2.2 s in Case I. The results indicate that mismatched discretization can lead to reduced reliability for stabilization of the Acrobot.

For the successful $(\theta_1^{\text{init}}, \theta_2^{\text{init}}) = (80^\circ, 80^\circ)$ example, Cases I, II, and III require approximately 3.1 h, 27.3 h, and 17.8 h. These illustrative runtimes depend on Δt_{SDP} and N_{SDP} , which determine each SDP size, and on the number of replanning iterations. A matched discretization case with $\Delta t_{\text{SDP}} = \Delta t_{\text{sim}} = \Delta t_{\text{ctrl}} = 0.0025 \text{ s}$ would require $N_{\text{SDP}} = 800$, which is not computationally feasible with current hardware. Consequently, the closed-loop implementation is only considered as an offline application instead of a real-time controller. However, the open-loop optimization using semidefinite relaxation is also not real-time capable, since the SDP solver requires approximately 15-31 minutes for the considered $N_{\text{SDP}} = 60$ problem.

7.2 Comparison with the State of the Art

The first contribution of this thesis is the Acrobot-specific LGVI in maximal coordinates (4.83), including its underactuation (4.87) and holonomic joint constraints. This formulation aligns with the preservation properties of [MW01, Lee08] and the maximal-coordinate formulations in [BM21b, BM21a]. Moreover, the reduction in Section 5.4 resulted in a smaller decision vector $\mathbf{z}$ and a quadratic polynomial model suitable for SDP relaxation (5.56).

The closest framework to this thesis is [TJVG25]. In that work, all reported trajectories are passed through a local-search step using IPOPT, including cases for which the SDP relaxation provides a small optimality gap or satisfies the rank condition. In contrast, the second-order relaxations of all considered 3D-pendulum problems in this thesis are numerically rank one and directly yield feasible trajectories

without subsequent NLP refinement. The agreement between the SDP lower bound and the objective of the extracted trajectory therefore certifies global optimality using the SDP solution alone. This demonstrates that, for the investigated single-body benchmark, the complete pipeline from the nonconvex LGVI formulation to a feasible and globally optimal trajectory can be performed without depending on the initialization or convergence of a local solver, in contrast to [TJVG25]. Nevertheless, the drone problems in [TJVG25] include translational dynamics and obstacle avoidance and are therefore more complex than the present pendulum problem. Whether similarly direct SDP extraction remains possible after introducing obstacle constraints for the 3D-pendulum has to be evaluated in future research. With respect to the research gap identified in Section 1.3, this thesis demonstrates that the complete LGVI-SDP framework can be transferred from a single rigid body to a constrained and underactuated multibody system of increased dynamical complexity. The results confirm that globally optimal trajectories can be certified when the relaxation is tight and that closed-loop feedback controls can still be extracted for some non-tight problems, for example the Acrobot stabilization. However, the transfer is not equally reliable for all configurations, since relaxation tightness, solver feasibility, and computational cost remain strongly problem-dependent. In particular, the closed-loop trajectories do not provide certificates of global optimality, and their computation is too slow for real-time application. The research gap is therefore addressed for the investigated single- and multibody models, but remains open for larger systems, contact and obstacle constraints, uncertain dynamics, and hardware experiments.

7.3 Potential Applications

The most direct application is offline trajectory generation. Tight SDP solutions can provide certified benchmarks for evaluating local optimizers, as initial guesses for related NLP problems, or as reference trajectories for online tracking controllers.

For non-tight problems, the SDP-MPC implementation can still generate feasible realized trajectories. The resulting state-control sequences could be stored as demonstrations for faster online MPC methods or used to construct warm starts from previously solved configurations. This is consistent with [KXS⁺26], where data-based warm starts substantially reduce repeated SDP solution times.

The trajectories could also provide training data for imitation learning or offline reinforcement learning. This is particularly relevant for underactuated systems such as the Acrobot, where swing-up requires coordinated actions over a longer time horizon and can be difficult to explore [DBD23]. A learned controller could approximate the SDP closed-loop policy while operating at lower online cost.

Chapter 8

Conclusion

This thesis analyzed the combination of Lie group variational integrators (LGVIs) with sparse Moment-SOS semidefinite relaxations for multi rigid-body trajectory optimization.

Firstly, Chapter 1 formulated the general optimization problem and identified following research gap: Although semidefinite relaxations provide globally optimal solutions for some low-dimensional rigid-body systems, their performance has not been analyzed for multibody systems of varying complexity. Thereby, the accuracy of lower bounds, relaxation tightness, and feasible extraction become problem-dependent as the dynamical complexity increases. This question is examined for a single rigid body and a constrained, underactuated multibody system.

In order to derive LGVIs, Chapter 2 briefly introduced the matrix Lie groups, Lie algebras, and local group variations required for the following derivations.

Chapter 3 established the continuous mechanical foundation. The Euler-Lagrange equations on a matrix Lie group are derived from Hamilton's principle using left-trivialized variations (3.8). Additionally, the Legendre transformation is used to express the dynamics in Hamiltonian form (3.12). This framework is applied to the unforced 3D pendulum on $SO(3)$, for which the continuous equations are derived (3.29). To validate the LGVI properties of the 3D pendulum, the momentum conservation associated with rotations about gravity is verified (3.38) using Noether's theorem.

Afterwards, Chapter 4 transferred the continuous Lagrangian variational construction to discrete time. Instead of discretizing the continuous equations of motion directly, the action integral is replaced by a discrete action sum (4.3). This yields the discrete Euler-Lagrange (4.9) and Hamiltonian equations (4.15), discrete Legendre transformations (4.12), and their forced extension (4.21). A computational approach is introduced to solve the implicit discrete equations of motion. Thereby, multiplicative kinematic updates preserved the Lie-group structure, while a Cayley parametrization and Newton iteration solved the implicit rotation update. In correspondence with the continuous case, the discrete unforced and controlled 3D-pendulum LGVIs are derived (4.41). This single rigid-body formulation is extended

to a multi-body framework for deriving LGVIs for coupled rigid bodies. Therefore, the Acrobot LGVI using maximal coordinates is derived as an example (4.83). Its two center-of-mass positions and $SO(2)$ attitudes are coupled by holonomic constraints and Lagrange multipliers.

To optimize these models trajectories, Chapter 5 transforms these discrete dynamical models into finite-horizon polynomial optimization problems (POPs). However, as these equations are nonconvex, semidefinite programming is introduced to convert them into convex optimization problems. Accordingly, the Moment-SOS hierarchy and sparse relaxations are introduced. Afterwards, the POP of the 3D-pendulum is defined on $SO(3)$ (5.46). For the Acrobot, the derived LGVI is reduced to a smaller variable size optimization problem. Thereby, the holonomic constraints are used to eliminate the position and velocity variables by expressing them through rotations, reducing the decision vector from $17N + 3$ to $13N - 1$ variables. Still, the quadratic structure of the dynamics is preserved. The resulting POP is defined in (5.56). In addition, chain-like cliques (5.58) are compared with a multibody-specific clique construction (5.60) to separate moment matrices for computational efficiency. Finally, the reduced Acrobot POP is embedded in an SDP-based model predictive control (MPC) loop that iteratively applied only the first extracted control after each SDP optimization.

Consequently, Chapter 6 evaluated the resulting models and optimization problems. Firstly, a numerical simulation of the 3D-pendulum LGVI and the Acrobot LGVI is implemented to verify the derived equations of motion. As a result, the unforced 3D-pendulum simulation remained on $SO(3)$, preserved the momentum, and conserved the energy over a long time horizon. Similarly, the Acrobot simulations preserved both $SO(2)$ attitudes and the base and elbow constraints. Secondly, the POPs are solved using SDP relaxations and evaluated for feasibility and optimality. All five evaluated 3D-pendulum relaxations are numerically rank one, gave feasible extractions, and are certified for their discrete POPs (see Table 6.2). For the 3D-pendulum, minimum degree chordal extension is used for the clique construction. Moreover, the Acrobot 0° and 45° targets are also certified using chain-like cliques and multibody-specific clique constructions. In contrast, the 90° , 135° , and 180° semidefinite relaxation cases produce higher-rank moment matrices and infeasible extractions (see Table 6.4-6.5). Therefore, no feasible solutions are obtained for these cases. Nevertheless, the closed-loop MPC implementation with repeated first-control extraction generated feasible simulated Acrobot swing-ups. With the 15 Nm bound, the two matched time-scale cases each stabilized 94 of 100 sampled states, compared with 34 of 100 for the mismatched case (see Fig.6.7).

Finally, Chapter 7 interpreted these findings in relation to the research question and the state of the art literature. Additionally, possible applications of optimization and learning-based methods are identified.

In relation to the research question, the results support the following analysis. The LGVI dynamics of both considered systems can be represented by low-degree polynomial constraints while retaining their Lie-group structure. The resulting

POPs can be solved by sparse Moment-SOS relaxations, which can provide certified global optima for some low-dimensional rigid-body problems. Here, sparse second-order Moment-SOS relaxation recovered feasible trajectories that are certified as globally optimal for the tight 3D-pendulum problems and the two simpler Acrobot optimization problems. However, the results for the Acrobot cases with $\theta_A^{\text{des}} \geq 90^\circ$ prevent this result from being generalized to complex multibody motions. Also, sparsity reduces the SDP computation time, but does not guarantee tightness or feasible extraction. The closed-loop SDP-MPC approach produces feasible trajectories even for complex optimization problems, for example the stabilization of an underactuated Acrobot, when the prediction is not rank one. However, it did not globally certify these solutions.

Limitations This thesis does not consider contact, collision avoidance, or obstacles. Additionally, hardware experiments are not performed.

Computational scaling is the strongest practical limitation because the moment matrix dimension grows with the variables in each clique. Without sparsity, the reported problems would require matrices of size $91,378 \times 91,378$ and $304,590 \times 304,590$. Even with sparsity, MOSEK required about 56 to 69 min for one 3D-pendulum problem. Also it requires 15 to 31 min for the special-multibody cliques, and 45 to 113 min for the complete chain-like cliques on eight CPU cores for the Acrobot. The successful initial $(80^\circ, 80^\circ)$ Acrobot MPC computations required 3.1, 27.3, and 17.8 hours in Cases I-III. The implementation is therefore an offline study. In general, larger time horizons, systems, or coupled constraints increase the computational expense significantly.

Future Work The results of this thesis suggest several directions for future work. Firstly, the POP can be reduced further before constructing the SDP relaxation. Especially, a unit-quaternion formulation should be compared with the rotation-matrix formulation used in this thesis. For $\text{SO}(3)$, replacing $\mathbf{R}_k$ and $\mathbf{F}_k$ by a corresponding unit quaternion reduces the number of scalar attitude variables from nine to four at each step [BM21b]. Therefore, the combined representation of $\mathbf{R}_k$ and $\mathbf{F}_k$ is reduced from 18 to 8 scalar variables per transition. In contrast, the $\text{SO}(2)$ representation used in this thesis already requires only the two variables c_k and s_k . Therefore, it must be analyzed whether the quaternion formulation for $\text{SO}(3)$ produces smaller sparse moment blocks while maintaining tight relaxations.

Secondly, the established framework should be transferred to additional multibody benchmarks. A cart-pole system provides an underactuated problem containing translational and rotational motion. Its LGVI and reduced POP could be derived, and its temporal and problem-specific sparsity could be identified. Additionally, an SDP-MPC closed-loop method could then be applied to determine whether the first-control strategy remains effective for various cases. Various multibody systems with different numbers of links, actuation, and constraints should be evaluated to determine whether the SDP-MPC approach can be generalized. More-

over, obstacles and contact constraints can be included to evaluate the performance of the SDP-MPC approach in more complex scenarios, and the approach can be evaluated on real hardware.

Another open issue concerns the computational backend. The second-order Moment-SOS relaxations in this thesis are solved using MOSEK, a robust and accurate interior-point conic solver. MOSEK could be compared with the GPU-accelerated first-order SDP solver CuADMM introduced in [KXS⁺26]. Here, first-order refers to the solver algorithm and not to the relaxation order. CuADMM can also solve the second-order relaxation used in this thesis. The results of [KXS⁺26] indicate that CuADMM scales more favorably with large sparse SDPs and can benefit substantially from warm starting, although it is not necessarily faster than MOSEK for small cold-started problems. Therefore, it could reduce the computational cost of longer horizons and make larger cliques or higher relaxation orders more accessible. However, first-order solvers generally achieve lower numerical accuracy than interior-point methods. Also for the SDP-MPC implementation, CuADMM could be warm-started using the shifted solution of the preceding MPC problem or previously computed solutions for nearby initial states.

Finally, future work could examine adaptive and iterative extraction. Rank metrics could identify locally tight moment blocks, after which only variables supported by verified rank-one blocks would be fixed and the remaining relaxation solved again. Clique merging or a higher relaxation order could be applied selectively where the rank condition fails. This procedure can improve constructing a feasible full-horizon trajectory.

Appendix A

Scalar Implementation of POPs

A.1 Scalar Implementation of the 3D-Pendulum Optimization Problem

This appendix expands the matrix formulation (5.46) into the scalar polynomial equations used to construct the semidefinite relaxation in [Els26]. Here, $r_{ij,k}$ and $f_{ij,k}$ denote the matrix components in row i and column j of $\mathbf{R}_k$ and $\mathbf{F}_k$, respectively, for $i, j = 1, 2, 3$. Let

$$\mathbf{J}_d = \text{diag}(j_{d,1}, j_{d,2}, j_{d,3}), \quad (\text{A.1})$$

and define

$$\begin{aligned} \boldsymbol{\rho}_c &= [\rho_1 \quad \rho_2 \quad \rho_3]^\top, & \bar{\mathbf{u}}_{p,k} &= [\bar{u}_{p1,k} \quad \bar{u}_{p2,k} \quad \bar{u}_{p3,k}]^\top, \\ \mathbf{u}_{p,k} &= u_{\max} \bar{\mathbf{u}}_{p,k}. \end{aligned} \quad (\text{A.2})$$

Objective Writing the components of $\mathbf{R}_N^{\text{des}}$ as $r_{ij,N}^{\text{des}}$ and the components of $\mathbf{F}^{\text{des}}$ as f_{ij}^{des} , the scalar objective polynomial of (5.46a) is

$$\begin{aligned} p_P &= w_R \sum_{i=1}^3 \sum_{j=1}^3 (r_{ij,N} - r_{ij,N}^{\text{des}})^2 + w_F \sum_{i=1}^3 \sum_{j=1}^3 (f_{ij,N-1} - f_{ij}^{\text{des}})^2 \\ &\quad + \sum_{k=0}^{N-1} \alpha_R \sum_{i=1}^3 \sum_{j=1}^3 (r_{ij,k} - r_{ij,N}^{\text{des}})^2 \\ &\quad + \sum_{k=0}^{N-2} \alpha_F \sum_{i=1}^3 \sum_{j=1}^3 (f_{ij,k} - f_{ij}^{\text{des}})^2 \\ &\quad + \sum_{k=1}^{N-1} w_u^{-1} (\bar{u}_{p1,k}^2 + \bar{u}_{p2,k}^2 + \bar{u}_{p3,k}^2). \end{aligned} \quad (\text{A.3})$$

Discrete Rotational Dynamics Since $\mathbf{R}_k^\top \mathbf{e}_3 = [r_{31,k} \ r_{32,k} \ r_{33,k}]^\top$, the gravity moment

$$\boldsymbol{\tau}_{g,k} = [\tau_{g1,k} \ \tau_{g2,k} \ \tau_{g3,k}]^\top,$$

equals

$$\tau_{g1,k} = mg (\rho_2 r_{33,k} - \rho_3 r_{32,k}), \quad (\text{A.4a})$$

$$\tau_{g2,k} = mg (\rho_3 r_{31,k} - \rho_1 r_{33,k}), \quad (\text{A.4b})$$

$$\tau_{g3,k} = mg (\rho_1 r_{32,k} - \rho_2 r_{31,k}). \quad (\text{A.4c})$$

Using the definition $\mathbf{u}_k = \mathbf{R}_k^\top \mathbf{e}_3 \times \mathbf{u}_{p,k}$ (4.44) with $\mathbf{u}_{p,k} = u_{\max} \bar{\mathbf{u}}_{p,k}$ (5.42), the three control components are

$$u_{1,k} = u_{\max} (r_{32,k} \bar{u}_{p3,k} - r_{33,k} \bar{u}_{p2,k}), \quad (\text{A.5a})$$

$$u_{2,k} = u_{\max} (r_{33,k} \bar{u}_{p1,k} - r_{31,k} \bar{u}_{p3,k}), \quad (\text{A.5b})$$

$$u_{3,k} = u_{\max} (r_{31,k} \bar{u}_{p2,k} - r_{32,k} \bar{u}_{p1,k}). \quad (\text{A.5c})$$

After multiplying (5.46b) by h , the three independent entries of the skew-symmetric dynamics constraint are

$$j_{d,2} f_{32,k+1} - j_{d,3} f_{23,k+1} - j_{d,3} f_{32,k} + j_{d,2} f_{23,k} - h^2 (\tau_{g1,k+1} + u_{1,k+1}) = 0, \quad (\text{A.6a})$$

$$j_{d,3} f_{13,k+1} - j_{d,1} f_{31,k+1} - j_{d,1} f_{13,k} + j_{d,3} f_{31,k} - h^2 (\tau_{g2,k+1} + u_{2,k+1}) = 0, \quad (\text{A.6b})$$

$$j_{d,1} f_{21,k+1} - j_{d,2} f_{12,k+1} - j_{d,2} f_{21,k} + j_{d,1} f_{12,k} - h^2 (\tau_{g3,k+1} + u_{3,k+1}) = 0. \quad (\text{A.6c})$$

for $k = 0, \dots, N-2$.

Discrete Kinematics The nine scalar equations (5.46c) corresponding to $\mathbf{R}_{k+1} = \mathbf{R}_k \mathbf{F}_k$ are

$$r_{11,k+1} - r_{11,k} f_{11,k} - r_{12,k} f_{21,k} - r_{13,k} f_{31,k} = 0, \quad (\text{A.7a})$$

$$r_{12,k+1} - r_{11,k} f_{12,k} - r_{12,k} f_{22,k} - r_{13,k} f_{32,k} = 0, \quad (\text{A.7b})$$

$$r_{13,k+1} - r_{11,k} f_{13,k} - r_{12,k} f_{23,k} - r_{13,k} f_{33,k} = 0, \quad (\text{A.7c})$$

$$r_{21,k+1} - r_{21,k} f_{11,k} - r_{22,k} f_{21,k} - r_{23,k} f_{31,k} = 0, \quad (\text{A.7d})$$

$$r_{22,k+1} - r_{21,k} f_{12,k} - r_{22,k} f_{22,k} - r_{23,k} f_{32,k} = 0, \quad (\text{A.7e})$$

$$r_{23,k+1} - r_{21,k} f_{13,k} - r_{22,k} f_{23,k} - r_{23,k} f_{33,k} = 0, \quad (\text{A.7f})$$

$$r_{31,k+1} - r_{31,k} f_{11,k} - r_{32,k} f_{21,k} - r_{33,k} f_{31,k} = 0, \quad (\text{A.7g})$$

$$r_{32,k+1} - r_{31,k} f_{12,k} - r_{32,k} f_{22,k} - r_{33,k} f_{32,k} = 0, \quad (\text{A.7h})$$

$$r_{33,k+1} - r_{31,k} f_{13,k} - r_{32,k} f_{23,k} - r_{33,k} f_{33,k} = 0. \quad (\text{A.7i})$$

for $k = 0, \dots, N-1$.

Discrete Rotation Constraints For every $k = 0, \dots, N$, the six independent scalar equations (5.46d) in $\mathbf{R}_k^\top \mathbf{R}_k = \mathbf{I}_3$ are

$$r_{11,k}^2 + r_{21,k}^2 + r_{31,k}^2 - 1 = 0, \quad (\text{A.8a})$$

$$r_{12,k}^2 + r_{22,k}^2 + r_{32,k}^2 - 1 = 0, \quad (\text{A.8b})$$

$$r_{13,k}^2 + r_{23,k}^2 + r_{33,k}^2 - 1 = 0, \quad (\text{A.8c})$$

$$r_{11,k}r_{12,k} + r_{21,k}r_{22,k} + r_{31,k}r_{32,k} = 0, \quad (\text{A.8d})$$

$$r_{12,k}r_{13,k} + r_{22,k}r_{23,k} + r_{32,k}r_{33,k} = 0, \quad (\text{A.8e})$$

$$r_{11,k}r_{13,k} + r_{21,k}r_{23,k} + r_{31,k}r_{33,k} = 0. \quad (\text{A.8f})$$

The remaining nine equations of (5.46d) for $\mathbf{R}_k$ are

$$r_{21,k}r_{32,k} - r_{31,k}r_{22,k} - r_{13,k} = 0, \quad (\text{A.9a})$$

$$r_{31,k}r_{12,k} - r_{11,k}r_{32,k} - r_{23,k} = 0, \quad (\text{A.9b})$$

$$r_{11,k}r_{22,k} - r_{21,k}r_{12,k} - r_{33,k} = 0, \quad (\text{A.9c})$$

$$r_{22,k}r_{33,k} - r_{32,k}r_{23,k} - r_{11,k} = 0, \quad (\text{A.9d})$$

$$r_{32,k}r_{13,k} - r_{12,k}r_{33,k} - r_{21,k} = 0, \quad (\text{A.9e})$$

$$r_{12,k}r_{23,k} - r_{22,k}r_{13,k} - r_{31,k} = 0, \quad (\text{A.9f})$$

$$r_{23,k}r_{31,k} - r_{33,k}r_{21,k} - r_{12,k} = 0, \quad (\text{A.9g})$$

$$r_{33,k}r_{11,k} - r_{13,k}r_{31,k} - r_{22,k} = 0, \quad (\text{A.9h})$$

$$r_{13,k}r_{21,k} - r_{23,k}r_{11,k} - r_{32,k} = 0. \quad (\text{A.9i})$$

Equivalently, for every $k = 0, \dots, N-1$, the corresponding six equations (5.46e) in $\mathbf{F}_k^\top \mathbf{F}_k = \mathbf{I}$ are

$$f_{11,k}^2 + f_{21,k}^2 + f_{31,k}^2 - 1 = 0, \quad (\text{A.10a})$$

$$f_{12,k}^2 + f_{22,k}^2 + f_{32,k}^2 - 1 = 0, \quad (\text{A.10b})$$

$$f_{13,k}^2 + f_{23,k}^2 + f_{33,k}^2 - 1 = 0, \quad (\text{A.10c})$$

$$f_{11,k}f_{12,k} + f_{21,k}f_{22,k} + f_{31,k}f_{32,k} = 0, \quad (\text{A.10d})$$

$$f_{12,k}f_{13,k} + f_{22,k}f_{23,k} + f_{32,k}f_{33,k} = 0, \quad (\text{A.10e})$$

$$f_{11,k}f_{13,k} + f_{21,k}f_{23,k} + f_{31,k}f_{33,k} = 0. \quad (\text{A.10f})$$

The remaining nine equations of (5.46e) for $\mathbf{F}_k$ are

$$f_{21,k}f_{32,k} - f_{31,k}f_{22,k} - f_{13,k} = 0, \quad (\text{A.11a})$$

$$f_{31,k}f_{12,k} - f_{11,k}f_{32,k} - f_{23,k} = 0, \quad (\text{A.11b})$$

$$f_{11,k}f_{22,k} - f_{21,k}f_{12,k} - f_{33,k} = 0, \quad (\text{A.11c})$$

$$f_{22,k}f_{33,k} - f_{32,k}f_{23,k} - f_{11,k} = 0, \quad (\text{A.11d})$$

$$f_{32,k}f_{13,k} - f_{12,k}f_{33,k} - f_{21,k} = 0, \quad (\text{A.11e})$$

$$f_{12,k}f_{23,k} - f_{22,k}f_{13,k} - f_{31,k} = 0, \quad (\text{A.11f})$$

$$f_{23,k}f_{31,k} - f_{33,k}f_{21,k} - f_{12,k} = 0, \quad (\text{A.11g})$$

$$f_{33,k}f_{11,k} - f_{13,k}f_{31,k} - f_{22,k} = 0, \quad (\text{A.11h})$$

$$f_{13,k}f_{21,k} - f_{23,k}f_{11,k} - f_{32,k} = 0. \quad (\text{A.11i})$$

Discrete Bounds The relative-rotation (5.46f) and control bounds (5.46g) become

$$f_{11,k} + f_{22,k} + f_{33,k} - 1 - 2\cos(\Delta\theta_{\max}) \geq 0, \quad k = 0, \dots, N-1, \quad (\text{A.12a})$$

$$1 - \bar{u}_{p1,k}^2 - \bar{u}_{p2,k}^2 - \bar{u}_{p3,k}^2 \geq 0, \quad k = 1, \dots, N-1. \quad (\text{A.12b})$$

Initial Values The fixed initial matrices (5.46h) are imposed entrywise as

$$r_{ij,0} = r_{ij}^{\text{init}}, \quad f_{ij,0} = f_{ij}^{\text{init}}, \quad i, j = 1, 2, 3. \quad (\text{A.13})$$

These equations give the complete scalar POP used for the controlled 3D-Pendulum relaxation.

A.2 Scalar Implementation of the Acrobot Optimization Problem

This appendix expands the reduced matrix formulation (5.56) into the scalar polynomial equations used to construct the sparse semidefinite relaxation. It uses exactly the notation of Section 5.4. Thereby, the normalized control and constraint multipliers are denoted by $\bar{u}_k$, $\bar{\lambda}_{0,k}$, and $\bar{\lambda}_{12,k}$. The center-of-mass positions $\mathbf{x}_{i,k}$ and velocities $\mathbf{v}_{i,k}$ are reconstructed from the rotations.

Reduced Variables For $i \in \{1, 2\}$, let

$$\mathbf{R}_{i,k} = \begin{bmatrix} c_{i,k} & -s_{i,k} \\ s_{i,k} & c_{i,k} \end{bmatrix}, \quad \mathbf{F}_{i,k} = \begin{bmatrix} a_{i,k} & -b_{i,k} \\ b_{i,k} & a_{i,k} \end{bmatrix}. \quad (\text{A.14})$$

For the uniform-rod geometry used in the implementation,

$$\boldsymbol{\rho}_{1,0} = \begin{bmatrix} 0 \\ l_1/2 \end{bmatrix}, \quad \boldsymbol{\rho}_{1,12} = \begin{bmatrix} 0 \\ -l_1/2 \end{bmatrix}, \quad \boldsymbol{\rho}_{2,12} = \begin{bmatrix} 0 \\ l_2/2 \end{bmatrix}. \quad (\text{A.15})$$

Therefore, the holonomic geometry reconstructs the center-of-mass positions as

$$\mathbf{x}_{1,k} = \mathbf{p}_0 + \begin{bmatrix} \frac{l_1}{2} s_{1,k} \\ -\frac{l_1}{2} c_{1,k} \end{bmatrix}, \quad (\text{A.16a})$$

$$\mathbf{x}_{2,k} = \mathbf{p}_0 + \begin{bmatrix} l_1 s_{1,k} + \frac{l_2}{2} s_{2,k} \\ -l_1 c_{1,k} - \frac{l_2}{2} c_{2,k} \end{bmatrix}. \quad (\text{A.16b})$$

Consequently, $\mathbf{x}_{1,k}$ and $\mathbf{x}_{2,k}$ are not optimization variables.

Objective Using $\mathbf{R}_{i,k+1} = \mathbf{R}_{i,k} \mathbf{F}_{i,k}$ and $\mathbf{R}_{i,k-1} = \mathbf{R}_{i,k} \mathbf{F}_{i,k-1}^\top$, the matrix $\mathbf{A}_{i,k} = \mathbf{F}_{i,k} - 2\mathbf{I}_2 + \mathbf{F}_{i,k-1}^\top$, introduced in the reduced translational dynamics of Section 5.4, contains only $\mathbf{F}_{i,k-1}$ and $\mathbf{F}_{i,k}$.

Defining

$$\begin{aligned} \mathbf{R}_i^{\text{des}} &= \begin{bmatrix} c_i^{\text{des}} & -s_i^{\text{des}} \\ s_i^{\text{des}} & c_i^{\text{des}} \end{bmatrix}, \\ \mathbf{F}_i^{\text{des}} &= \begin{bmatrix} a_i^{\text{des}} & -b_i^{\text{des}} \\ b_i^{\text{des}} & a_i^{\text{des}} \end{bmatrix}. \end{aligned} \quad (\text{A.17})$$

the structured $SO(2)$ matrices are $\|\mathbf{R} - \mathbf{R}^{\text{des}}\|_F^2 = 2[(c - c^{\text{des}})^2 + (s - s^{\text{des}})^2]$. Therefore, the scalar objective polynomial (5.56a) is

$$\begin{aligned} p_A &= 2w_R \sum_{i=1}^2 [(c_{i,N} - c_i^{\text{des}})^2 + (s_{i,N} - s_i^{\text{des}})^2] \\ &\quad + 2w_F \sum_{i=1}^2 [(a_{i,N-1} - a_i^{\text{des}})^2 + (b_{i,N-1} - b_i^{\text{des}})^2] \\ &\quad + 2\alpha_R \sum_{k=0}^{N-1} \sum_{i=1}^2 [(c_{i,k} - c_i^{\text{des}})^2 + (s_{i,k} - s_i^{\text{des}})^2] \\ &\quad + 2\alpha_F \sum_{k=0}^{N-2} \sum_{i=1}^2 [(a_{i,k} - a_i^{\text{des}})^2 + (b_{i,k} - b_i^{\text{des}})^2] + w_u^{-1} \sum_{k=1}^{N-1} \bar{u}_k^2. \end{aligned} \quad (\text{A.18})$$

Reduced Discrete Translational Dynamics The normalized variables are mapped to the physical quantities exactly as in Section 5.4:

$$u_k^{\text{phys}} = u_{\max} \bar{u}_k, \quad \lambda_{0,k}^{(j)} = \lambda_{\max} \bar{\lambda}_{0,k}^{(j)}, \quad \lambda_{12,k}^{(j)} = \lambda_{\max} \bar{\lambda}_{12,k}^{(j)}, \quad j = 1, 2. \quad (\text{A.19})$$

For the geometry in (A.15) and the gravity direction $\mathbf{e}_2 = [0 \ -1]^\top$, the four scalar components of the reduced translational dynamics (5.56b)-(5.56c) are

$$m_1 \frac{l_1}{2} [s_{1,k}(a_{1,k} + a_{1,k-1} - 2) + c_{1,k}(b_{1,k} - b_{1,k-1})] - h^2 \lambda_{\max} (\bar{\lambda}_{0,k}^{(1)} + \bar{\lambda}_{12,k}^{(1)}) = 0, \quad (\text{A.20a})$$

$$- m_1 \frac{l_1}{2} [c_{1,k}(a_{1,k} + a_{1,k-1} - 2) - s_{1,k}(b_{1,k} - b_{1,k-1})] + h^2 m_1 g - h^2 \lambda_{\max} (\bar{\lambda}_{0,k}^{(2)} + \bar{\lambda}_{12,k}^{(2)}) = 0, \quad (\text{A.20b})$$

$$m_2 \left[l_1 (s_{1,k}(a_{1,k} + a_{1,k-1} - 2) + c_{1,k}(b_{1,k} - b_{1,k-1})) + \frac{l_2}{2} (s_{2,k}(a_{2,k} + a_{2,k-1} - 2) + c_{2,k}(b_{2,k} - b_{2,k-1})) \right] + h^2 \lambda_{\max} \bar{\lambda}_{12,k}^{(1)} = 0, \quad (\text{A.20c})$$

$$- m_2 \left[l_1 (c_{1,k}(a_{1,k} + a_{1,k-1} - 2) - s_{1,k}(b_{1,k} - b_{1,k-1})) + \frac{l_2}{2} (c_{2,k}(a_{2,k} + a_{2,k-1} - 2) - s_{2,k}(b_{2,k} - b_{2,k-1})) \right] + h^2 m_2 g + h^2 \lambda_{\max} \bar{\lambda}_{12,k}^{(2)} = 0, \quad (\text{A.20d})$$

for $k = 1, \dots, N - 1$.

Discrete Rotational Dynamics For the geometry (A.15), the scalar moments generated by the constraint forces are

$$\tau_{\lambda,1,k}^{(0)} = -\frac{l_1}{2} \lambda_{\max} (c_{1,k} \bar{\lambda}_{0,k}^{(1)} + s_{1,k} \bar{\lambda}_{0,k}^{(2)}), \quad (\text{A.21a})$$

$$\tau_{\lambda,1,k}^{(12)} = \frac{l_1}{2} \lambda_{\max} (c_{1,k} \bar{\lambda}_{12,k}^{(1)} + s_{1,k} \bar{\lambda}_{12,k}^{(2)}), \quad (\text{A.21b})$$

$$\tau_{\lambda,2,k}^{(12)} = -\frac{l_2}{2} \lambda_{\max} (c_{2,k} \bar{\lambda}_{12,k}^{(1)} + s_{2,k} \bar{\lambda}_{12,k}^{(2)}). \quad (\text{A.21c})$$

Here, $\tau_{\lambda,1,k}^{(0)} = \boldsymbol{\rho}_{1,0} \times (\mathbf{R}_{1,k}^\top \boldsymbol{\lambda}_{0,k})$, $\tau_{\lambda,1,k}^{(12)} = \boldsymbol{\rho}_{1,12} \times (\mathbf{R}_{1,k}^\top \boldsymbol{\lambda}_{12,k})$, and $\tau_{\lambda,2,k}^{(12)} = \boldsymbol{\rho}_{2,12} \times (\mathbf{R}_{2,k}^\top \boldsymbol{\lambda}_{12,k})$. Accordingly, the moment convention in the matrix POP is $\tau_{\lambda,1,k} = \tau_{\lambda,1,k}^{(0)} + \tau_{\lambda,1,k}^{(12)}$ and $\tau_{\lambda,2,k} = -\tau_{\lambda,2,k}^{(12)}$. Therefore, the two independent scalar rotational equations (5.56d)-(5.56e) are

$$\text{tr}(\mathbf{J}_{d,1})(b_{1,k-1} - b_{1,k}) + h^2 (\tau_{\lambda,1,k} - u_{\max} \bar{u}_k) = 0, \quad (\text{A.22a})$$

$$\text{tr}(\mathbf{J}_{d,2})(b_{2,k-1} - b_{2,k}) + h^2 (u_{\max} \bar{u}_k + \tau_{\lambda,2,k}) = 0, \quad (\text{A.22b})$$

for $k = 1, \dots, N - 1$. Equations (A.20) and (A.22) are all quadratic polynomials in the reduced decision variables.

Discrete Kinematics For each link, $\mathbf{R}_{i,k+1} = \mathbf{R}_{i,k} \mathbf{F}_{i,k}$ gives the two independent scalar kinematic equations

$$c_{i,k+1} - c_{i,k} a_{i,k} + s_{i,k} b_{i,k} = 0, \quad (\text{A.23a})$$

$$s_{i,k+1} - s_{i,k} a_{i,k} - c_{i,k} b_{i,k} = 0, \quad (\text{A.23b})$$

for $i = 1, 2$ and $k = 0, \dots, N - 1$.

Discrete Rotation Constraints Using the structured parameterization (A.14), the $SO(2)$ constraints (5.56g)-(5.56h) reduce to

$$c_{i,k}^2 + s_{i,k}^2 - 1 = 0, \quad i = 1, 2, \quad k = 0, \dots, N, \quad (\text{A.24a})$$

$$a_{i,k}^2 + b_{i,k}^2 - 1 = 0, \quad i = 1, 2, \quad k = 0, \dots, N - 1. \quad (\text{A.24b})$$

Discrete Bounds The increment (5.56i), control (5.56j), and componentwise multiplier bounds (5.56k) are

$$a_{i,k} - \cos(\Delta\theta_{i,\max}) \geq 0, \quad i = 1, 2, \quad k = 1, \dots, N - 1, \quad (\text{A.25a})$$

$$1 - \bar{u}_k^2 \geq 0, \quad k = 1, \dots, N - 1, \quad (\text{A.25b})$$

$$1 - (\bar{\lambda}_{0,k}^{(j)})^2 \geq 0, \quad j = 1, 2, \quad k = 1, \dots, N - 1, \quad (\text{A.25c})$$

$$1 - (\bar{\lambda}_{12,k}^{(j)})^2 \geq 0, \quad j = 1, 2, \quad k = 1, \dots, N - 1. \quad (\text{A.25d})$$

Thus, the physical quantities satisfy $|u_k^{\text{phys}}| \leq u_{\max}$ and $|\lambda_{0,k}^{(j)}|, |\lambda_{12,k}^{(j)}| \leq \lambda_{\max}$.

Initial Values Finally, defining the initial rotation matrices as

$$\mathbf{R}_i^{\text{init}} = \begin{bmatrix} c_i^{\text{init}} & -s_i^{\text{init}} \\ s_i^{\text{init}} & c_i^{\text{init}} \end{bmatrix}, \quad \mathbf{F}_i^{\text{init}} = \begin{bmatrix} a_i^{\text{init}} & -b_i^{\text{init}} \\ b_i^{\text{init}} & a_i^{\text{init}} \end{bmatrix}. \quad (\text{A.26})$$

The implementation fixes $\mathbf{R}_{i,1} = \mathbf{R}_i^{\text{init}}$, $\mathbf{F}_{i,0} = \mathbf{F}_i^{\text{init}}$, and $\mathbf{R}_{i,0} = \mathbf{R}_i^{\text{init}}(\mathbf{F}_i^{\text{init}})^\top$. Entrywise, this is

$$c_{i,1} = c_i^{\text{init}}, \quad s_{i,1} = s_i^{\text{init}}, \quad a_{i,0} = a_i^{\text{init}}, \quad b_{i,0} = b_i^{\text{init}}, \quad (\text{A.27a})$$

$$c_{i,0} = c_i^{\text{init}} a_i^{\text{init}} + s_i^{\text{init}} b_i^{\text{init}}, \quad s_{i,0} = s_i^{\text{init}} a_i^{\text{init}} - c_i^{\text{init}} b_i^{\text{init}}, \quad (\text{A.27b})$$

for $i = 1, 2$. For the initial rest condition used in the numerical study, $a_i^{\text{init}} = 1$ and $b_i^{\text{init}} = 0$, such that $\mathbf{R}_{i,0} = \mathbf{R}_{i,1} = \mathbf{R}_i^{\text{init}}$. Together with the objective (A.18), equations (A.20)- (A.27) give the complete scalar POP implemented for the Acrobot relaxation.

Appendix B

Validation of the Reduced Acrobot Dynamics

The reduction used in Section 5.4 is validated against two references. The first reference is the full maximal-coordinate LGVI used in the numerical simulation of Section 6.1.2, which propagates $\mathbf{x}_{i,k}$, $\mathbf{v}_{i,k}$, and $\mathbf{R}_{i,k}$ while enforcing the base and elbow constraints. The second comparison model is the continuous joint-coordinate Acrobot model integrated with DOP853 [HNW93]. It serves as a high-accuracy numerical approximation of the physical dynamics.

The reduced LGVI eliminates the position and velocity variables, as derived in Section 5.4.

For the discretization time steps $h \in \{0.1, 0.05\}$ s, a constant torque is applied over $T_c = 0.1$ s. Another setting uses $h = 0.0025$ s and $T_c = 0.025$ s for all cases,

$$u \in \{-20, -10, -5, 0, 5, 10, 20\} \text{ N m.}$$

The terminal and trajectory errors are

$$e_{\theta,f} = \sqrt{\Delta\theta_{1,N_c}^2 + \Delta\theta_{2,N_c}^2}, \quad e_{\theta,\text{RMSE}} = \sqrt{\frac{1}{2(N_c + 1)} \sum_{k=0}^{N_c} (\Delta\theta_{1,k}^2 + \Delta\theta_{2,k}^2)}, \quad (\text{B.1})$$

where $N_c = T_c/h$. Table B.1 reports the mean over all tested inputs. Since the full and reduced LGVIs coincide within the reported precision, one common error pair is shown.

The reduction from $h = 0.1$ s to $h = 0.05$ s over the same physical interval decreases both errors, although deviations of several degrees remain for larger inputs. $h = 0.0025$ s additionally reduces the errors to the order of 10^{-3} degrees.

Table B.2 separates the influence of the input size. For $|u| > 0$, the values are averaged over the corresponding positive and negative torques. The error metrics are reported in degrees.

The error increases with $|u|$, because stronger torques produce larger intra-step changes that are represented less accurately by a coarse finite step. For $h = 0.0025$ s

Table B.1: Mean angular errors of the full and reduced maximal-coordinate LGVIs relative to the continuous physical reference DOP853.

h [s]	T_c [s]	Steps	$e_{\theta,f}$ [°]	$e_{\theta,\text{RMSE}}$ [°]
0.1000	0.100	1	7.758	3.879
0.0500	0.100	2	4.202	1.752
0.0025	0.025	10	0.003	0.001

Table B.2: Mean angular errors of the full and reduced LGVIs grouped by control size relative to the continuous reference, with configuration sets (h, T_c) .

$ u $ [N m]	(0.05 s, 0.1 s)		(0.0025 s, 0.025 s)	
	$e_{\theta,f}$ [°]	$e_{\theta,\text{RMSE}}$ [°]	$e_{\theta,f}$ [°]	$e_{\theta,\text{RMSE}}$ [°]
0	0.041	0.017	0.000026	0.000008
5	1.860	0.773	0.001144	0.000346
10	4.074	1.693	0.002512	0.000749
20	8.752	3.658	0.006794	0.001953

and $T_c = 0.025$ s, the mean terminal error remains below 0.01° , even at $|u| = 20$ N m. These results verify that the reduction preserves the full maximal-coordinate LGVI and that the selected fine simulation scale provides close agreement with the continuous Acrobot dynamics.

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Acronyms

CPU	central processing unit
CS	correlative sparsity
CSP	correlative sparsity pattern
FASRC	Faculty of Arts and Sciences Research Computing
GPU	graphics processing unit
IPOPT	Interior Point OPTimizer
LGVI	Lie group variational integrator
LQR	linear-quadratic regulator
MD	minimum degree
MF	minimum fill
MPC	model predictive control
NLP	nonlinear programming
POP	polynomial optimization problem
PRM	probabilistic roadmap
RAM	random-access memory
RK4	fourth-order Runge–Kutta method
RMSE	root-mean-square error
RRT	rapidly exploring random tree
SDP	semidefinite programming
SOS	sum of squares
SPOT	Sparse Polynomial Optimization Toolbox
TS	term sparsity

Notation

x	italic lowercase letter: scalar
$\boldsymbol{x}$	bold lowercase letter: vector
$\mathbf{X}$	bold uppercase letter: matrix
$\mathcal{X}$	calligraphic uppercase letter: set, map, or graph object
g, f, h	lowercase letter: element of an abstract Lie group
G, H	uppercase letter: abstract matrix Lie group
$\mathfrak{g}$	Lie algebra of G , with $\mathfrak{g} = T_e G$
$\widehat{\mathfrak{g}^*}$	dual space of the Lie algebra
$(\cdot)$	hat map from vector coordinates to a skew-symmetric matrix
$(\cdot)^\vee$	inverse of the hat map
$(\cdot)^\top$	matrix or vector transpose
$\dot{x}$	time derivative of x
δx	infinitesimal variation of x
$(\cdot)_{i,k}$	body or component index i and discrete-time index k
$(\cdot)^{\text{init}}, (\cdot)^{\text{des}}$	initial and desired values
$(\cdot)^*$	optimal value or optimizer
$\bar{x}$	normalized or scaled quantity
$\hat{x}$	estimated or extracted candidate value
$(\cdot)^{\text{P}}, (\cdot)^{\text{A}}$	3D-pendulum and Acrobot quantities
$\ \boldsymbol{x}\ _2, \ \mathbf{X}\ _F$	Euclidean and Frobenius norms
$\langle \boldsymbol{\mu}, \boldsymbol{\xi} \rangle$	duality pairing or matrix trace inner product
$\mathbf{X} \succeq \mathbf{0}$	$\mathbf{X}$ is positive semidefinite

List of Symbols

t, t_0, t_f, t_k	continuous time, interval endpoints, and discrete time instant
h	integration or discretization time step
N	number of discrete intervals or optimization horizon length
i, j, k	body, MPC-iteration, and discrete-time indices, respectively
$n_b, n_c, n_{cl}, N_{cl}, n_z$	numbers of bodies, scalar constraints, variables per clique, cliques, and decision variables
d, n	spatial or algebraic dimensions; d also denotes a polynomial degree locally
$GL(n), SO(n), SE(n)$	general linear, rotation, and rigid-motion groups
$G, \mathfrak{g}, \mathfrak{g}^*$	matrix Lie group, its Lie algebra, and the algebra dual
$\mathbf{g}_k \in G$	generic Lie-group configuration at time step k
$\mathbf{f}_k \in G$	relative group update, $\mathbf{g}_{k+1} = \mathbf{g}_k \mathbf{f}_k$
ξ	left-trivialized, body-fixed velocity in $\mathfrak{g}$
$\eta_k, \mathbf{X}_k, \epsilon$	Lie-algebra variations of $\mathbf{g}_k$ and $\mathbf{f}_k$, and the variation parameter
$\mathbb{L}_h, T_e \mathbb{L}_g, T_e^* \mathbb{L}_g$	left translation, its differential, and its dual
$\text{Ad}_g, \text{ad}_\xi$	adjoint operator and its Lie-algebra derivative; a star denotes the dual
L, L_d	continuous and discrete Lagrangians
$\mathcal{S}, S_d$	continuous action integral and discrete action sum
T, U, V, E_k	kinetic, potential, and total mechanical energy
$\mathbb{F}L, \mathbb{F}^\pm L_d$	continuous and discrete Legendre transformations
$\mu_k \in \mathfrak{g}^*$	generic body-fixed momentum
$\Theta_L, \Omega_L, \Theta_{L_d}^\pm, \Omega_{L_d}$	continuous and discrete Lagrangian one-forms and symplectic two-forms
J_L	Lagrangian momentum map associated with a symmetry
$\mathbf{R}_{i,k}$	orientation of body i , mapping body-frame vectors to the space frame

$\mathbf{F}_{i,k}$	relative rotation, $\mathbf{R}_{i,k+1} = \mathbf{R}_{i,k}\mathbf{F}_{i,k}$
q_k	stacked maximal-coordinate configuration at time step k
$Q_{\max}, Q_c$	unconstrained maximal-coordinate manifold and constrained configuration set
$\mathbf{x}_{i,k}, \mathbf{v}_{i,k}$	center-of-mass position and velocity of body i
$\phi(q), \phi_0, \phi_{12}$	holonomic constraint map, base constraint, and elbow constraint
$\mathbf{G}_{x_i,k}$	Jacobian of the holonomic constraints with respect to $\mathbf{x}_{i,k}$
$\lambda_k, \lambda_{0,k}, \lambda_{12,k}$	Lagrange multipliers for all, base, and elbow constraints
ρ_c	body-fixed vector from the pendulum pivot to its center of mass
$\rho_{1,0}, \rho_{1,12}, \rho_{2,12}$	body-fixed vectors from a link center of mass to a joint
l, l_i, r	pendulum length, Acrobot link length, and cylinder radius
p_0	fixed Acrobot base position in the space frame
$m_i, \mathbf{M}_i$	mass and translational mass matrix of body i
$\mathbf{J}_i, \mathbf{J}_{\text{com}}, J_i^{\text{com}}, \mathbf{J}_{d,i}, j_{d,1}, j_{d,2}, j_{d,3}$	physical, center-of-mass, and nonstandard inertias; $j_{d,1}, j_{d,2}, j_{d,3}$ are the diagonal entries of $\mathbf{J}_d$
$\omega_k, \omega_{i,k}$	3D body-fixed angular-velocity vector and planar link angular velocity
$\Pi_{i,k}$	discrete body-fixed angular momentum
$g, \mathbf{e}_2, \mathbf{e}_3$	gravitational acceleration and gravity-direction unit vectors in 2D and 3D
$\mathbf{u}_k, \mathbf{u}_{p,k}, u_k$	3D body moment, its control parameter, and Acrobot elbow torque
$\tau_{\lambda,i,k}$	constraint-induced moment on Acrobot link i at step k
$\tau_{g,k}$	body-fixed moment due to gravity
$\mathbf{A}_{i,k}$	relative-rotation difference matrix in the reduced Acrobot translation dynamics
$u_{\max}, \lambda_{\max}, \Delta\theta_{\max}, \Delta\theta_{i,\max}$	control, multiplier, and per-step rotation bounds
$\mathbf{Y}_k, \mathbf{U}_k, \mathbf{U}_{\min}, \mathbf{U}_{\max}$	general robot state, control input, and component-wise control bounds
$\Psi, \psi, \mathcal{D}_i, \mathcal{A}_k, \mathcal{Y}_i, \mathcal{H}$	terminal and stage costs, dynamics, admissible set, state constraints, and inequalities
$\mathbf{z} \in \mathbb{R}^{n_z}$	complete polynomial-optimization decision vector
$p(\mathbf{z}), p^*$	polynomial objective and its global optimum
$h_l(\mathbf{z}), g_j(\mathbf{z}), \mathcal{K}$	polynomial equalities, inequalities, and their feasible set

$\mathcal{B}, \mathcal{E}, \mathcal{J}$	body, equality-constraint, and inequality-constraint index sets
$\mathbf{X}, \mathbf{C}, \mathbf{A}_i, \mathcal{A}, \mathbf{b}$	standard SDP variable, cost and constraint matrices, linear map, and right-hand side
$\mathbf{z}^\alpha, [\mathbf{z}]_d, s(n_z, d)$	monomial, monomial basis, and its dimension
$\sigma_j, \chi_l, \mathbf{Q}, \Sigma[\mathbf{z}]$	SOS and equality multipliers, Gram matrix, and the SOS cone
κ	Moment–SOS relaxation order
$y_\alpha, \mathbf{y}, \mathcal{L}_y$	scalar moment, moment sequence, and Riesz functional
$\mathbf{M}_\kappa(\mathbf{y}), \mathbf{M}_{\kappa,k}$	order- κ moment or localizing matrix and its clique-specific block
$\gamma_\kappa^*, \beta_\kappa^*$	SOS and moment lower bounds at relaxation order κ
$\hat{\mathbf{z}}$	extracted feasible trajectory or candidate decision vector
$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \mathcal{C}$	correlative-sparsity graph and one of its cliques
$\mathbf{z}(\mathcal{I}_k), \mathbf{z}_P(\mathcal{I}_k^P), \mathbf{z}_A(\mathcal{I}_k^A)$	generic, pendulum, and Acrobot local decision vectors selected by clique index sets
$c_{i,k}, s_{i,k}, a_{i,k}, b_{i,k}$	cosine-sine coordinates of $\mathbf{R}_{i,k}$ and $\mathbf{F}_{i,k}$
p_P, p_A, p_P^*, p_A^*	pendulum and Acrobot POP objective polynomials and their optimal values
$w_R, w_F, \alpha_R, \alpha_F, w_u, \mathbf{Q}_x, \mathbf{Q}_u$	terminal, stage, and control-effort objective weights and weighting matrices
$\bar{\mathbf{u}}_{p,k}, \bar{u}_k, \bar{\boldsymbol{\lambda}}_k$	normalized control parameters, torque, and constraint multipliers
$\Delta t_{\text{SDP}}, \Delta t_{\text{sim}}, \Delta t_{\text{ctrl}}$	prediction-model, simulation, and control-update intervals
$N_{\text{SDP}}, T_{\text{pred}}, n_{\text{sim}}$	prediction nodes, prediction horizon, and simulation steps per control interval
$k \mid j$	prediction node k in the SDP solved at MPC iteration j
$\delta_k, \delta_1, \delta_{\max}$	moment-matrix eigenvalue-ratio diagnostics for clique k , the first clique, and the worst clique
$\nu_{1,k}, \nu_{2,k}$	largest and second-largest eigenvalues of clique moment matrix k
ε_N	Newton-iteration convergence tolerance
$\epsilon_{\text{warm}}, \epsilon_{\text{cold}}$	relative SDP–NLP suboptimality gaps using warm- and cold-start upper bounds
$J_{\text{SDP}}, J_{\text{NLP}}$	SDP lower-bound and feasible NLP objective values
$\mathbf{z}_{A,k}, \mathbf{A}_{\text{acro}}, \mathbf{A}_{T,k}, \mathbf{A}_{R,k}$	Acrobot simulation unknown, nonlinear residual, and its translational and rotational blocks

$\theta_{i,k}, \Delta\theta_{i,k}, \theta_{x,\text{des}}, \theta_i^{\text{des}}, \theta_A^{\text{des}}$	link and relative-step angles, 3D-pendulum target angle, and Acrobot link and common target angles
$\mathbf{f}, \mathbf{a}, \mathbf{A}(\mathbf{f})$	Cayley vector, coefficient vector, and nonlinear residual for the 3D-pendulum update
$\Delta E_k, \Delta\mu_k, \Delta\mathbf{x}_k$	energy deviation, momentum error, and trajectory-position difference
$e_{\text{orth},i,k}, e_{\text{orth}}^{\text{P,max}}, e_{\text{orth}}^{\text{A,max}}$ $e_{R,k}^{\text{P}}, e_{R,k}^{\text{A}}, e_{R,j}, e_{F,j}$	per-step and maximum rotation-orthogonality errors open-loop pendulum and Acrobot attitude errors and MPC attitude and relative-step errors
$\varepsilon_R, \varepsilon_F, n_{\text{stable}}, N_{\text{MPC}}$	stabilization tolerances, dwell count, and MPC iteration limit
$T_c, N_c, e_{\theta,f}, e_{\theta,\text{RMSE}}$	validation interval and step count, terminal error, and root-mean-square angle error

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